

वार्षिक प्रतिवेदन ANNUAL REPORT 2020



भा.कृ.अनु.प.-राष्ट्रीय पादप जैवप्रौद्योगिकी संस्थान
लाल बहादुर शास्त्री भवन, पूसा परिसर, नई दिल्ली-110012

ICAR-NATIONAL INSTITUTE FOR PLANT BIOTECHNOLOGY
Lal Bahadur Shastri Centre, Pusa Campus, New Delhi - 110012 (India)

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आमुख

भा. कृ.अ.प.—राष्ट्रीय पादप जैवप्रौद्योगिकी संस्थान (भा.कृ.अ.प.—एनआईपीबी) भारतीय कृषि को अति वांछित गति प्रदान करने के लिए कृषि जैवप्रौद्योगिकी का एक प्रमुख अनुसंधान संस्थान है। यह संस्थान पादप आण्विक विज्ञान और जैवप्रौद्योगिकी के क्षेत्र में कार्मिकों के अनुसंधान, शिक्षण एवं प्रशिक्षण में कार्यरत है। पांच अधिदेशित फसलों नामतः चावल, गेहूं, चना, अरहर और सरसों में मौलिक, कार्यनीतिपरक और व्यावहारिक अनुसंधान में उल्लेखनीय योगदान दिया जा रहा है। अजैविक एवं जैविक प्रतिबलों के लिए नए प्रमोटरों, पराजीनियों और जीनों की पहचान; जीनोमिक और आण्विक मार्करों, उत्पादकता बढ़ाने के लिए जैवप्रौद्योगिकीय युक्तियों पर विशेष बल दिया गया है। वैश्विक महामारी के बावजूद वर्ष 2020 में इस संस्थान के अनेक योगदान हैं।

भा.कृ.अ.प.—एनआईपीबी में चावल वैज्ञानिकों द्वारा नगीना 22 में उत्परिवर्तनों के लक्षण—वर्णन पर ध्यान केन्द्रित किया जा रहा है और इसके अंतर्गत सात पुष्पगुच्छ आर्कीटेक्चर उत्परिवर्तनों की पहचान कर ली गई हैं। वन्य तथा बाउ चावल से प्रवणता के प्रति सहिष्णु, पत्राच्छद और जलमग्नता के प्रति सहिष्णुता वाले चावल के वंशक्रमों की पहचान की गई है और उनका लक्षण—वर्णन किया जा रहा है। नाइट्रोजन की चिरकालिक कम आपूर्ति और सूखा प्रतिबल की अवस्था

के अंतर्गत पौध अवस्था में नगीना 22 और आईआर 64 से डीईजी, क्यूटीएल और प्रत्याशी जीन पहचाने गए हैं तथा अकेले अथवा इन दोनों के संयोग से चावल सुधार तथा कार्यात्मक विज्ञान के लिए प्रमुख संसाधन सिद्ध हो सकते हैं। गेहूं की फसल पर पर्यावरण पर पड़ने वाले विभिन्न प्रभावों से निपटने के लिए संस्थान में अनेक अध्ययन किए जा रहे हैं, पोषक उपयोग की दक्षता, जैवप्रौद्योगिकी कार्यनीतियों के माध्यम से अजैविक और जैविक प्रतिकूल स्थितियों से निपटने पर विशेष ध्यान दिया जा रहा है। सरसों में फसल सुधार के लिए गहन अनुसंधान किए जा रहे हैं। ऐसे पादप प्रकारों के विकास पर अनुसंधान हो रहे हैं जो भारतीय सरसों के प्रमुख कीट—पीड़कों व रोगजनकों के प्रतिरोधी हैं। इस दिशा में आरएनएआई का उपयोग करके माहू और आल्टर्नैरिया पत्ती धब्बा रोग के प्रति प्रतिरोध, पराजीनियों के माध्यम से प्रोटीएजन्ट इनहिबिटर के समाहन और वन्य *डिफ्लोटेक्सिस एरुकोइडेस* के दोहन पर अनुसंधान किए जा रहे हैं। इसके अतिरिक्त जैवप्रौद्योगिकीय कार्यनीतियों का उपयोग करके सरसों के रोगजनकों *एस. स्कलेरोटियोसम* और *एल्ब्यूगो कैडिडा* के नियंत्रण पर भी बल दिया जा रहा है। भा. कृ.अ.प.—एनआईपीबी ने चने की पूसा चिकपी 20211 (पूसा चना मानव) के विकास में भी भागेदारी की है जिसे भारतीय कृषि अनुसंधान संस्थान के सहयोग



से देश के मध्य अंचल के लिए जारी किया गया है। इस अंचल में गुजरात, महाराष्ट्र और मध्य प्रदेश राज्य आते हैं। यह किस्म पूसा 391 के समाहन वंशक्रम (डब्ल्यूआरआईएल-3) मुझान प्रतिरोधी है तथा इसमें एमएबीसी के माध्यम से डब्ल्यूआर 315 से विल्ट क्यूटीएल 1, 3, 4 और 5 एलजी 2 पर समाहित किए गए हैं। लेपिडोप्टेरन पीड़क, *हैलिकोवर्पा आर्मीजेरा* के प्रबंधन में cry1I-टाइप और vip3Aa-टाइप जीन के नए युग्मविकल्पी विलगित किए गए हैं और उनका लक्षण—वर्णन किया गया है। इन जीनों में पीड़क नियंत्रण की पर्याप्त क्षमता है।

विभिन्न अधिदेशित फसलों पर चल रहे कार्य के अतिरिक्त इस संस्थान ने अनेक जीनोमिक कार्यक्रमों

जैसे ग्वार, ओराइज़ा कोएरेक्टेटा, एक विकल्पी लवण रागी, भारतीय चाय (सी. असामिका) के लिए क्लेरोप्लास्ट (cp) तथा कैमेलिया जातियों के लिए माइटोकॉन्ड्रियाई (mt) पर अनेक जीनोमिक कार्यक्रम में सक्रिय रूप से भाग लिया है। भा.कृ.अ.प.—एनआईपीबी ने TeaMiD का सफलतापूर्वक विकास किया है जो चाय के साधारण क्रम रिपीट मार्कर का एक बृहत डेटाबेस है। साथ ही TEnGExA एक R पैकेज आधारित युक्त साधन है जिसके द्वारा किसी भी जाति के लिए जीनों या ट्रांसक्रिप्टों की कितनी ही संख्या का ऊतक समृद्ध विश्लेषण किया जा सकता है।

जैवप्रौद्योगिकी क्षेत्र में नए प्रोटोकाल के विकास पर संस्थान ने सदैव विशेष ध्यान दिया है। कार्यात्मक जीनोमिक्स के सफल उपयोग के लिए अलसी (फ्लेक्स) में सीड्स मैरिस्टेम—लक्षित इन प्लांटा रूपांतरण प्रोटोकाल को उपयुक्ततम बनाया गया है। यह विधि एक आशाजनक युक्ति है और आनुवंशिक रूपांतरण के माध्यम से फ्लेक्स के किस्म सुधार कार्यक्रमों में इसे लागू किया जा सकता है। बीज शुद्धता के लिए किस्मों में मिलावट व दोहराव को हटाने और शुद्धता के मूल्यांकन के लिए बीज की किस्मों को सटीक रूप से वर्गीकृत करने के लिए आईआईटी, दिल्ली के सहयोग से संस्थान में एक अतिउत्कृष्ट ऑप्टिकल कोहेरन्स टोमोग्राफी (ओसीटी) तकनीक विकसित की गई है।

भा.कृ.अ.प.—एनआईपीबी पादप आण्विक विज्ञान तथा जैवप्रौद्योगिकी के क्षेत्र में मानव संसाधन के विकास में निरंतर कार्यशील है। 58वें दीक्षांत समारोह 2020 में पीएच.डी. के सात तथा एम.एससी. के पांच छात्रों को क्रमशः डॉक्टरल और स्नातकोत्तर उपाधियां प्रदान की गईं। डॉ. शिखा दीक्षित को उनके डॉक्टरल अनुसंधान कार्य के लिए भा.कृ.अ.सं. प्रतिभा पदक से सम्मानित किया गया। इसके अलावा भा.कृ.अ.प.—एनआईपीबी प्रसार तथा आउटरीच गतिविधियों जैसे मेरा गांव मेरा गौरव (एमजीएमजी) और अनुसूचित जाति उप योजना (एससीएसपी) में भी सक्रिय रूप से भाग ले रहा है। देश में वैश्विक महामारी की स्थिति के चलते वर्ष 2020 में ये कार्यक्रम वर्चुअल माध्यम से आयोजित किए गए।

भविष्य में भी हम नई चुनौतियां लेते रहेंगे तथा नए जीनों और प्रमोटरों की पहचान करके उनका लक्षण—वर्णन करेंगे, जीनोम को रूपांतरित करने के लिए नई युक्तियां विकसित करेंगे, बेहतर जलवायु समुत्थानशीलता के लिए इन जीनोमों का उपयोग करेंगे; बेहतर स्वास्थ्य और रोगरोधिता के लिए पोषणिक जीनोमिक और नए अणुओं का विकास करेंगे और इसके साथ ही देश की भावी आवश्यकता पूरी करने के लिए अत्यधिक शिक्षित व कुशल जनशक्ति तैयार करेंगे।

मैं भा.कृ.अ.प.—एनआईपीबी के सभी वैज्ञानिक, तकनीकी व प्रशासनिक स्टाफ, छात्रों, अनुसंधान अध्येताओं और संविदा पर कार्यरत सहायी स्टाफ को संस्थान की विभिन्न गतिविधियों में योगदान देने के लिए धन्यवाद देता हूँ। इन गतिविधियों में हिन्दी चेतना मास और स्वच्छ भारत अभियान विशेष रूप से शामिल हैं। मैं डॉ. मोनिका दलाल, डॉ. अमिता मित्रा सेवंथी, डॉ. प्रशांत दाश, डॉ. रोहिणी श्रीवत्स, डॉ. नवीन गुप्ता, डॉ. अमोल सोलंके, डॉ. आर.एस. निरंजन, डॉ. दीपक बिष्ट और डॉ. एम.एस. निम्मी को यह रिपोर्ट तैयार करने में सहायता देने के लिए विशेष रूप से धन्यवाद देना चाहूंगा।

मैं कृषि अनुसंधान एवं शिक्षा विभाग के सचिव तथा भा.कृ.अ.प. के महानिदेशक डॉ. टी. महापात्र, भा.कृ.अ.प. के उप महानिदेशक (फसल विज्ञान), डॉ. टी.आर. शर्मा और सहायक महानिदेशक (बीज) डॉ. डी.के. यादव का संस्थान के समस्त कार्य संचालन में मार्गदर्शन देने और सहायता प्रदान करने के लिए हार्दिक आभार व्यक्त करता हूँ।

अजित्

दिनांक: 14 जुलाई, 2021

(अजित कुमार शासनी)

PREFACE

The ICAR-National Institute for Plant Biotechnology (ICAR-NIPB) is a premier research institute with a vision to impart agricultural biotechnology, a much-needed thrust to Indian Agriculture. The institute is engaged in research, teaching and training of personnel in the area of plant molecular biology and biotechnology. Immense contributions are being made in the five mandated crops, rice, wheat, chickpea, pigeonpea and mustard in the areas of basic, strategic and applied research. Emphasis has been on identification of novel genes, promoters, transgenics and gene editing for abiotic and biotic stress; genomics and molecular markers; biotechnological approaches for increasing productivity. Despite the pandemic, the institute has several contributions to its credit in the year, 2020.

The rice group at ICAR-NIPB is focussing on the characterization of mutants in Nagina 22 and the major achievements are identification of seven panicle architecture mutants. Rice lines tolerant to salinity, sheath blight and submergence tolerance have been identified from wild and Bao rice and are being characterized. DEGs, QTLs and candidate genes identified from Nagina 22 and IR64 at seedling

stage under chronic low N supply and drought stress singly and in combination can form major resources for rice improvement and functional biology. Studies are underway in the institute to tackle various environmental impact faced by wheat. Various aspects of nutrient use efficiency, mitigation of abiotic and biotic stresses through biotechnological strategies are being focussed. Extensive research is being carried out for crop improvement in mustard. Research is underway for developing plant types that are resistant to major insect-pests and pathogens of Indian mustard. Towards this, resistance to aphids and Alternaria leaf spot disease are being attained by deploying RNAi, introgression of protease inhibitors through transgenics and exploitation of wild *Diplotaxis erucoides*. Additionally, emphasis is also being given on the control of mustard pathogens *S. sclerotiorum* and *Albugo candida* using biotechnological strategies. ICAR-NIPB participated in the development of a chickpea variety, Pusa Chickpea 20211 (Pusa Chickpea Manav) which was released for central zone comprising Gujarat, Maharashtra and Madhya Pradesh in collaboration with IARI. This variety is a wilt resistant introgression line (WRIL-3) of Pusa 391, possessing the Wilt QTLs



1, 3, 4 and 5 on LG 2, introgressed from WR 315 through MABC. Towards the management of the lepidopteran pest, *Helicoverpa armigera*, novel alleles of cry1I-type and vip3Aa-type genes have been isolated and characterized. These genes have potential for deployment for pest control.

In addition to the ongoing work in various mandate crops, the institute has been actively participating and leading in several genomics programmes like clusterbean, *Oryza coarctata*, an obligate halophyte, chloroplast (cp) genome for Indian tea (*C. assamica*) and mitochondrial (mt) genome for *Camellia* species. ICAR-NIPB has also successfully developed TeaMiD: a

comprehensive database of simple sequence repeats markers of tea and TEnGExA, an R package-based tool which allows tissue-enrichment analysis for any number of genes or transcripts for any species.

Development of novel protocols in the field of biotechnology has always been the focus of the institute. For successful exploitation of functional genomics, an apical meristem-targeted in planta transformation protocol has been optimized in flax. The method is a promising tool and can be incorporated in varietal improvement programs in flax through genetic modification. The institute has developed a state of art optical coherence tomography (OCT) technique in collaboration with IIT Delhi to accurately classify seed varieties for the removal of varietal duplication and assessment of seed purity.

ICAR-NIPB is involved in Human Resource Development in the area of Plant Molecular

Biology and Biotechnology (MBB). Seven Ph.D. and five M.Sc. students were awarded doctoral and master's degrees in the 58th Convocation 2020, respectively and Dr. Shikha Dixit was bestowed IARI Merit Medal for her Doctoral research work. Additionally, ICAR-NIPB has been actively participating in extension and outreach activities like Mera Gaon Mera Gaurav (MGMG) and Scheduled Castes Sub Plan (SCSP). Considering the ongoing pandemic situation across the country, the programmes in 2020 were conducted virtually.

In future, we will together take up new challenges to identify and characterize novel genes and promoters, develop novel tools to modify genomes, utilize genomes for better climate resilience; nutritional genomics and novel molecules for better health and immunity in addition to developing highly trained skilled manpower for the future need of the country.

I would like to extend my sincere thanks

to all scientific, technical and administrative staff, students, research fellows and contractual supporting staff of ICAR-NIPB for their contributions in various institute activities including Hindi Chetna Maas and Swacchh Bharat Abhiyan. Special thanks are due to Dr. Monika Dalal, Dr. Amitha Mithra Sevanthi, Dr. Amolkumar U. Solanke, Dr. Prasanta Dash, Dr. Rohini Sreevathsa, Dr. Naveen Gupta, Dr. R. S. Niranjan, Dr. Deepak Bisht and Dr. M. S. Nimmy for their help in compilation of this report.

I am grateful to Dr. T. Mohapatra, Secretary DARE and Director General, ICAR, Dr. T. R. Sharma (Deputy Director General (Crop Sciences) ICAR and Dr. D.K. Yadav, Assistant Director General (Seeds), ICAR, for their constant support and help in overall functioning of the institute.

Date: 14 July, 2021


(Ajit K Shasany)

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EXECUTIVE SUMMARY

विशिष्ट सारांश

राष्ट्रीय पादप जैवप्रौद्योगिकी संस्थान भा.कृ. अ.प. का प्रमुख अनुसंधान संस्थान है जिसका सुपरिभाषित अधिदेश एनएआरईएस प्रणाली में फसल सुधार के लिए जैवप्रौद्योगिकी युक्तियों व तकनीकों का विकास करना व उनका उपयोग करना है। संस्थान के वैज्ञानिक पादप आण्विक जीवविज्ञान तथा जैवप्रौद्योगिकी के क्षेत्र में दिए जाने वाले शिक्षण एवं प्रशिक्षण में भी शामिल हैं। मेरा गांव मेरा गौरव (एमजीएमजी) और अनुसूचित जाति उप योजना (एससीएसपी) कार्यक्रमों के माध्यम से हमारा संस्थान गांव में रहने वाले हजारों किसानों तक पहुँचा है। वर्ष 2020 में संस्थान की प्रमुख उपलब्धियां संक्षेप में नीचे दी जा रही हैं।

उच्च गैलेक्टोमेनान से युक्त ग्वार की लोकप्रिय किस्म, आरजीसी-936 के जीनोम अनुक्रमण कर न्यूक्लियोटाइड के सटीक क्रम का पता लगाया गया। उच्च-घनत्व के एसएनपी मानचित्र के प्रति इन स्कैफोल्ड की एंकरिंग द्वारा अंतिम जीनोम असेम्बली प्राप्त की गई। इसके अलावा तीन विकासात्मक अवस्थाओं पर आरजीसी-936 (अधिक मात्रा में गोंद वाली) तथा एम-83 (कम मात्रा में गोंद वाली) किस्मों के बीच एक तुलनात्मक ट्रांसक्रिप्टोम अध्ययन किया गया जिसका उद्देश्य अधिक मात्रा में गोंद वाली प्रजाति में ऐसे जीनों का पता लगाना था जिनका गैलेक्टोमेनान के जैवसंश्लेषण में महत्वपूर्ण भूमिका हो।

ओराइज़ा कोएरेक्टाटा (जीनोम आकार:665 एमबी) तटवर्ती क्षेत्रों में पाए जानी वाली धान की एक लवण मृदोदभिद वन्य प्रजाति है। भा.कृ.अ.प.—एनआईपीबी द्वारा पहली बार रिपोर्ट किया गया कि ओ. कोएरेक्टाटा त्रिगुणित है न कि चतुर्गुणित और इसकी पुष्टि जड़ की नोकों से माइटोसिस मेटाफेस प्लेट तैयार करके तथा फलो सयाटोमीटर, दोनों के द्वारा की गई है।

भा.कृ.अ.प.—एनआईपीबी में जिनोम अनुक्रमण की नवीनतम विधियों का उपयोग करके कैमेलिया जातियों के माइटोकोन्ड्रियाई (mt) और भारतीय चाय (पी. एनेमिका) के क्लोरोप्लास्ट (cp) जीनोम को अनुक्रमण कर न्यूक्लियोटाइड के सटीक क्रम का पता लगाया है। क्लोरोप्लास्ट जीनोम 157,353 bp लंबा है जबकि उज जीनोम 707,441bp लंबा है। इसके अलावा हमने विकासात्मक अध्ययन करके, साधारण क्रम रिपीट (एसएसआर) की तुलना करके और उनके बीच वितरण के पैटर्न का उपयोग करके कोडान के लिए कैमेलिया वंश के अंतर्गत विभिन्न क्लोरोप्लास्ट (cp) जीनोम के तुलनात्मक विश्लेषण के माध्यम से भारतीय चाय के उद्गम को समझने का प्रयास किया है। हमने हाल ही में प्रकाशित चाय जीनोम, RNAseq, GSS, ESTs और उपांग जीनोम (क्लोरोप्लास्ट तथा माइटोकोन्ड्रियाई) आंकड़ों और इसके साथ ही चाय के एसएसआर मार्करों के कैटालॉग के सत्यापित सैट पर

प्रकाशित साहित्य के माध्यमसे एसएसआर के खनन द्वारा चाय के साधारण क्रम रिपीट मार्करों का एक बृहत डेटाबेस: TeaMiD भी विकसित किया है। इसके अलावा R पैकेज आधारित सॉफ्टवेयर TEnGExA भी विकसित किया गया है जिससे इनपुट के रूप में रीड-काउंट या एफपीकेएम-मान मेट्रिक्स से किसी भी जाति के लिए कितने ही जीनों या ट्रांसक्रिप्टों के लिए ऊतक-समृद्धि विश्लेषण किया जा सकता है।

भा.कृ.अ.प.—एनआईपीबी ने आईआईटी दिल्ली के सहयोग इमेजिंग की एक त्वरित, सम्पर्क विहीन और गैर आक्रामक तकनीक, गहरे अधिगम सहायी ऑप्टिकल कोहेरन टोमोग्राफी (ओसीटी) तकनीक विकसित की हैं जिससे चावल की विभिन्न किस्मों में भेद करने के लिए उप छायांकन किया जा सकता है। इस विद्यमान तकनीक से बीज की किस्मों को सटीक रूप से वर्गीकृत किया जा सकता है।

भा.कृ.अ.प.—एनआईपीबी में 'चावल के EMS प्रेरित उत्परिवर्तकों की राष्ट्रीय रिपोजिटरी' में नगीना 22 के EMS प्रेरित उत्परिवर्तक उपलब्ध हैं जिनका उपयोग IR छायांकन का इस्तेमाल करते हुए सूखा सहिष्णुता के लिए उत्परिवर्तकों के उच्च थ्रूपुट मूल्यांकन की विधि के विकास में हुआ है। इस विधि द्वारा पिछले वर्ष लीफ और पैनिकल ब्लास्ट (झुलसा रोग) दोनों के

प्रति सहनशील चार उत्परिवर्तकों की भी पहचान की गई थी। इसके अलावा इस संसाधन से सात पुष्पगुच्छ आर्कीटेक्चर उत्परिवर्तक पहचाने गए तथा दो स्थानों पर पुष्पगुच्छ आर्कीटेक्चर संबंधित प्राचलों तथा इसकी वृद्धि संबंधी गतिकी के लिए इनका विस्तारपूर्वक लक्षण-वर्णन किया गया। इसी प्रकार, लवणता और जलमग्नता की प्रतिकूल स्थितियों के प्रति सहिष्णु वन्य चावल तथा बाउ चावल वंशक्रमों की पहचान की गई। कुल 4 जीनप्ररूप लवणीय जल के अंतर्गत अवायवीय अंकुरण की दृष्टि से सर्वाधिक सहिष्णु पाए गए हैं।

चावल के दो जीनप्ररूपों नामतः N22 और IR64 को पौध अवस्था पर अलग-अलग और एक साथ नाइट्रोजन की चिरकालिक कमी तथा सूखा की प्रतिकूल स्थितियों में विश्लेषण किया गया। कुल मिलाकर एन22 में दोनों प्रतिकूल दशाओं के अंतर्गत बेहतर निष्पादन दिया जिसका कारण बेहतर जड़ आर्कीटेक्चर, क्लोरोफिल और पोरफाइरीन संश्लेषण तथा ऑक्सीकारक प्रतिबल प्रबंधन था। इसके साथ ही IR64 और N-22 से व्युत्पन्न 253 पुनर्संयोगी अंतःप्रजात वंशक्रमों का उपयोग करके बीज और भूसे में नाइट्रोजन अंश के लिए 12 क्यूटीएल और 63 प्रत्याशी जीनों की जनन अवस्था के दौरान पहचान की गई।

चावल में लवणता के प्रति संवेदी किस्म IR-29 में 0.01 प्रतिशत की दर से एलनटॉइन का पूरक उपयोग करके इसकी लवण सहिष्णुता में सुधार देखा गया और यह पाया गया कि एलेनटॉइन संकेतनकारी अणु/प्रेरक के रूप में कार्य कर सकता है जिससे वे अन्य चयापचयजी/हार्मोन पथ सक्रिय को प्रभावित कर सकता है जो आईआर 29 को लवण सहिष्णुता

प्रदान करते हैं। *ओराइजा सेटाइवा* उपजाति *इंडिका* में MIR319 के होमोलॉग गूणसूत्र 1 में 22.8 Mbp की दूरी पर स्थित पाए गए। *OsBAK1* की अतिअभिव्यक्ति द्वारा पराजीनी पौधे विकसित किए गए और पुष्पगुच्छ प्रध्वंस ट्रांसक्रिप्टोम से उक्त जीन की पहचान की गई। *OsLRK10*—जैसे प्रध्वंस प्रेरणशील जीन के विभिन्न वैकल्पिक स्लाइस किए गए ट्रांसक्रिप्ट पहचाने गए तथा दो वैरियंट द्विपदीय वाहकों में क्लोन किए गए। कैल्सियक सिलिकेट से उपचारित चावल के पौधों में जीन से संबंधित प्रतिरक्षा की अभिव्यक्ति की जांच टेपे और एचपी 2216 में प्रध्वंस संक्रमण के पश्चात् जांच की गई। इससे प्रध्वंस के प्रति विरोधी अनुक्रिया प्रदर्शित हुई।

ट्रांसजेनिक गेहूं के पौधों को उष्मा इंड्यूसिबल RRM समृद्ध RNA binding प्रोटीन के साथ विकसित किया गया है और वर्तमान में T_3 पीढ़ी का आण्विक तथा गुणप्ररूपी विश्लेषण किया जा रहा है। गेहूं के पौधे में गृहीत तथा पारपीढ़ीगत ताप सहिष्णुता का पता लगाने के लिए ऑक्सिन, जीबरलिन हार्मोनों तथा ऊष्मा प्राइमिंग का उपयोग किया गया और पौधे की वृद्धि और विकास पर इनके विभिन्न प्रभावों का विश्लेषण किया गया। TaNHX1 जीन का उपयोग करके गेहूं के पराजीनी उत्पन्न किए गए और लवण प्रतिबल उपचार का उपयोग करते हुए सस्यविज्ञानी प्राचलों का मूल्यांकन किया गया। गेहूं के PM19 जीन के पूर्ण लंबाई वाले व आंशिक प्रमोटर खंडों से युक्त तम्बाकू के पराजीनी पौधे विकसित किए गए। टी1 पराजीनी पौध की दो पत्ती और चार पत्ती वाली अवस्था पर स्थानिक-कालिक अभिव्यक्ति का विश्लेषण किया

गया जिससे यह स्पष्ट हुआ कि ये दोनों प्रमोटर खण्ड पत्तियों में अतिअभिव्यक्त होते हैं जबकि वृद्धि की सामान्य दशाओं के अंतर्गत जड़ों में इनकी निम्न अभिव्यक्ति प्रदर्शित हुई।

ब्रेड गेहूं की चाइनीज स्प्रिंग किस्म के जीनोमवार विश्लेषण द्वारा कुल 46 और 8 NRT2 व NAR2 जीनों की पहचान की जा चुकी है। समूह विशिष्ट ट्रांसक्रिप्ट अभिव्यक्ति से गेहूं की Ks907 किस्म के जड़ ऊतकों तथा चायनीज़ स्प्रिंग किस्म के जड़ व प्ररोह ऊतकों में नाइट्रेट से बचाव की अनुक्रिया का सुझाव मिलता है। ब्रेड गेहूं की दो घटक वाली उच्च संलयता नाइट्रेट परिवहन प्रणाली के साझेदार प्रोटीनों की पहचान के लिए स्प्लिट-ईबीक्यूटिन प्रोटीन-प्रोटीन अंतरक्रिया मूल्यांकन किया गया जिससे पूर्व में क्लोन किए गए TaNRT2.1/2.2 और TaNAR2.2/2.2 की भौतिक अंतरक्रिया की पुष्टि हुई। ब्रेड गेहूं तथा इसमें विद्यमान एक खरपतवार, नामतः *फ़ैलेरिस माइनर* की जड़-जड़ अंतरक्रिया पर किए गए एक अध्ययन से यह प्रदर्शित हुआ कि *पी. माइनर* की उपस्थिति से जड़ प्रणाली का आर्कीटेक्चर माड्युलेट होता है, गेहूं की जड़ों में जैवमात्रा आबंटन होता है जिसके परिणामस्वरूप उपयुक्ततम तथा नाइट्रोजन विहीन, दोनों दशाओं के अंतर्गत गेहूं में नाइट्रोजन-अंतरग्रहण कम हो जाता है।

GWAS के लिए जीनप्ररूपी आंकड़ों के साथ (35K के प्रजनक ऐरे का उपयोग करके एसएनपी जीनप्ररूपण) गेहूं के 300 से अधिक जीनप्ररूपों (यूके और भारत दोनों के) का रूपात्मक एवं जैवरासायनिक गुणप्ररूपण किया गया। प्राथमिक विश्लेषण से यह स्पष्ट हुआ कि प्ररोह लंबाई (Chr 2A) के लिए कुछ

उल्लेखनीय क्यूटीएल/एमटीए, रंजक अंश(Chr 1B), NR (Chr 5A), GS (Chr 1B, 2A, 2B, 3A, 4A और 7A) और एएलटी (Chr 3A) नाइट्रोजन-प्रतिबल दशाओं के अंतर्गत थे।

प्रतिपोषणिक घटकों से संबंधित जीनों की पहचान के उद्देश्य से ड्यूरेम गेहूं और चपाती गेहूं के आरएनए क्रम आंकड़ों का विश्लेषण किया गया। ये दोनों गेहूं की भिन्न जातियां हैं। कुल 20506 जीन भिन्न रूप से व्यक्त होने वाले पाए गए। गेहूं की विभिन्न टेट्रामेरिक एमाइलेज निरोधक उप इकाइयों के तुलनात्मक विश्लेषण से N-टर्मिनल सिग्नल पेप्टाइड क्रम में अत्यधिक संरक्षित 40–50 एमिनो अम्ल अपशिष्टों की पुष्टि हुई। ये संरक्षित क्रम टेट्रामेरिक प्रोटीन की स्थायी चतुर्दिक संरचना और क्रिया के लिए उत्तरदायी हो सकते हैं। आप्विक कार्य से संबंधित तीन प्रमुख जीनों के होमियोलॉगस ट्रांसक्रिप्ट का विश्लेषण षटगुणित ब्रेड गेहूं और इसकी द्विगुणित संततियों में दाना विकास की अवस्था के दौरान तीन वर्गों नामतः पोषण रिजर्ववायर (NR), कार्बोहाइड्रेट चयापचयन (CM) और प्रतिरक्षा प्रोटीन (DP) के लिए किया गया।

बी. जंसिया और बी. ओलिरसिया में अगुणित इंड्यूसर वंशक्रम के विकास के लिए इन दोनों जातियों में CRISPR/Cas9 वाहक लक्षित CENH3 पैरालॉग विकसित किए गए हैं। चूंकि CENH3 नॉकआउट भ्रूण के लिए घातक है इसलिए CENH3 नॉकआउट वंशक्रमों को बचाने के लिए एक जीएपी-टेल स्वेप वैक्टर डिज़ाइन किया गया। ये दोनों वाहक बी. जंसिया की वरुणा और आएलएम-198 तथा बी. ओलिरसिया किस्म बोटाइटिस की पूसा मेघना

किस्म में सहरूपांतरित किए गए। बी. ओलिरसिया में अभी तक कोई भी पराजीनी प्राप्त नहीं किया जा सका है। तथापि, बी. जंसिया में कुछ प्ररोह पुनर्जनन कराया गया है। BJKX3 जीन के लिए विकसित CRISPR/Cas9 कांस्ट्रक्ट एग्रोबैक्टीरियम-मध्यित रूपांतरण से बी. जंसिया (वरुणा) में रूपांतरित किए गए तथा पुनर्जनित पौधे ऊतक संवर्धन दशा के अंतर्गत विभिन्न विकासात्मक अवस्थाओं में उग रहे हैं।

खाद्य तेलों की उत्पादकता में वृद्धि, तेल के आयात को कम करने का एकमात्र विकल्प है। इस उद्देश्य से भारतीय सरसों में प्रमुख कीट पीड़कों और रोगजनकों के विरुद्ध प्रतिरोधी पादप किस्मों का विकास किया जा रहा है। माहू जीनों के RNAi मध्यित जीन मूककरण से व्युत्पन्न पोषक का उपयोग करके माहू के विरुद्ध आनुवंशिक प्रतिरोध में धीरे-धीरे सुधार हो रहा है। आल्टर्नेरिया पत्ती धब्बा रोग के विरुद्ध प्रतिरोध के लिए कृष्य बी. जंसिया में वन्य क्रूसीफेरी कुल की जाति डिप्टोटेक्सिस एरुकाईडिस से मात्रात्मक प्रतिरोध का समाहन किया गया है। इन समाहनशील वंशक्रमों को पीढ़ी-दर-पीढ़ी स्थिर भी बनाया गया है।

ब्रूमरेप हरियाणा के चरखादादरी जिले के अनेक गांवों में प्रमुख समस्याओं में से एक है। सर्वेक्षण के दौरान ब्रूमरेप की एक प्रमुख जाति, फैलीफैंके एगिप्टियाका भारतीय सरसों पर आक्रमण करता हुआ पाया गया। ब्रैसिका के विभिन्न जननद्रव्यों की हरियाणा के हॉट-स्पॉट में पी. एगिप्टियाका के विरुद्ध प्रतिरोध के लिए छंटाई की गई।

पिछले वर्षों के दौरान अनेक पुनःसंश्लेषित ब्रैसिका (आरबीजे) वंशक्रम विकसित किए गए।

आरंभिक पीढ़ी के संश्लेषी वंशक्रमों का विश्लेषण S² पीढ़ी में किया जा रहा है। जनक जातियों, बी. रापा और बी. नाइग्रा से उत्पन्न पुनः संश्लेषित बी. जंसिया वंशक्रमों (आरबीजे 1 से आरबीजे-92) का सस्यविज्ञानी गुणों के लिए बहुस्थानिक मूल्यांकन किया गया। वर्ष 2020 के दौरान वन्य जातियों (बी. फ्रुटिकुलो, इरुकस्ट्रम कैनारिएंस, ई. कार्डामिनॉयडेस, ई. कैलिकम, इनेथोकोर्पस लायरेटस, बी. टाउर्नफोर्टी) और बी. रापा का उपयोग करके नए कृत्रिम उभयगुणित विकसित किए गए। जिन 90 पुनरसंश्लेषित ब्रैसिका (आरबीजे) का मूल्यांकन किया गया, में से केवल आरबीजे 18 वंशक्रम में Ac-Del, Ac-Ldh, Ac-Bhrtr, Ac-Mrn और Ac-Smstr विलगकों के विरुद्ध पूर्ण प्रतिरोध देखा गया। रोगजनक के 8 विलगकों के विरुद्ध रोगरोधी के रूप में (पीडीआई=0) 160 प्रगत समाहन वंशक्रमों (बीसी₂एफ₁₀) में से रोगरोधी के रूप में केवल ईआरजे-40 की भी पहचान की गई। गोविंद बल्लभ पंत कृषि एवं प्रौद्योगिकी विश्वविद्यालय, पंतनगर; भा.कृ.अ.सं.-क्षेत्रीय केन्द्र, पूसा; भा.कृ.अ.सं. अनुसंधान फार्म, दिल्ली में प्राकृतिक दशाओं के अंतर्गत और भा. कृ.अ.प.-एनआईपीबी दिल्ली में नियंत्रित दशाओं के अंतर्गत आल्टर्नेरिया प्रतिरोध के लिए प्रगत समाहन वंशक्रमों (ईआरजे वंशक्रमों) (बीसी₂एफ₁₀) की छंटाई की गई। कुल 14 ऐसे आशाजनक वंशक्रम पहचाने गए जिनमें आल्टर्नेरिया झुलसा के विरुद्ध उच्च स्तर का प्रतिरोध था।

जिन 40 ब्रैसिका वन्य संबंधियों का परीक्षण किया गया उनमें से डी. फ्रक्टीकुलोसा, केमेलिना सेटाइवा, डिप्लोटेक्सिस एसुरजेंस, डी. कैथोलिका, डी. क्रेटेसिया,

डी. एरुकांडेस, डी. क्यूरेलिस, डी. सेईटियाना, डी. टेनुइसिलिक्वी, डी. विमिनिया, एरुकैस्ट्रम लाइरेटस, ई. एबाइसिनिकम, ई. कैनारिएस, ई. कार्डामिनोयडेस, क्रम्बे एबिसिनिका पांच प्रविष्टियां और एरुका सेटाइवा की चार प्रविष्टियां रोगजनक के छह विलगकों नामतः Ac-Del, Ac-Ldh, Ac-Pnt, Ac-Amb, Ac-Rnc vkSj Ac-Wltn के विरुद्ध रोगरोधी (पीडीआई =0) पाई गई हैं। इस मौसम के दौरान ब्रैसिका प्रविष्टियों (30), समाहित वंशक्रमों (160) और पुनर्संश्लेषित वंशक्रमों (92) की प्रक्षेत्र छंटाई भारतीय कृषि अनुसंधान संस्थान, राष्ट्रीय पादप आनुवंशिक संसाधन ब्यूरो, नई दिल्ली तथा भा. कृ.अ.सं. क्षेत्रीय केन्द्र, पूसा, बिहार के खेतों में की गई। इनमें से बी. जंसिया की पांच किस्में, बी. कैरिनाटा की तीन, 5 आईएल और 4 आरबीजे वंशक्रम खेत की प्राकृतिक दशाओं के अंतर्गत सफेद रतुआ रोगजनक के विरुद्ध प्रतिरोधी पाए गए।

एल्यूगो कैंडिडा के 30 आइसोलेट्स भारत में सरसों के प्रमुख उत्पादन स्थानों से एकत्र किए गए तथा उन्हें शुद्ध करके -200 से. तापमान पर अनुरक्षित करते हुए अलग से भंडारित किया गया। ब्रैसिका जननद्रव्य/प्रविष्टियों/समाहित वंशक्रमों/पुनर्संश्लेषित वंशक्रमों की कुल 480 प्रविष्टियों को एल्यूगो कैंडिडा के 10 विद्यमान विलगकों से कृत्रिम रूप से संरोपित किया गया। इसके अलावा भा.कृ.अ.प. -राष्ट्रीय पादप आनुवंशिक संसाधन ब्यूरो, नई दिल्ली से प्राप्त की गई एरुका सेटाइवा और लैपिडियम जातियों से युक्त ब्रैसिका जातियों की 65 प्रविष्टियों का मूल्यांकन कृत्रिम संरोपित दशाओं के अंतर्गत प्रतिरोध के लिए किया गया और बी. कैरिनाटा की केवल एक

प्रविष्टि की सभी 10 विलगकों के विरुद्ध रोगरोधी होने के रूप में पहचान की गई।

एस. स्कलेरोटियोरम ईएसआर-01 विलगक जीनोम के जीनोम-व्यापी सैक्रेटोम विश्लेषण, CAZymes पूर्वानुमान, रोगजनक-पोषक अंतरक्रिया से संबंधित जीनोम और KEGG विश्लेषण से कुल 57 प्यूटेटिव मोलेक्यूलज का पता लगाया है। सरसों में स्कलेरोटेनिया टॉलरेंस को ट्रांसफर करने के लिए 3 RILs और 2 NILs का मूल्यांकन किया जा रहा है।

सैरीन और सिस्टेइन-प्रकार के कुल 35 प्रत्याशी कैजानस कैजन प्रोटीएज़ निरोधक (CcaPI) जीनों की ड्राफ्ट अरहर जीनोम से पहचान की गई। होमोलॉजी मॉडलिंग से कीटों के रिसेप्टरों की एमिनोअम्ल अपशिष्टों के साथ प्रोटीएज़ निरोधक एमिनो अम्ल अपशिष्टों की अंतरक्रिया प्रदर्शित हुई। इसके अतिरिक्त माहू जैव मूल्यांकन तथा लैपिडोप्टेरन जैवमूल्यांकन से भी प्रोटीन सोयाबीन ट्रिप्सिन निरोधक से तुलना करने पर उच्च मृत्यु दर प्रदर्शित हुई।

चना भारतीय आहार में प्रोटीन के समृद्ध स्रोत के रूप में एक महत्वपूर्ण दलहनी फसल है। इस वर्ष चने की पूसा चिकपी 20211 (पूसा चिकपी मानव) किस्म भा. कृ.अ.प. (आईएआरआई) और भा.कृ.अ.प.-एनआईपीबी के सहयोग से देश के मध्य अंचल के लिए जारी की गई है। इस अंचल में गुजरात, महाराष्ट्र और मध्य प्रदेश राज्य आते हैं। यह किस्म पूसा 391 का एक मुर्झान प्रतिरोधी समाहन वंशक्रम (डब्ल्यूआरआईएल-3) है जिसमें एमएबीसी के माध्यम से डब्ल्यूआर 315 से मुर्झान प्रतिरोध से युक्त एलजी 2 स्थल पर मुर्झान क्यूटीएल 1, 3, 4 और 5 मौजूद हैं।

चने को बहुभोजी लेपिडोप्टेरन कीट पीड़क हेलिकोवर्पा आर्मीजेरा (कपास का गुला कृमि/चने का फली छेदक) से बहुत क्षति होती है। cry1I-प्रकार और vi3Aa- प्रकार के जीनों के नए युग्मविकल्पी विलगित और लक्षण-वर्णित किए गए हैं जो cry1Ac1 जीन से भिन्न हैं और इसलिए इनमें पीड़क नियंत्रण के लिए उपयोग किए जाने की बहुत क्षमता है।

वाहिका मुर्झान रोगजनक, फ्यूजेरियम यूडम के पूर्ण लंबाई के माइटोकॉन्ड्रियाई जीनोम (mt-जीनोम) को असेम्बल करके एनोट किया गया। फ्यूजेरियम यूडम के उज. जीनोम से फ्यूजेरियम जाति संकुल में अन्य पादप रोगजनक फ्यूजेरियम जातियों के साथ विकासात्मक संबंध को बेहतर रूप से समझा जा सकेगा।

चने में 48 MATE Domain से युक्त प्रोटीन वाले MATE जीन कुल का मिश्रित लक्षण-वर्णन किया जा चुका है, जिससे विकास और विविधीकरण में इनकी भूमिका को समझने में मदद मिल सकी, इसके साथ ही एक अन्य जीन कुल ASR-Abscisic acid and stress ripening का भी विस्तृत विश्लेषण किया गया।

सी. इस्काराबीओइडेस में पुष्पन में शामिल lncRNAs तथा miRNAs की पहचान करके इनका लक्षण-वर्णन किया गया। इस अध्ययन में कलिका और पत्ती ऊतकों के ट्रांसक्रिप्टोम का उपयोग करके कुल 1,672 lncRNAs और 57 miRNAs की पहचान की गई। केवल एक lncRNA कलिका विशिष्ट और 10 पत्ती विशिष्ट पाए गए। इनटरेक्टोम विश्लेषण से यह स्पष्ट हुआ है कि miRNA, miRNA, mRNA तथा lncRNA के बीच सभी अंतरक्रियाओं की जड़ों में विद्यमान हैं और मुख्य भूमिका निभाते हैं। कुल 17

Cas-IncRNA, Csa-miRNA में बाधा डालते हुए पाए गए और इसलिए ये महत्वपूर्ण eTMs के रूप में कार्य कर रहे हैं।

उपरोक्त जीन कुल के कार्यात्मक और विकासात्मक महत्व को समझने के लिए अरहर (*कैजानस कैजान*) से प्राप्त किए गए GA-Oxidase के तीन वर्गों का वृहत अध्ययन किया गया। गुणवत्ता संबंधी विशेषकों की उप परियोजना के अंतर्गत अरहर के कुल 75 जीनप्ररूपों की ट्रिप्सिन निरोधक क्रिया के लिए छंटाई की गई।

एग्रोबैक्टीरियम ट्यूमेफेसियंस मध्यित रूपांतरण के लिए फ्लेक्स की उपयुक्तता को *इन-प्लांटा* रूपांतरण विधि द्वारा उपयुक्ततम बनाया गया। यह विधि आशाजनक युक्ति है और इसका उपयोग आनुवंशिक रूपांतरणों के द्वारा फ्लेक्स में किस्मगत सुधार कार्यक्रम में लागू किया जा सकता है। ऊतक संवर्धन द्वारा उगाए गए पौधों के लिए राष्ट्रीय प्रमाणीकरण के अंतर्गत ISSR मार्करों का उपयोग करके अंगूर और अनार में दो

और आनुवंशिक फीडेलिटी परीक्षण प्रोटोकॉल विकसित किए गए। वर्तमान में वाणिज्यिक स्तर पर आनुवंशिक फीडेलिटी केले, आलू, गन्ना, बांस और खजूर ताड़ में औद्योगिक उपयोग के लिए ऊतक संवर्धन द्वारा उगाए गए पौधों पर किया जाता है।

भा.कृ.अ.प.—एनआईपीबी अपनी स्थापना के समय से ही पादप आण्विक जैविकी तथा जैवप्रौद्योगिकी क्षेत्र में मानव संसाधन विकास में सक्रिय रूप से रत है। वर्तमान में, संस्थान में पादप आण्विक जैविकी तथा जैवप्रौद्योगिकी विषय में 32 पीएच.डी. और 22 एम.एससी. छात्र पंजीकृत हैं। वर्ष 2020 में आयोजित 58वें दीक्षांत समारोह में 7 पीएच.डी. और 5 एम.एससी. छात्रों को क्रमशः डॉक्टरल और स्नातकोत्तर उपाधियां प्रदान की गईं। डॉ. शिखा दीक्षित को उनके डॉक्टरल अनुसंधान कार्य के लिए भा.कृ.अ.सं. प्रतिभा मैडल प्रदान किया गया। एनएएचईपी—सीएएएसटी के अंतर्गत 17 फरवरी से 1 मार्च 2020 तक एनएआरईएस प्रणाली

के छात्रों, वैज्ञानिकों और संकाय सदस्यों के लिए एक प्रशिक्षण कार्यक्रम आयोजित किया गया।

भा.कृ.अ.प.—एनआईपीबी भारत सरकार की दो प्रमुख पहलों नामतः मेरा गांव मेरा गौरव (एमजीएमजी) और अनुसूचित जाति उप योजना (एससीएसपी) के माध्यम से प्रसार एवं आउटरीच गतिविधियां आयोजित करता है। चूंकि वर्ष 2020 में कोरोना विषाणु महामारी का प्रकोप रहा इसलिए संस्थान के सभी आउटरीच कार्यक्रम ऑन लाइन मोड में संचालित किए गए। वर्ष 2020 के दौरान जो गतिविधियां चलाई गईं उनमें 19 दिसम्बर 2020 को 'ऑन लाइन कृषक-वैज्ञानिक परिचर्चा; 23 दिसम्बर 2020 को किसान दिवस का आयोजन प्रमुख हैं। साथ ही खरीफ मौसम के दौरान किसानों के बीज और रोपण सामग्री वितरित किए गए। हमने सभी सावधानियां बरतते हुए हिंदी चेतना मास, सतर्कता जागरूकता सप्ताह आदि जैसी अन्य गतिविधियां भी चलाईं।

EXECUTIVE SUMMARY

National Institute for Plant Biotechnology is a premier research institution of ICAR, with well-defined mandates for development and applications of biotechnology tools and techniques for crop improvement in NARES system. The scientists from the institute are also involved in the teaching and training of plant molecular biology and biotechnology. Through Mera Gaon Mera Gaurav (MGMG) and Scheduled Castes Sub Plan (SCSP) programmes, our institute reached to thousands of farmers living in the villages. The salient achievements of the institute during the year 2020 are summarized below:

- A chromosome-scale reference genome assembly of clusterbean (Guar), from a high galactomannan containing popular cultivar, RGC-936, was completed. The initial assembly contained 1580 scaffolds with an N50 value of 7.12 Mbp. The final genome assembly was obtained by anchoring these scaffolds to a high-density SNP map. A comparative transcriptome study was performed between RGC-936 (high gum) and M-83 (low gum) varieties at three developmental stages. Differentially

expressed genes (DEGs) analysis indicated a total of 5,147 shared unigenes between the two genotypes. Besides this, candidate unigenes that encode for enzymes involved in the biosynthesis of galactomannan were identified and analysed, and 15 key enzyme genes were experimentally validated by quantitative real-time PCR.

- *Oryza coarctata* (genome size: 665Mb) is a wild obligate halophyte found in the coastal regions. ICAR-NIPB first time reported that *O. coarctata* is triploid, not tetraploid which was determined by both preparing mitosis metaphase plates from root tips and by flow cytometer.
- ICAR-NIPB decoded the chloroplast (cp) genome for Indian tea (*C. assamica*) and mitochondrial (mt) genome for *Camellia* species using PacBio and Illumina Mate-pair reads. The chloroplast genome was 157,353 bp long while the mt genome was 707,441 bp long. Further, we made an attempt to understand the origin of Indian tea through the comparative analysis of different chloroplast (cp) genomes under the *Camellia* genus by performing evolutionary study

and comparing simple sequence repeats (SSRs) and codon usage distribution pattern among them. We also developed TeaMiD: a comprehensive database of simple sequence repeats markers of tea, by mining SSRs from the recently published tea genomes, RNAseq, GSS, ESTs and organelle genomes (chloroplasts and mitochondrial) data as well as from the published literature to catalogue validated set of tea SSR markers. TEnGExA, a R package-based tool was developed, which allows tissue-enrichment analysis for any number of genes or transcripts for any species provided only a read-count or FPKM-value matrix as input.

- ICAR-NIPB in collaboration with IIT Delhi has developed a fast, non-contact and non-invasive technique, deep learning assisted optical coherence tomography (OCT) technique of sub surface imaging to distinguish different rice varieties. The current technique can accurately classify seed varieties irrespective of the morphological similarities.
- EMS induced mutants of Nagina22 available at the 'National Repository of EMS induced

mutants of rice' at ICAR-NIPB, is being used to develop a high throughput evaluation of mutants for drought tolerance using IR imaging. From this resource, four mutants tolerant to both leaf and panicle blast were also identified last year. Further, seven panicle architecture mutants were identified from this resource and thoroughly characterized for panicle architecture related parameters and its growth dynamics in two locations. Similarly, wild rice and Bao rice lines tolerant to salinity and submergence stress were identified. A total of 4 genotypes had been found as most tolerant for anaerobic germination under saline water.

- Chronic low N stress and drought stress were imposed separately as well as together to two rice genotypes namely, Nagina22 and IR64 at seedling stage and RNAseq was done. Overall, N22 performed better under dual stress conditions owing to its better root architecture, chlorophyll and porphyrin synthesis and oxidative stress management. Further, 12 QTLs and 63 candidate genes for seed and straw N content using 253 recombinant inbred lines derived from IR64 and N22 were identified at reproductive stage.
- Supplementing allantoin @ 0.01% to a salinity sensitive genotype of rice, IR-29 under salinity improved its salinity

tolerance and found that allantoin may act as a signalling molecule/inducer which may activate other metabolic/hormone pathways that provide salt tolerance to IR-29. In *Oryza sativa* subspecies *indica*, two homologs of MIR319 were found and both are located on chromosome 1 in a distance of 22.8 Mbp. Transgenic plants are developed in rice by overexpressing *OsBAK1*, a gene identified from panicle blast transcriptome. Different alternative spliced transcripts of *OsLRK10*-like blast inducible gene were identified and two variants were cloned in binary vectors. Expression of defence related genes in calcium silicate treated rice plants was checked after blast infection in Tetep and HP2216 showing contrasting response to blast.

- Towards elucidating acquired and transgenerational thermotolerance in wheat plant, hormones auxin, gibberellin and heat priming were used and various effects on plant growth and development were analysed. The transgenics of wheat produced using *TaNHX1* gene were evaluated for agronomic parameters following salt stress treatment.
- A total of 46 and 8 *NRT2* and *NAR2* genes have been identified by genome-wide analysis in bread wheat Chinese Spring cultivar reference genome sequence. Group specific transcript expression suggests

nitrate-deprivation response in root and shoots tissues of Chinese Spring and root tissue of K9107 wheat cultivars. In order to identify partner proteins of two component high affinity nitrate transport system of bread wheat, split-ubiquitin protein-protein interaction assay confirmed the physical interaction of previously cloned *TaNRT2.1/2.2* and *TaNAR2.2/2.2*. In a separate study of root-root interaction of bread wheat and one of its prevalent weeds, i.e., *Phalaris minor*, demonstrated that presence of *P. minor* modulates root system architecture, biomass allocation to root of wheat that causes reduced N-uptake (total N content and $\delta^{15}\text{N}$) in wheat both under optimum and deprived N-conditions.

- Morphological and biochemical phenotyping of >300 wheat genotypes (including those from UK and India) were used along with the genotypic data (SNP genotyping using 35K Breeder's array) for genome wide association studies (GWAS). Preliminary analysis revealed some significant QTLs/MTAs for Shoot length (Chr 2A), Pigment content (Chr 1B), NR (Chr 5A), GS (Chr 1B, 2A, 2B, 3A, 4A and 7A) and Alt (Chr 3A) under N-stress condition.
- The RNAseq data of durum wheat and bread wheat was analysed with the purpose of identifying genes related to anti-nutritional factors which are different in these two-

wheat species. A total of 20506 genes were found differentially expressed. The comparative analysis of different wheat tetrameric amylase inhibitors subunits reveals highly conserved 40-50 amino acid residues in N-terminal signal peptide sequence. These conserved sequences may account for stable quaternary structure and activity of the tetrameric protein. The homeologous transcripts of major genes related to molecular function were analysed for the three classes viz. nutrient reservoir (NR), carbohydrate metabolism (CM), and defence protein (DP) in hexaploid bread wheat and its diploid progenitors during grain development.

- For developing haploid inducer line in *B. juncea* and *B. oleracea*, a CRISPR/Cas9 vector targeting *CENH3* paralog(s) in both these species have been developed. Since *CENH3* knockout is embryo lethal, a GFP-tail swap vector was designed to rescue these *CENH3* knockout lines. These two vectors were co-transformed in Varuna and RLM-198 varieties of *B. juncea* and Pusa Meghna variety of *B. oleracea* var. *botrytis*. In *B. oleracea* till now no transgenic could be recovered. However, in *B. juncea* some shoot regeneration has been obtained. CRISPR/Cas9 constructs developed for the *BjCKX3* gene were transformed into *B. juncea* (Varuna) through *Agrobacterium*-mediated

transformation and regenerating plants are at different developmental stages growing under tissue culture condition.

- Increasing the domestic productivity of edible oil is the only option to minimize its import. Towards this aim research is underway for developing plant types that are resistant to the major insect-pest and pathogen of Indian mustard. Incremental improvement in genetic resistance against aphid has been attained by deploying host delivered RNAi mediated gene silencing of aphid genes. For resistance to *Alternaria* leaf spot disease, quantitative resistance has been introgressed from wild cruciferous species *Diplotaxis erucoides* into cultivated *B. juncea*. The introgression lines have been stabilized over the generations.
- The broomrape is one of the major problems in many villages in the Charkha Dadri district of Haryana. During the survey, *Phelipanche aegyptiaca*, the major broomrape species was found attacking Indian mustard. Different Brassica germplasm was screened for *P. aegyptiaca* resistance at a hot spot in Haryana.
- In previous years, large number of Resynthesized Brassica (RBJ) lines were developed. The analysis of early generation synthetic lines is being done in S2 generation. Resynthesized *B. juncea* lines (RBJ 1 to RBJ-92) derived from its parental species; *B.*

rapa and *B. nigra*, were subjected for multi-location evaluation for the agronomic traits in S6 generations. New Synthetic amphidiploids were developed using wild species (*B. fruticulosa*, *Erucastrum canariense*, *E. cardaminoides*, *E. gallicum*, *Enathocarpus lyratus*, *B. tournefortii*) and *B. rapa*; during 2020. Among 90 Resynthesized Brassica (RBJ) lines evaluated, complete resistance against Ac-Del, Ac-Ldh, Ac-Bhrtr, Ac-Mrn and Ac-Smstr isolates was observed only in RBJ 18 line. Of the 160 advanced introgressed lines (BC2F10), ERJ-40 was identified as immune (PDI=0) against 8 isolates of the pathogen. The advanced introgression lines (ERJ Lines) (BC2F10) were screened for the *Alternaria* resistance under the natural condition at GBPUAT Pantnagar; IARI-RS, Pusa, IARI research farm, Delhi and also under controlled conditions at ICAR-NIPB, Delhi. Total 14 promising lines were identified which are giving a good level of resistance against *Alternaria* blight.

- Among 40 Brassica wild relative tested; *B. fruticulosa*, *Camelina sativa*, *Diplotaxis assurgens*, *D. catholica*, *D. cretacea*, *D. erucoides*, *D. muralis*, *D. siettiana*, *D. tenuisiliqua*, *D. viminea*, *Erucastrum lyratus*, *E. abyssinicum*, *E. canariense*, *E. cardaminoides*, *Crambe abyssinica* five accessions and *Eruca sativa* four accessions were found immune (PDI=0) against six isolates namely Ac-Del, Ac-Ldh,

Ac-Pnt, Ac-Amb, Ac-Rnc & Ac-Wltn of the pathogen. During this season, field screening and evaluation of Brassica accessions (30), introgressed lines (160) and resynthesized lines (92) were done at IARI, NBPGR, Delhi and IARI Pusa-Bihar fields. Among these, 5 accessions of *B. juncea*, 3 of *B. carinata*, 5 ILs and 4 RBJ lines were found resistant against the white rust pathogen under natural field conditions.

- Thirty isolates of *Albugo candida* were collected from major mustard growing locations in India and purified, maintained and stored separately at -20°C. A total of 408 Brassica germplasm/accessions/introgressed lines/resynthesized lines were artificially inoculated with 10 prevalent isolates of *Albugo candida*. Along with these 65 accessions of Brassica species consisting of *Eruca sativa* and *Lepidium* species obtained from ICAR-NBPGR, New Delhi were evaluated for resistance under artificial inoculated conditions and only one accession of *B. carinata*, EC206642 was identified as immune to all ten isolates.
- Genome-wide secretome analysis, CAZymes prediction, pathogen-host interaction related genes and KEGG analysis in *S. sclerotiorum* ESR-01 isolate genome has been completed and the total 57 putative effector candidates were predicted. For *Sclerotinia* stem rot disease tolerance, three different

RILs (~650) in their F3 and two NILS in BC2F1 (~100) generation are being screened and evaluated.

- A total of 35 candidate *Cajanus cajan* Protease Inhibitor (CcaPI) genes of serine and cysteine-type were identified from the draft pigeonpea genome. The homology modelling showed the interaction of amino acid residues of the protease inhibitors with the amino acid residues of the receptors of the insects. Further, aphid bioassay and lepidopteran bioassay also showed high mortality when compared with reference protein soybean trypsin inhibitor
- Chickpea is an important pulse crop and a rich source of protein in Indian diet. This year, a variety of chickpea Pusa Chickpea 20211 (Pusa Chickpea Manav) is released for central zone comprising Gujarat, Maharashtra and Madhya Pradesh by ICAR-IARI and ICAR-NIPB as a collaborator. This variety is a wilt resistant introgression line (WRIL-3) of Pusa 391, possessing the Wilt QTLs 1, 3, 4 and 5 on LG2 locus having wilt resistance introgressed from WR 315 through MABC.
- Chickpea is damaged by a polyphagous lepidopteran insect pest *Helicoverpa armigera* (cotton bollworm/gram pod borer). Novel alleles of cry1I-type and vip3Aa-type genes have been isolated and characterized which are different from cry1Ac1 gene and

therefore, have potential for deployment in pest control.

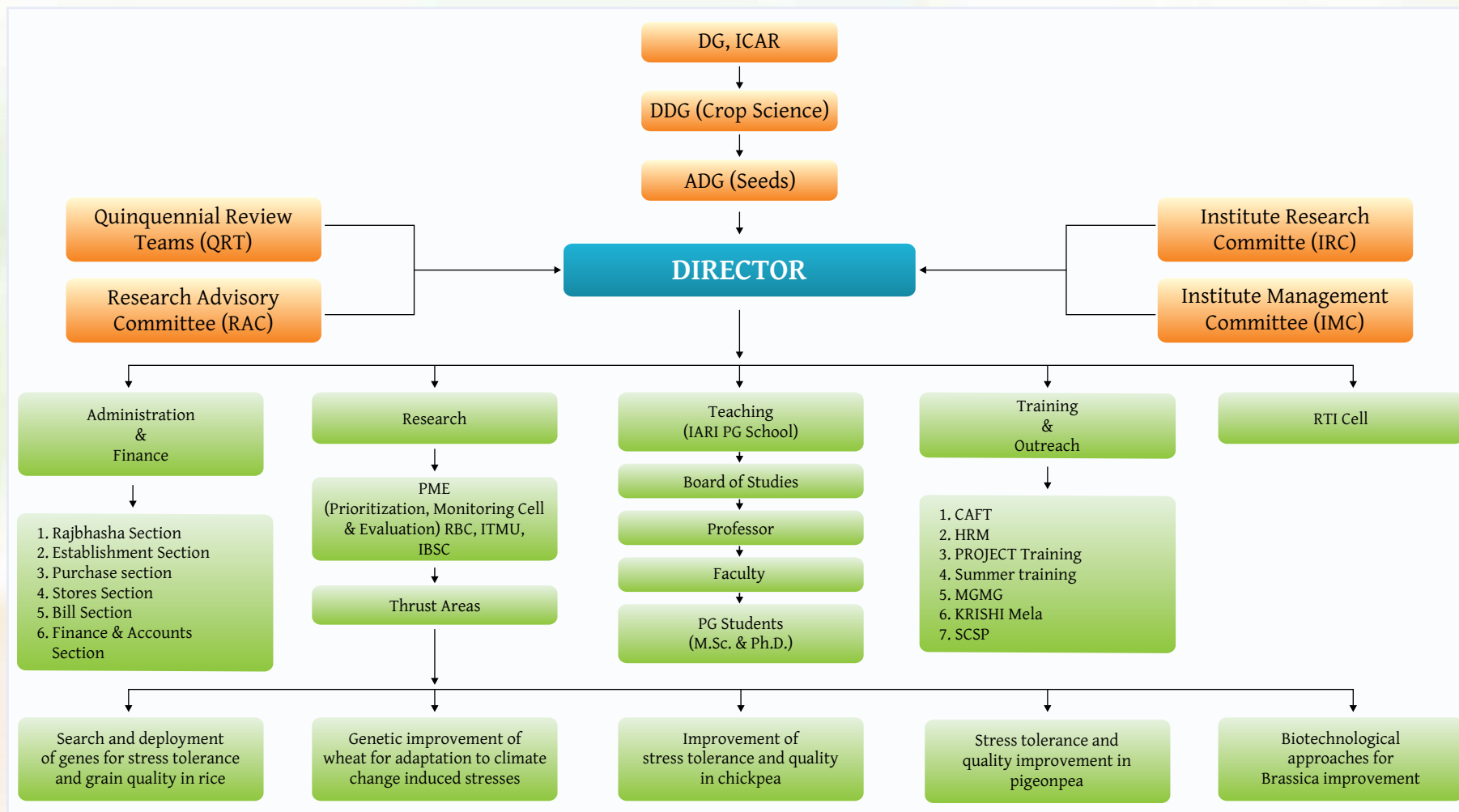
- The full-length mitochondrial genome (mt-genome) of the vascular wilt pathogen, *Fusarium udum* was assembled and annotated. The mt-genome of *Fusarium udum* would likely facilitate in better understanding of its evolutionary relationship with other plant pathogenic species in the *Fusarium* species complex.
- A detailed characterization of the MATE gene family consisting of 48 MATE-domain containing proteins in chickpea has been carried out to elucidate its evolution and diversification. This study gives a first insight on chickpea MATE family. Along with this chickpea ASR (Absciscic acid and stress ripening) gene family is also characterized under drought stress condition.
- Identification and characterization of lncRNAs and miRNAs involved in flowering in *C. scarabaeoides* were carried out. In this study, a total of 1,672 lncRNAs and 57 miRNAs were identified utilizing transcriptome data of the bud and leaf tissues. Only one lncRNA was found to be bud specific and 10 were leaf specific. Interactome analysis reveals that miRNA is the key player and stands in the roots of all interactions between miRNA, mRNA and lncRNA. 17 Csa-lncRNA were found to be interfering with Csa-miRNA, therefore acting as important eTMs.

- A comprehensive study of three classes of GA oxidases from pigeonpea (*Cajanus cajan*) was done to understand the functional and evolutionary significance of this gene family. Under quality traits subproject, a total of 75 pigeon pea genotypes were screened for Trypsin inhibitors activity.
- The amenability of flax to *Agrobacterium tumefaciens* mediated transformation was optimized by *In-planta* transformation method. The method is a promising tool and can be incorporated in varietal improvement programs in flax through genetic modification. Under the National certification system for tissue culture raised plants, two more genetic fidelity testing protocols were developed in guava and pomegranate using ISSR markers. Presently, commercial level genetic fidelity is carried out for industrial tissue culture plants in banana, potato, sugarcane, bamboo and date palm with marker system developed by ICAR-NIPB.
- The institute has been actively engaged in Human Resource Development in the area of Plant Molecular Biology and Biotechnology (MBB) since its inception. Currently 32 Ph.D. and 22 M.Sc. students are registered in the discipline of MBB at the institute. Seven Ph.D. and five M.Sc. students were awarded doctoral and master's degrees in the 58th Convocation 2020, respectively. Dr. Shikha Dixit was awarded IARI Merit Medal for her Doctoral research work. A training program was conducted for students and scientific and teaching faculty of the NARES system under NAHEP-CAAST between 17th February to 1st March 2020.
- ICAR - NIPB organizes extension and outreach activities through two major initiatives of Government of India namely, Mera Gaon Mera Gaurav (MGMG) and Scheduled Castes Sub Plan (SCSP). Since year 2020 was marred by Corona virus pandemics, all the institute's outreach programs were conducted in online mode. The activities carried out during the year 2020 were 'Online Farmers-Scientist interaction' on December 19, 2020; Kisan Diwas Celebration on December 23, 2020. Seed and planting material was distributed to farmers during Kharif season. We also organized other activities such as Hindi Chetna Maas, Vigilance Awareness week, etc. with all necessary precautions.

- **NIPB AND THE MANDATE**
- **STAFF POSITION**
- **FINANCIAL STATEMENT**
- **RESOURCE GENERATION**
- **PERSONNEL**

ABOUT THE INSTITUTE

ORGANOGRAM



ICAR-NIPB

National Institute for Plant Biotechnology (NIPB) is the premiere research institution of the Indian Council of Agricultural research (ICAR), engaged in molecular biology and biotechnology research. The Biotechnology Centre, established in 1985, as part of the Indian Agricultural Research Institute (IARI), was upgraded to a National Research Centre on Plant Biotechnology in the year 1993, with a vision to impart the biotechnology advantage to the National Agricultural Research System (NARS). In February 2019, NRCPB was elevated to the level of National Institute for Plant Biotechnology (ICAR-NIPB). This brings us much greater national responsibility to conduct basic and applied research, and human resource development in plant biotechnology. NIPB has acquired an excellent infrastructure in terms of equipment and other physical facilities and also a high degree of scientific competence. Development of transgenic crops for biotic and abiotic stress management,

exploitation of heterosis through marker and genomic approaches, marker assisted selection and molecular breeding of major crops for productivity and quality enhancement, search for novel genes and promoters for efficient native and transgene expression are the major activities taken up by the Institute. There is now considerable emphasis on structural and functional genomics of crop species such as rice, wheat, chickpea, pigeonpea, and mustard in the Institute. In addition to research, the Institute is contributing significantly to competent human resource development by way of offering regular M.Sc. and Ph.D. programmes by partnering with PG School, IARI.

Mandate

- To undertake basic plant molecular biology research for understanding molecular mechanisms underlying basic biological processes

- To develop capabilities of devising tools and techniques of biotechnology and genetic engineering for crop improvement
- To use the knowledge gained and technologies developed for advancing agriculture development
- To serve as a national lead centre for plant molecular biology and biotechnology research and to create trained manpower in the areas of plant biotechnology and genetic engineering

Staff Strength of the Institute

Staff	Sanctioned	Filled	Vacant
Scientific	36+1	33	02
Technical	14	09	05
Administrative	18	13	05
Skilled Supporting Staff	02	-	02
Total Strength	70+1	55	14

Financial Statement 2020-21

(Rs. In lakhs)

Institute Grant		
	Allocation	Utilization
Capital	69.58	65.74
Revenue		
Establishment	960.00	959.80
Pension & other retirement benefits	98.57	55.00
Travelling Allowances	6.70	6.33
Research and Operational Expenses	287.65	286.07
Administrative Expenses	521.27	488.03
Miscellaneous Expenses	3.50	1.24
Total	1947.27	1862.21

Resource Generation

Sales of Farm Produce	0.00
License Fee	0.00
Leave Salary and Pension Contribution	0.00
Interest Earned on Short Term Deposits	9.19
Income Generated from Internal Resource Generation	12.80
Miscellaneous Receipts	8.13
Total	30.12

Network projects on transgenic crops (NPTC)

	Allocation	Utilization
Capital	30.00	0.93
Travelling Allowances	0.00	0.00
Research and Operational Expenses	400.00	370.10
Total	430.00	371.03

Fund Received through Externally Funded Projects

Externally Funded Projects	616.25
Consultancy Projects	0.00
Total	616.25

Scientific Staff

Director		
Dr. Ajit Kumar Shasany		
National Professor B. P. Pal Chair		
Dr. Nagendra Kumar Singh		
Principal Scientist	Senior Scientist	Scientist (S. S.)
Dr. Sarvjeet Kaur	Dr. S.V.Amitha Charu Rama Mithra Dr. Amolkumar U. Solanke	Dr. Navin Chandra Gupta
Dr. Anita Grover		Dr. Ramavatar Nagar
Dr. Rekha Kansal		Dr. Mahesh Rao
Dr. Sanjay Singh		Dr. Nimmy M.S.
Dr. Jasdeep Chatrath		Dr. Anshul Watts
Padaria		Dr. Deepak Singh Bisht
Dr. Ram Charan Bhattacharya		
Dr. Debasis Pattanayak		
Dr. Pranab Kumar Mandal		
Dr. Pradeep Kumar Jain		
Dr. Kishor Gaikwad		
Dr. Sharmistha Barthakur		
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- 1. SEARCH AND DEPLOYMENT OF GENES FOR STRESS TOLERANCE AND GRAIN QUALITY IN RICE**
- 2. GENETIC IMPROVEMENT OF WHEAT FOR ADAPTATION TO CLIMATE CHANGE INDUCED STRESSES**
- 3. IMPROVEMENT OF STRESS TOLERANCE AND QUALITY TRAITS IN CHICKPEA**
- 4. STRESS TOLERANCE AND QUALITY IMPROVEMENT IN PIGEONPEA**
- 5. BIOTECHNOLOGICAL APPROACHES FOR BRASSICA IMPROVEMENT**
- 6. NATIONAL PROFESSOR -B. P. PAL CHAIR**

RESEARCH ACHIEVEMENTS

1. SEARCH AND DEPLOYMENT OF GENES FOR STRESS TOLERANCE AND GRAIN QUALITY IN RICE



Screening of rice landraces for anaerobic germination under saline water

Abhishek Mazumder, Megha Rohilla, Swati Mishra, Tapan Kumar Mondal

Coastal areas are more vulnerable than inland areas for saline submergence and hence, identification of genotype(s) tolerant to anaerobic germination and their utilization in crop improvement can mitigate this problem. A total of 100 rice genotypes representing different coastal regions had been screened till date for anaerobic germination under saline water inside greenhouse at phenotyping facility, NIPB, New Delhi in the years 2017, 2018, 2019 and 2020. Seeds were allowed to be imbibed in freshwater and subsequently germinated in plastic glass (20 cm height) with 5 cm of soil depth. Then glasses had been filled with low saline ($EC = 5dSm^{-1}$; ~ 41.6 mM NaCl) and high saline ($EC = 10dSm^{-1}$; ~ 83.3 mM NaCl) water as well as a control set. After 20 days of treatment, germination, survival percentage

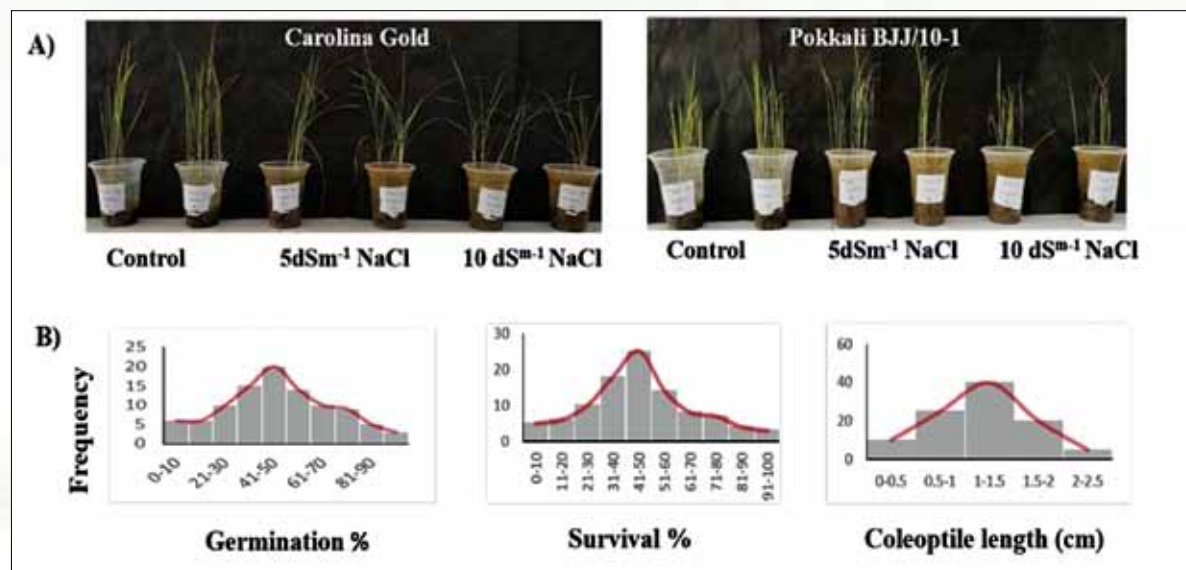


Fig. 1.1: Germination under saline submergence A) Carolina Gold and Pokkali-BJJ/10-1, two of the tolerant genotypes germinated and grown under saline water in anaerobic condition for 21days. B) Screening of 100 rice genotypes for anaerobic germination under saline water for three different related traits under high salinity (10 dS/m NaCl; data represents 4 years average).

and coleoptile length were measured and compared with control. A total of 4 genotypes had been found to be most tolerant for anaerobic germination under saline water (Fig 1.1 A and B).

Phenotyping of deep-water rice landraces under deep water stress

Megha Rohilla, Abhishek Mazumder, Swati Mishra, Nagendra Kumar Singh, Tapan Kumar Mondal

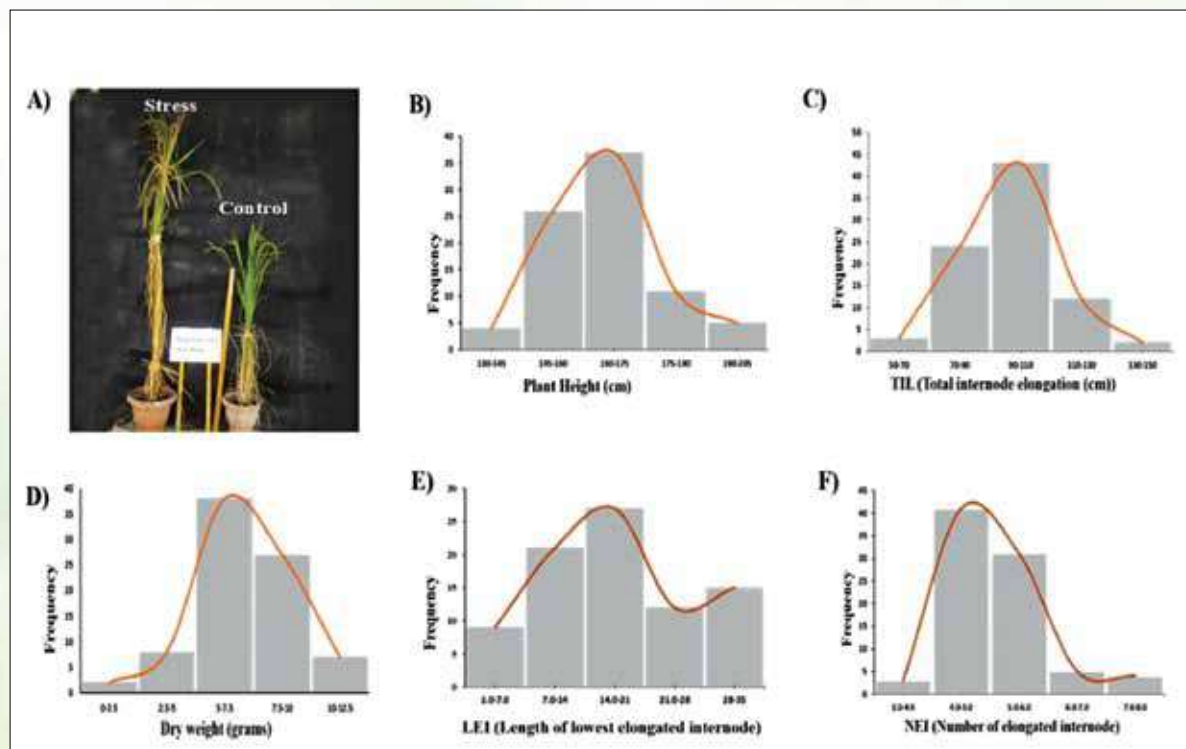


Fig. 1.2: Phenotyping of deep-water rice of Assam. A) Growth of deep water rice under stress condition and control condition B) Frequency distribution of Plant height (PH), C) Frequency distribution of TIL (cm), D) Frequency distribution of dry weight (grams), E) Frequency distribution of LEI, F) Frequency distribution of NEI.

In Assam, deep water rice is locally known as Bao dhan, has potential for rapid internode elongation under deep water stress. A set of 94 deep-water landraces of Assam, India were screened for phenotypic traits under deep-water condition at NIPB, New Delhi and Regional Agricultural Research Station, North Lakhimpur, Assam for 3 subsequent years i.e., 2018, 2019 and 2020. Plants were pre-germinated,

transplanted and given the stress at sixth leaf emergence. On the 15th day of treatment, we measured plant height (PH), total internode elongation (TIL), length of lowest elongated internode (LEI), the number of elongated internodes (NEI), leaf and culm dry weight (Fig. 1.2). Two genotypes, Negheri Bao 1 and Panikekowa Bao 1 were found to be tolerant to deep water stress.

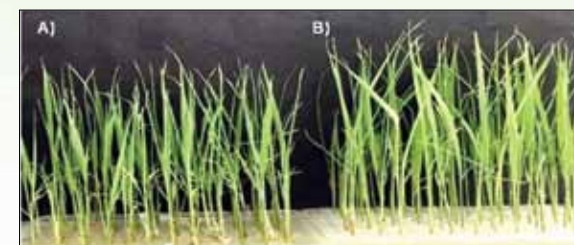
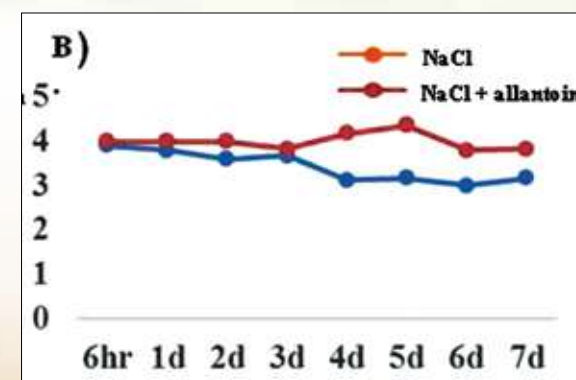
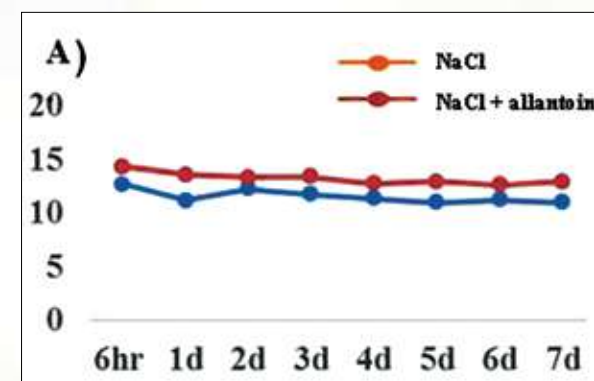


Fig. 1.3: IR-29 plants grown hydroponically. (A) with NaCl (100mM) and (B) with NaCl (100 mM) + Allantoin (0.01mM)



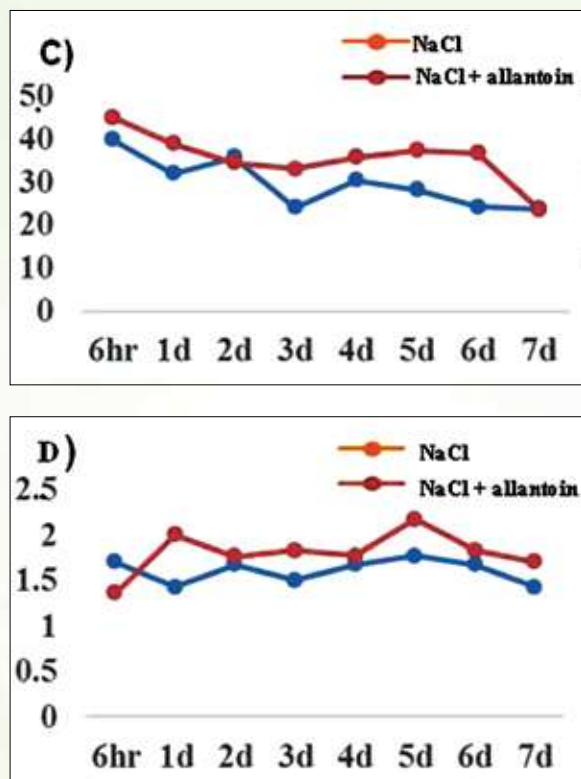


Fig. 1.4: Phenotypic changes in IR-29 plants grown hydroponically for seven days under NaCl and NaCl + allantoin. A) Shoot length, B) Root length, C) Dry weight of shoot, and D) Dry weight of root

Study of allantoin-mediated salinity tolerance in salinity stress sensitive genotype IR-29

Rekha Mahato, Jyoti Nishad, Soni Chowrasia, Tapan Kumar Mondal

Native accumulation of allantoin in plant tissues under different abiotic stress conditions

is well-known. However, the impact of exogenous application of allantoin to improve salinity tolerance has not yet been reported. A salt sensitive rice genotype IR-29 was evaluated under 100mM NaCl with and without allantoin (0.01mM). The allantoin treatment rendered tolerance to salinity for a week (Fig. 1.3) with the maximum differences in growth parameters observed at 5d between control and treated samples (Fig. 1.4). So, our hypothesis is that exogenous application of allantoin induces molecular pathways providing salt tolerance while the endogenous content had no role.

RNAseq approach is being used to understand the molecular mechanism of role of allantoin in imparting salinity tolerance.

Oryza coarctata, the only halophyte in the genus of *Oryza* is a triploid species

Soni Chowrasia, Jyoti Nishad, Tapan Kumar Mondal

Chowrasia *et al.* 2021 Plant Science

Oryza coarctata is a wild species of rice which is a tetraploid and obligate halophyte. Recently, the ploidy nature of *O. coarctata* was determined in our lab. From the ten metaphase plates of

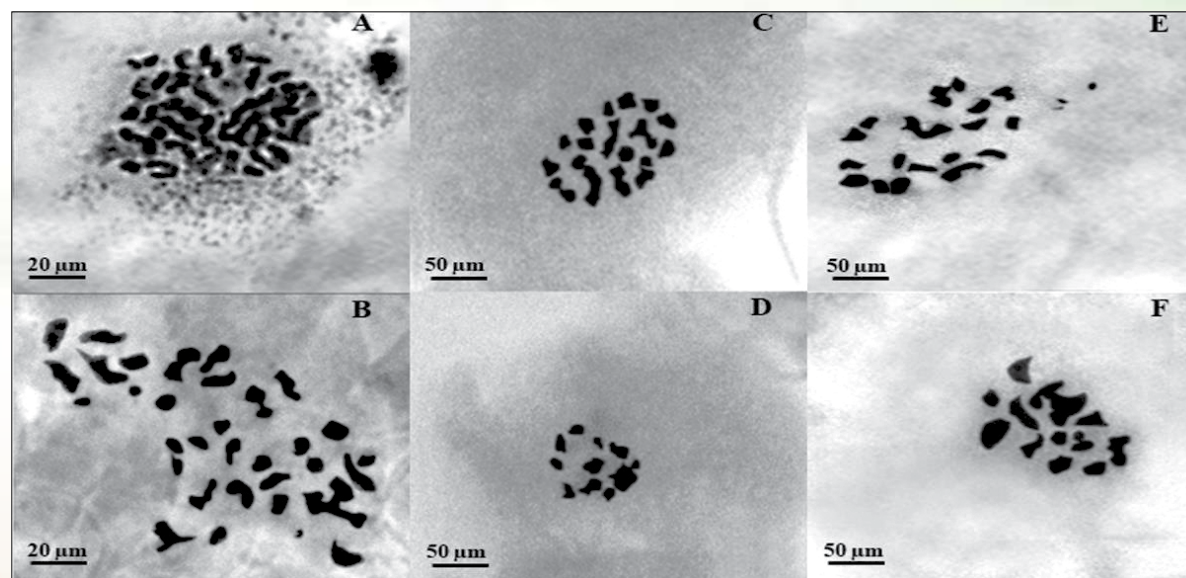


Fig. 1.5: Chromosomal study of *O. coarctata*. (A-B): Two independent metaphase plates from root tips showing the chromosome number of 36 ($2n = 3x = 36$). C): MI stage of meiosis from flower buds of *O. coarctata* showing 24 chromosomes. D): The PMC at MI stage showing 12 univalents. E): The PMC at MI stage consisting of 20 chromosomes. F): The PMC at MI stage showing 16 chromosomes. MI- Metaphase I and PMC- Pollen mother cell.

different roots tips, *O. coarctata* was found to be a triploid ($2n=3x=36$) (Fig. 1.5). Further, the meiotic behaviour of chromosomes in pollen mother cells exhibited various abnormalities (trivalent, bivalent and univalent) in metaphase I and II (Fig. 1.5 C-F). Simultaneously, ploidy level in *O. coarctata* was also confirmed by flow cytometer where its ploidy integer value was found as 3.02 that matched with other triploid value of plants.

TEnGExA: A R package-based tool for tissue enrichment and gene expression analysis

Hukam C Rawal, Tapan Kumar Mondal

Rawal *et al.* 2021 Briefings in Bioinformatics

RNA-seq data analysis provides large number of transcripts or genes from multiple tissues and we need to identify tissue-specific or tissue-enriched transcripts. A tool for quickly performing tissue-enrichment and gene expression analysis irrespective of number of input genes or tissues at various fragments per kilobase of transcript per million fragments mapped (FPKM) thresholds is all what needed. To fulfill this need, we developed a freely available R package (<https://bioconductor.org/packages/TENGEXA>) and web-interface tool (http://webtom.cabgrid.res.in/tissue_enrich/), TEnGExA, which allows tissue-enrichment analysis for any number of genes or transcripts for any species provided only a read-count or FPKM-value matrix as input (Fig. 1.6).

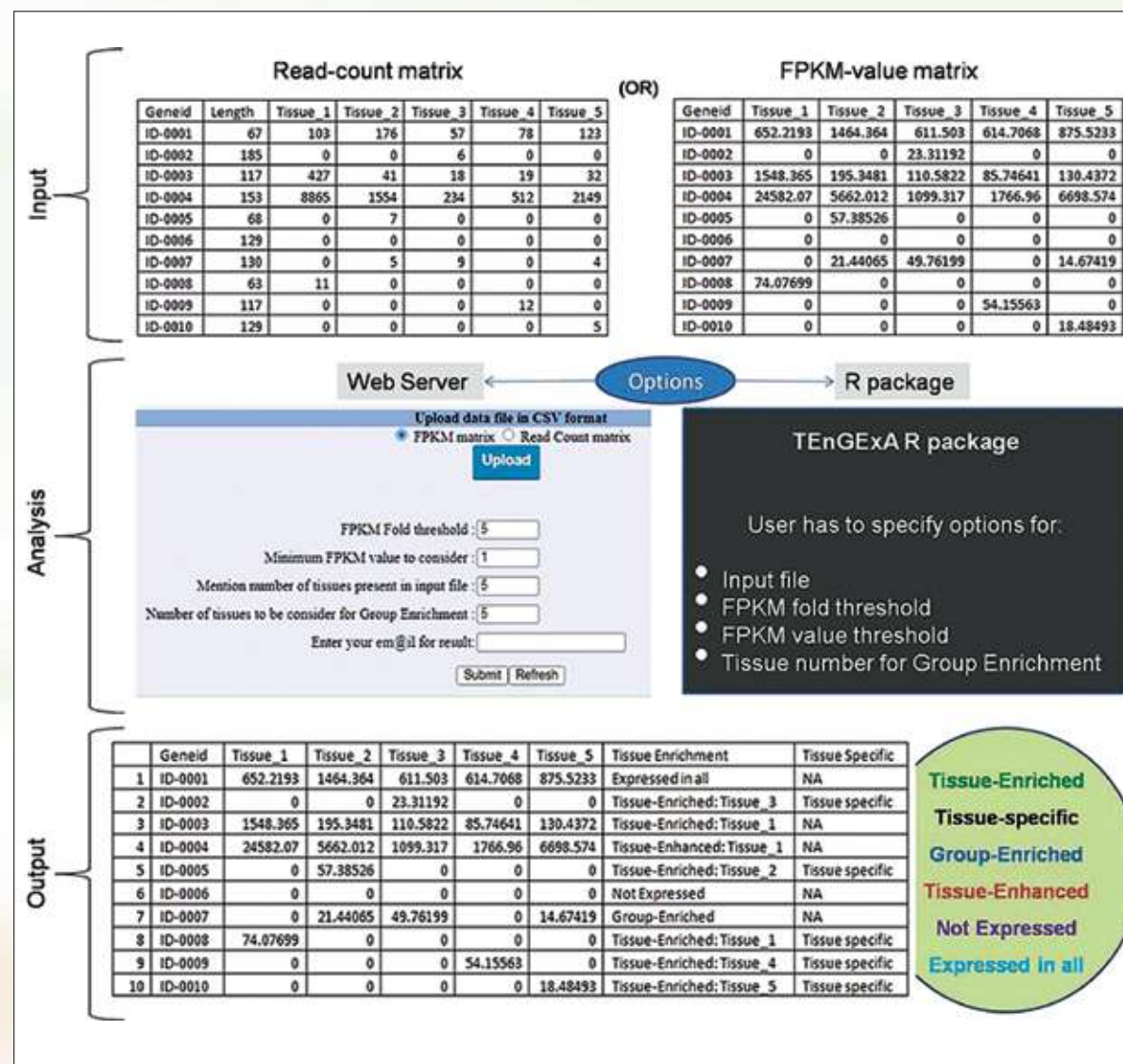


Fig. 1.6: The workflow of TEnGExA tool analysis

Decoding of nuclear and organelle genomes (chloroplast and mitochondria) of popular Indian tea genotype, TV1 [*Camellia assamica* L. (O). Kunze]

Hukam C Rawal, Nagendra Kumar Singh, Tapan Kumar Mondal

Using PacBio and Illumina Mate-pair reads, we successfully assembled and analyzed first ever reported chloroplast (cp) (157,353 bp) and mitochondrial (mt) genome (707,441

bp) of Indian tea (*C. assamica*). The cp genome comprised a typical circular quadripartite structure with 2 inverted repeats (IRa and IRb, 26,031bp each), a large single copy (LSC) and a small single copy (SSC) region of 87,213bp and 18,078 bp, respectively (Fig. 1.7a). While the cp genome comprised 126 genes, the mt genome had 66 genes (Fig. 1.7b). The LSC-IR and SSC-IR junctions along with 6 randomly selected cp genes were validated and confirmed by PCR.

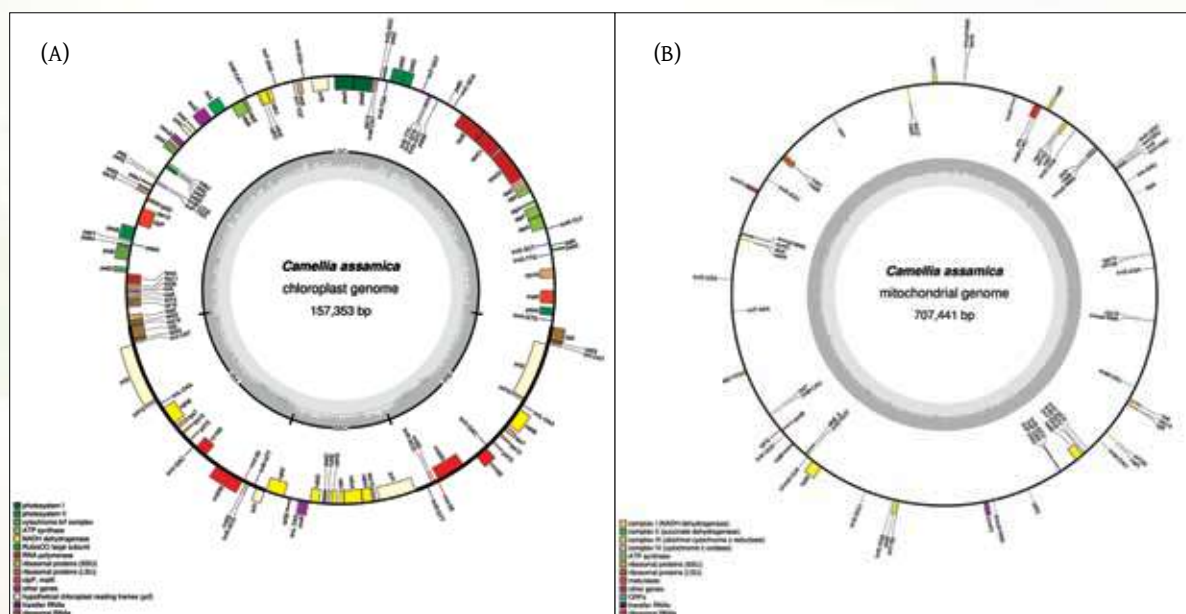


Fig. 1.7: Map of the cp genome (a) and mt genome (b) of Indian tea showing annotated genes with different functional groups that are color-coded on outer circle as transcribed clock-wise (outside) and transcribed counter clock-wise (inside). The inner circle indicates the GC content as dark grey plot

Comparative analysis of chloroplast genomes indicates different origin for Indian tea (*Camellia assamica* cv. TV1) as compared to Chinese tea

Hukam C Rawal, Tilak Raj Sharma, Nagendra Kumar Singh, Tapan Kumar Mondal

Rawal *et al.* 2021 Scientific Reports

Based upon the morphological characteristics, tea is classified botanically into 2 distinct main types i.e., Assam and China. The general consensus of origin of tea is India, Burma and China adjoining area, but specific origin of China and Assam type are yet to be made clear. To unravel the origin of Indian tea, we performed chloroplast genome based phylogenetic analysis and indicated that Indian Tea, TV-1 is distinct from China tea (Fig. 1.8). Hence, supporting

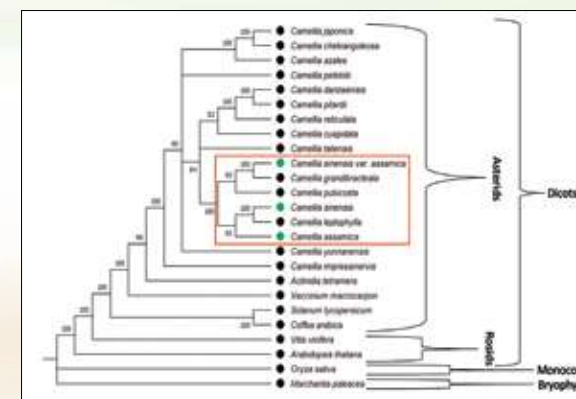


Fig. 1.8: Chloroplast (cp) genome sequence based ML tree showing the phylogenetic relation between China tea, China Assam tea and Indian Assam tea (green bullet) to confirm their different origins

the possible different domestication origins of Indian and China Assam tea with the existence of Indian Assam tea prior to *C. sinensis* and China Assam tea.

TeaMiD: a comprehensive database of simple sequence repeats markers of tea

Himanshu Dubey, Hukam C Rawal, Megha Rohilla, Nagendra Kumar Singh, Tapan Kumar Mondal

Dubey et al., 2020 Database

Information about simple sequence repeat (SSR) in tea is scanty. We identified a total of 935, 547 SSRs from the RNA-seq, GSS, ESTs and organelle genomes (chloroplasts and mitochondrial) along with published literature to catalog the validated set of tea SSR markers (Fig. 1.9). Further 82 SSRs were validated among a diverse set of tea genotypes. A user-friendly

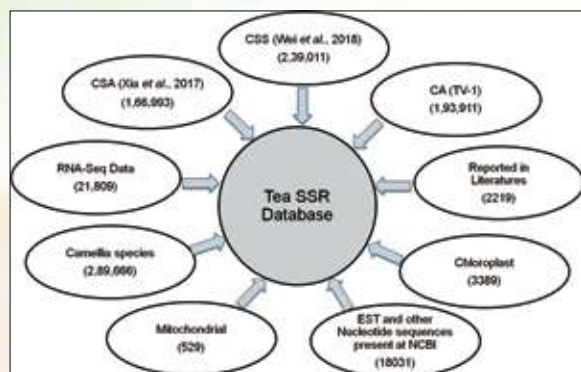


Fig. 1.9: Summary of SSR database. Numbers of SSR identified and their sources used for database development

database (TeaMiD; <http://indianteagenome.in:8080/teamid/>) was developed to host SSR from all the six resources including three nuclear genomes of tea and transcriptome sequences of 17 *Camellia* wild species. Database URL: <http://indianteagenome.in:8080/teamid/>.

Investigation of the plant U-Box family of proteins (PUB) in rice (*Oryza sativa*) with emphasis to their roles in abiotic stress responses

Harmeet Kaur, Akansha Chaurasia and Tapan Kumar Mondal

Plant U Box ligases (PUB) are a class of E3 ligases involved in the ubiquitin proteasome pathway. Abiotic stress responsive PUB63 which encodes for a CDS of 1032bp with a predicted protein of 344 amino acid residues was cloned for further characterization in yeast and Arabidopsis. PUB63 cloned in the entry

vector has also been transferred to pGBKT7 yeast vector for the identification of potential interacting partners and another vector with GFP tag. *In-silico* prediction of subcellular localization revealed a high probability of cytosolic and nuclear localization. Localization in onion peel via agroinfiltration revealed that the PUB63 protein was localized in the cytosol as well as nucleus (Fig. 1.10)

Down Regulation of *Rhizoctonia solani* endogenous genes by host-Delivered RNA interference to enhance sheath blight disease resistance in Rice

Ila Mukul Tiwari, Deepak Singh Bisht, Tilak Raj Sharma and Tapan Kumar Mondal

Development of sheath blight (ShB) resistance by silencing the target *Rhizoctonia solani* pathogenicity genes essential for disease development is the objective of this project. For



Fig. 1.10: Localization of OsPUB63:GFP showing nuclear and cytosolic signals in infiltrated onion peel cells



Fig. 1.11: Different strains of *Rhizoctonia solani* maintained on potato dextrose agar

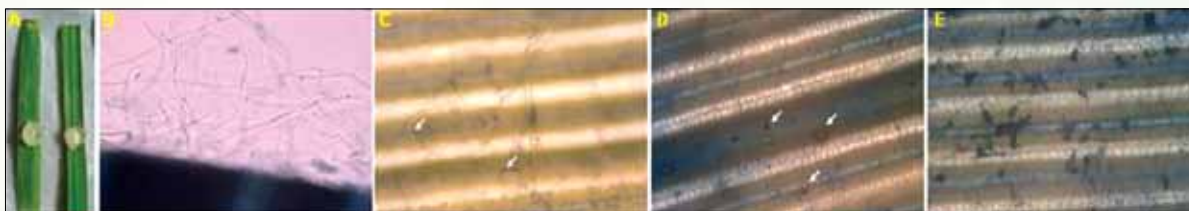


Fig. 1.12: Disease progression in rice leaves post inoculation with *Rhizoctonia solani* (A) Inoculation of leaves with fungal mycelial disks (B) Attachment of fungal hyphae to leaf surface (C & D) Formation of infection structures and fungal penetration inside leaf tissues (E) Formation of infection cushions and subsequent development of lesions over leaf surface, leading to necrosis of tissues

genetic transformation experiments, *in vitro* somatic embryogenesis protocol was standardized. Mature seeds of susceptible cultivar against sheath blight were inoculated on callus induction medium to generate embryogenic calli. To silence the target genes, genetic transformation experiments were initiated and the transformed callus is now at selection phase. For challenging the transgenic plants with the pathogen many strains of *R. solani* are being maintained (Fig. 1.11).

Infection process of *R. solani* involves growth of fungus over the plant and then attachment of mycelia over plant surface. After that infection structures are formed which leads to penetration of fungus inside the plant tissues. After tissue colonization disease symptoms appears on plants surface (Fig. 1.12). To study the disease progression, rice leaves were challenged with *R. solani* using detached leaf assay. Development of fungal structures as well as infection cushions was observed.



Role of miRNA (*MIR319*) in growth and yield

Prasanta K Dash, Arora Gauri

Amongst myriad miRNAs in plants, MIR319 is the first miRNA experimentally characterized in Arabidopsis and found to be most conserved miRNA families in plants. Information regarding the evolution of MIR319. miRNA genes in Poaceae is lacking. The comparative genomics was carried out to identify MIR319 homologs and its homeologs in *Oryza* and non-*Oryza* species. Our comparative genomics data revealed that in *Oryza sativa* ssp. indica, two homologs of MIR319 i.e. MIR319A

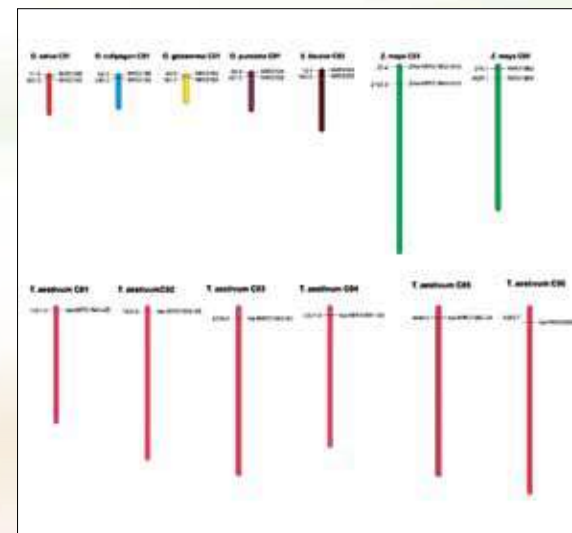


Fig 1.13: Genomic location of *MIR319* homologs (*MIR319A* and *MIR319B*) and homeologs (*MIR319 A1*, *A2*, *A3* and *MIR319 B1*, *B2*, *B3*) of different Poaceae members

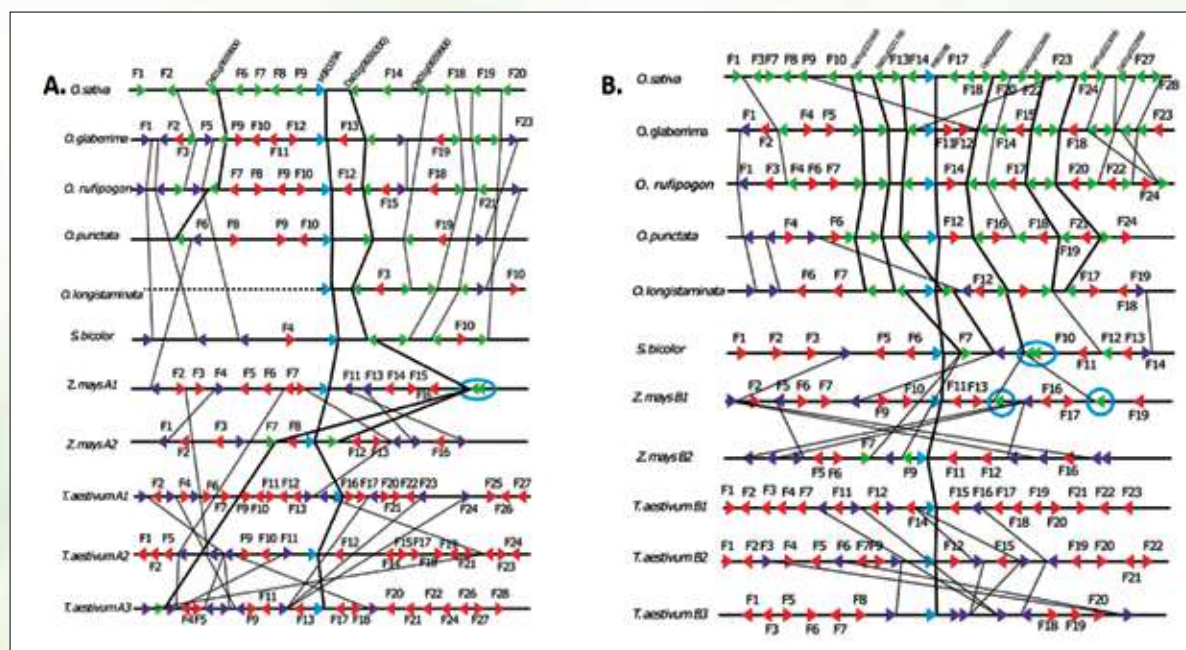


Fig 1.14: Microsynteny analysis of 100kb genomic segments flanking MIR319A (A) and MIR319B (B) across different Poaceae members. This region is marked by various gene retentions, duplications, losses and rearrangements (encircled). Blue arrow: rMIR319A; green arrow: *O. sativa* homologs; purple arrow: genes conserved among other genomes and sub genomes; red arrow: genome/ subgenome specific gene

and MIR319B were found and both are located on chromosome 1 in a distance of 22.8 Mbp. While two homologs of MIR319 were detected in rice, Sorghum, maize and wheat i.e. MIR319A and MIR319B; two homeologs for MIR319A (MIR319A1 and MIR319A2) and MIR319B (MIR319B1 and MIR319B2) in maize and three homeologs each of MIR319A (MIR319A1, MIR319A2 and MIR319A3) and MIR319B

(MIR319B1, MIR319B2 and MIR319B3) were detected in wheat (Fig. 1.13). Gene organization analysis of 100 kb region harboring MIR319 revealed 14 genes in *O. sativa*. Further, analysis of the gene content and gene density among 100kb segments of eight poaceae members revealed that highest percentage of *O. sativa* homologs were conserved in *O. punctata* (40% i.e., 4 out of 10 genes) and the lowest conservation in *T.*

aestivum A1 (3.8% i.e., 1 out of 21 genes). Similarly, maximum gene density was found in *T. aestivum* 4D (1gene/ 43.8kb) and minimum density in *O. longistaminata* (1gene/12.5kb) (Fig.1.14A). In the 100kb segment harboring MIR319A, only V-type proton ATPase Subunit E was found to be conserved in all poaceae members. Similarly, one homolog each of MIR319B was detected in *O. rufipogon*, *O. glaberrima*, *O. longistaminata*, *O. punctata* and *S. bicolor*, while two homeologs of MIR319B were detected in case of *Zea mays* (B1 and B2) and three in *T. aestivum* (B1, B2 and B3). Maximum density of gene was observed in *T. aestivum* B1 (1 gene/ 4.7kb) and minimum in *Z. mays* (1 gene/ 7.4 kb) (Fig. 1.14B). A total of 23 genes were detected in 100 kb region of *O. sativa*, of which six genes were found to be conserved in all *Oryza* species. A vacuolar channel protein having higher selectivity for malate viz. Aluminum-activated malate transporter 9-like was found to be conserved while V-type proton ATPase subunit E was present in tandem in *O. sativa*, *O. glaberrima*, and *O. rufipogon*.

Development of transgenics in flax

Prasanta K Dash, Soumyadeep Mukherjee Karthik

Kesiraju et al. 2021 Frontiers in Plant Science

Flax is a dual-purpose crop that has industrial as well as nutraceutical value. Development of flax cultivar with high fiber

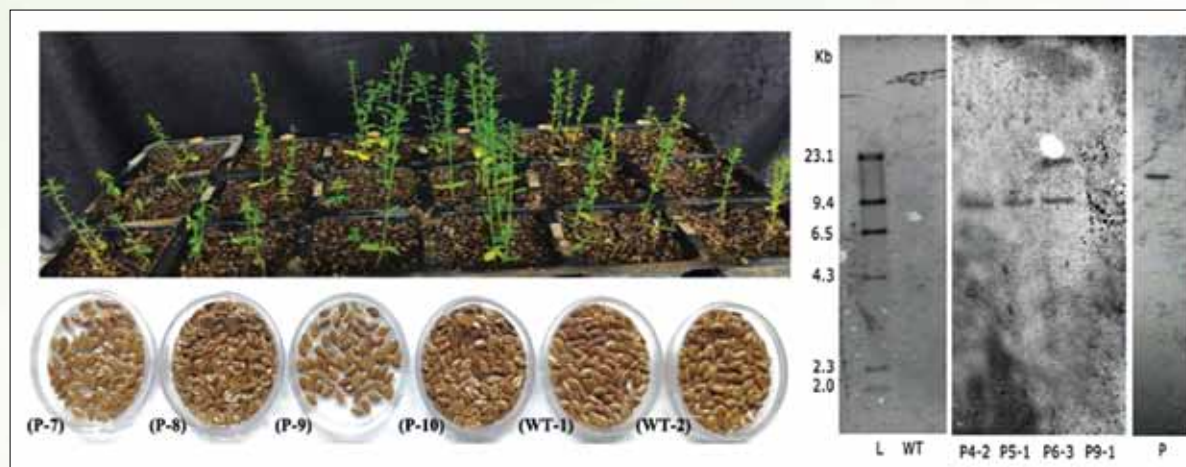


Fig 1.15: Overview of the variable response of flax transgenics (T_1) exhibiting resistance and susceptibility to kanamycin and seed pool collected from 10 transgenic and wild-type plants. Molecular detection of transgene by Southern analysis

yield is an industrial requirement and scientific endeavor. Since flax is a tissue-culture recalcitrant plant, its amenability to *Agrobacterium tumefaciens* mediated transformation was optimized by *in planta* transformation method. The amenability, authenticity, and reproducibility of the flax transformation was demonstrated, verified at the molecular level, by GUS histochemical analysis, gene-specific PCR, Southern hybridization for stable integration of T-DNA, and expression analysis of transgenes by sqRT-PCR (Fig. 1.15). The method is a promising tool and can be incorporated in varietal improvement programs in flax through genetic modification.



Integration of drought and low nitrogen dual stress transcriptomes and nitrogen use efficiency QTLs in rice

Amitha Mithra SV, Subodh Kumar Sinha, Sureshkumar V, Manju Rani, Manish Ranjan Saini, Sapna Kumari, Megha Kaushik, Chandra Prakash, and Pranab Kumar Mandal

Sevanthi *et al.* 2021, Rice

We used Nagina 22 (N22), a poor N use but drought tolerant genotype and IR64 an efficient N use but drought sensitive genotype for understanding the impact of drought, low N and their combined stress in rice. Overall,

N22 performed better under dual stress conditions owing to its better root architecture, chlorophyll and porphyrin synthesis and oxidative stress management. Nearly double the number of differentially expressed genes (DEGs) were found in the shoot tissues compared to the root tissues under N-W+, N+W- and N-W- treatments (Fig. 1.16). The negative regulators of N- stress, e.g., *NIGT1*, *OsACTPK1* and *OsBT* were downregulated in IR64 while in N22, *OsBT* was not downregulated. We also identified 12 QTLs for seed and straw N content using 253 recombinant inbred lines derived from IR64 and

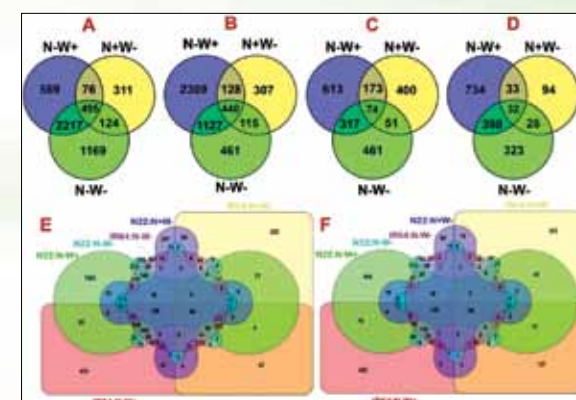


Fig. 1.16: Comparison of common and unique DEGs identified across low N (N-W+), low water (N+W-) and dual stress (N-W-) treatments compared to optimal (N+W+) conditions in root (R) and shoot (S) tissues of two rice genotypes, IR64 and N22. A: DEGs in shoot tissues of IR64 across treatments; B: DEGs in shoot tissues of N22 across treatments; C: DEGs in root tissues of IR64 across treatments; D: DEGs in root tissues of N22 across treatments; E: DEGs identified in shoot tissues of both the genotypes; F: DEGs identified in root tissues of both the genotypes

N22 and a 5K SNP array data. Three of the 12 QTLs co-localized on chromosome 6 spanning 61 genes, of which, five were DEGs including a UDP-glucuronosyl/UDP-glucosyltransferase family protein, serine-threonine protein kinase, anthocyanidin 3-O-glucosyltransferase protein and a nitrate-induced protein. The DEGs, QTLs and candidate genes identified in this study can serve as a major resource for both rice improvement and functional biology.

Evaluation of drought tolerance in the EMS induced mutants of Nagina 22 by IR imaging

Neera Yadav, Ramkumar, Ajaykumar, Vasuda Jadon, Rakesh Pandey, Nagendra Kumar Singh and Amitha Mithra SV

Nearly 1100 EMS induced mutants of Nagina 22 (N22) from the 'National Repository of EMS induced mutants of rice' at ICAR-NIPB, along with 10 check varieties were screened for drought tolerance at active tillering stage by withholding water for 17 days. Canopy temperature (CT) measurements were recorded by fluke thermal imaging (°C) using Fluke Temperature Measurement tools on day 0, day 6 and day 17 and the images were visualized by Smart view software. Based on the least differences in the CT, 45 mutants were selected for further studies. The frequency distribution of the mutants based on the temperature measurements on Day 17 are presented in Fig.

1.17 and a representative imaging of the crop canopy is presented in Fig. 1.18.

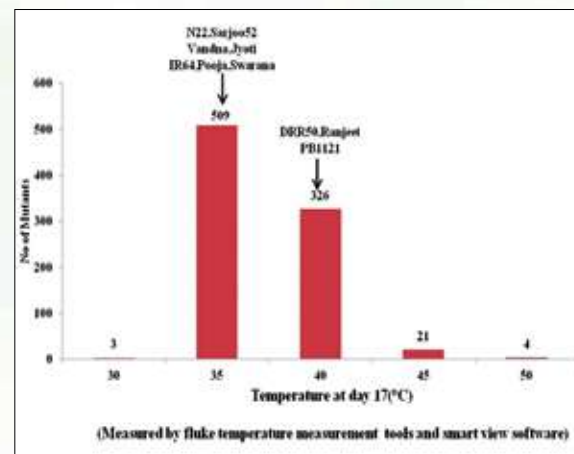


Fig. 1.17: Frequency distribution of the 1100 EMS induced mutants of Nagina 22 based on the thermal imaging recorded on Day 17 of imposing drought stress in the rainout shelter, ICAR-NIPB

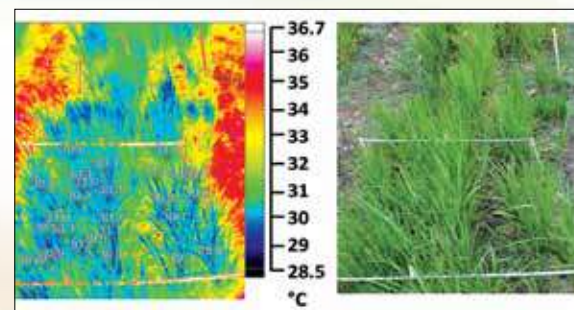


Fig. 1.18: A representation of the thermal imaging data recorded by Fluke Temperature Measurement tools and visualized by Smart view software in the 1100 EMS induced mutants of Nagina 22 for evaluation of their performance under drought stress

Growth and grain filling dynamics in the selected EMS induced Nagina22 mutants and mapping of the causal genes

Jeet Roy, Ajaykumar, Ramkumar, Vasuda Jadon, Neera Yadav, Amolkumar Solanke, Singh NK and Amitha Mithra SV

Seven EMS induced mutants of N22 with varying panicle architecture were selected for the phenotype and molecular genetic analysis. The four semi-dwarf mutants, MG010, MG029, MG044, MG046 showed early heading like N22 while the tall mutants MG092, MG469 and MG502 showed delayed heading by ~1-2 weeks. MG010 had the highest number of tillers and panicles across locations but had the least value for all other panicle architecture traits while MG502 and MG469 had the least number of tillers and panicles but had the highest values for panicle length, rachis length, number and length of primary and secondary branches, grain number and fresh and dry weight of panicle (Figs. 1.19 and 1.20). Whole genome resequencing (WGS) data generated at ~50x for the mutants showed variants affecting 0.041-0.16% of the genome with 38-39% of them resulting in non-synonymous (NS) substitutions with the least changes in MG010 and the most changes in MG092. Splice junction variants in mutants ranged from 149-868 while stop codon variants ranged from 26-175. MG502 followed by MG469 had the highest number of genes showing NS changes (22 and 20) while MG029

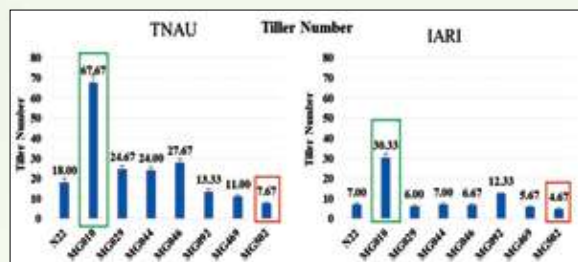


Fig. 1.19: Performance of the seven selected mutants and Nagina 22 for tiller number at two locations, Coimbatore (TNAU farm), and New Delhi (IARI farm)

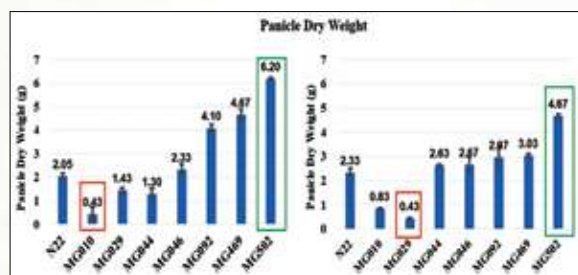


Fig. 1.20: Performance of the seven selected mutants and Nagina 22 for panicle dry weight at two locations, Coimbatore (TNAU farm), and New Delhi (IARI farm)

followed by MG046 had the least changes (7 and 9). The genes with higher number of NS changes included *IPA1*, *GIF1*, *SLG7*, *OsFH15*, *Hd1* and *An-1*. The phenotype of the erect and longer panicles of the mutants MG046, MG092 and MG502 corresponded with the higher number of NS changes in *IPA1* as compared to no changes in other mutants and 4 changes in MG469. The latter had sorghum like panicle and was also accompanied by 3 NS changes in *LAX2* gene. The four high grain number mutants also had multiple NS changes in *GIF1* and *SLG7*.



Cloning and characterization of panicle blast inducible *OsBAK1* gene

Amolkumar U Solanke,
Vishesh Kumar, Suhas G
Karkute, Mayuri Mahalle, SV

Amitha Mithra and TR Sharma

In the previous year, to explore the comparative panicle blast responsive transcriptome, we used a well-known blast resistant cultivar Tetep and a susceptible cultivar HP2216. We have also proposed a model where crucial pathways and genes have been depicted culminating in a strong defense response against panicle blast disease in Tetep.

BRASSINOSTEROID INSENSITIVE 1-associated receptor kinase 1 (BAK1), a central regulator of plant immunity was found to be highly up-regulated in Tetep. We first confirmed the expression of *OsBAK1* gene in both Tetep and HP2216 and further cloned it from Tetep. Also confirmed it by sequencing in binary vector pCAMBIA1300. This vector is transformed in the *Agrobacterium* strain LBA4404 for rice plant transformation. We generated many calli on selection media in TP309. Few putative transformants are now shifted to rooting media and ready for molecular characterization and blast phenotyping. Relevant figures are presented in Fig. 1.21.

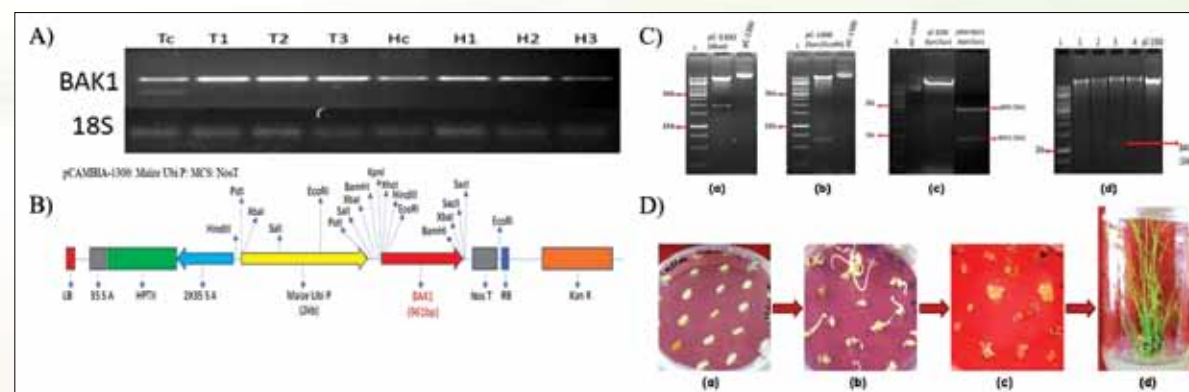


Fig. 1.21: (A) Semi-quantitative PCR of *BAK1* gene in Tetep (T) and HP2216 (H), C: control, 1, 2 and 3 are 48, 72 and 96 hours post infection. (B) and (C) Vector map and restriction analysis of binary vector pCAMBIA-1300:BAK1, (D) rice transformation steps and putative transgenic plants in rooting media after regeneration

Validation of EMS induced rice mutants of N22 for leaf and panicle blast resistance

Amolkumar U Solanke, SV Amitha Mithra, Mohd. Tasleem, Meena Adhikari, Umakanta Ngangkham, Someshwar Bhagat

Rice blast is the most devastating disease of rice. In 2017 and 2018, we identified 60 blast resistant lines from 1.5 kg M₂ generation N22 mutant seed nursery (approximately 1 lakh seeds) raised for blast screening in the blast nursery at NRRI-Central Rainfed Upland Rice Research Station (CRURRS) Hazaribagh. In 2019 and 2020, these 60 mutants were again screened for leaf and panicle blast resistance at ICAR-NIPB and for leaf blast at CRURRS and ICAR-Research Complex for NEH Region, Umiam-793103, Meghalaya which revealed varied reaction to blast infection. This year, we screened three segregating progeny lines of each mutant with artificial evaluation using *Magnaporthe oryzae* strain *Mo-ni-25*.) NBM13C, 13D, 16A, 55C, 56A, 56B and 56C were found to be resistant for both

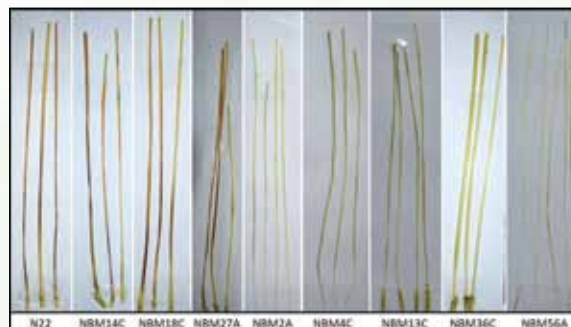


Fig. 1.22: Panicle blast disease screening of N22 mutants using artificial inoculation of *M. oryzae*

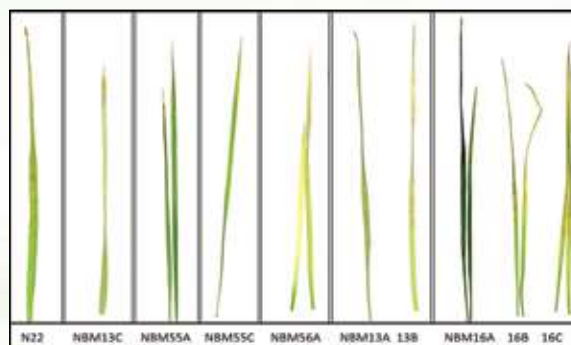


Fig. 1.23: Leaf blast disease screening of N22 mutants using artificial inoculation of *M. oryzae*

leaf and panicle blast (Table 1.1, Fig. 1.22 and 1.23).

Cloning of different transcript variants of *OsLRK10-like* in binary vector

Amit Pareek and Amolkumar Solanke

To investigate the importance of alternative splicing in blast disease resistance in rice, different variants of *LRK10* like gene as *LRK10L1.1* and *LRK10L1.2* including numerous alternative splicing variants of *LRK10L1.2* have been identified, cloned and confirmed by sequencing. Further, to determine the role of individual transcript in blast disease resistance, all these variants were amplified from genomic DNA and cDNA of leaf tissue of rice plants. Amplified products were cloned into binary vector and cloning was confirmed by restriction digestion with specific restriction enzymes and further confirmed by sequencing (Figs. 1.24 and 1.25).

Table 1.1: Details of the N22 mutants identified for blast resistance from the EMS induced N22 mutant resource

No. of mutants	Mutant IDs	Disease reaction
16	NBM 2A, 4C, 7A, 13C, 16A, 16B, 16C, 50C, 51C, 52A, 52C, 55A, 55C, 56A, 56B, 56C	Highly resistant to panicle blast
15	NBM 4A, 4B, 6C, 7B, 8C, 9A, 13D, 15A, 22C, 23A, 23B, 36B, 36C, 32A, 53C	Moderately resistant to panicle blast
14	NBM 5C, 14C, 18C, 21A, 21B, 25C, 27A, 27B, 37A, 41C, 43B, 45C, 47A, 60C	Susceptible to panicle blast
10	NBM 13C, 13D, 16A, 22B, 53A, 53B, 55C, 56A, 56B, 56C	Resistant to leaf blast
7	NBM 6C, 7B, 13B, 16C, 22C, 35A and 53C	Susceptible to leaf blast

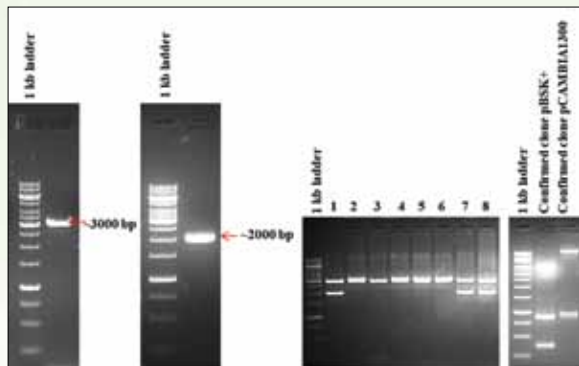


Fig. 1.24: (A) and (B) PCR amplification of 3kb and 2kb transcript from genomic DNA and cDNA, (C) Confirmation of 2kb cDNA into pBSK+ vector by restriction digestion with *Bam*H1/*Sac*1, (D) Confirmation of 2kb cDNA cloned into pBSK+ vector and pCAMBIA1300 vector by restriction digestion with *Eco*R1

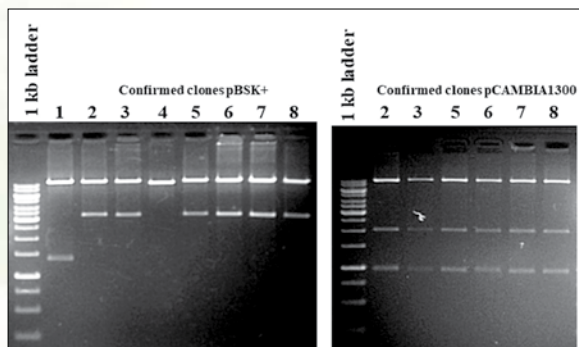


Fig. 1.25: (A) Confirmation of 3 kb clone into pBSK+ vector by restriction digestion with *Bam*H 1/*Sac* 1, (B) Cloning confirmation into pCAMBIA1300 vector by restriction digestion with *Eco*R1

Understanding molecular mechanism of Silicon mediated resistance against biotic stress using rice blast as a system

Gitanjali Jiواني and Amolkumar Solanke

Silicon (si) application in plants provides protection against a broad range of pathogens by acting as a physical barrier or by the soluble Si modifying the host resistance system. To better understand this exogenous application of Si under leaf blast infection is being investigated. To validate and establish the effect of calcium silicate on providing resistance to blast infection in rice, a set of defense related genes *PR-1*, *PR-10*, *PR-17*, β -1,3 *glucanase*, *ERF* (Ethylene response factor) and a silicon transporter, *Lsi6* were selected for gene expression analysis. β -1,3 *glucanase*, *PR1*, and *PR-10* and *ERF* showed very high expression in response to infection as well as silicate treatment in cv. HP2216 but not in Tetep. The silicon transporter *Lsi6* was downregulated in Tetep in response to infection as well as silicate treatment while in HP2216, it did not show any change in expression under blast infection but was slightly down regulated under silicate treatment.

Development of Standard Operating Protocols (SOP) for genetic fidelity testing of Guava and Pomegranate

Amolkumar U. Solanke, Amitha Mithra SV, Mohd. Tasleem, Mayuri Mahalle, Ram Sevak Tomar, Dipti Dhumale

We develop Standard Operating Protocols (SOP) for genetic fidelity testing of tissue culture raised plants as Referral Centre for Genetic

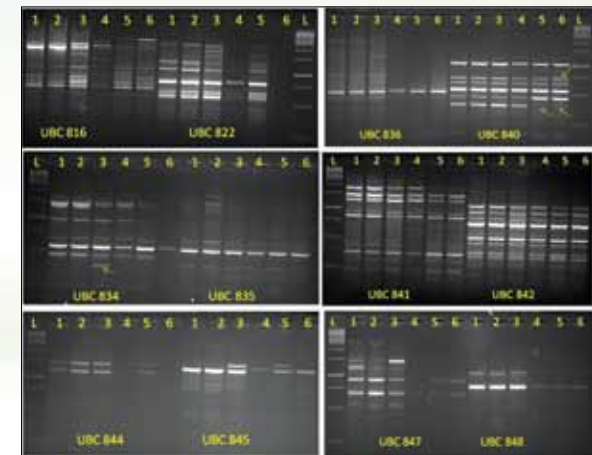


Fig. 1.26: Genetic fidelity testing of guava samples using 12 ISSR markers. Sample 1-2: Guava variety L49 from Pusa farm; 3-4: variety L49 from Sawai Madhopur and 5-6: variety Allahabad Safeda

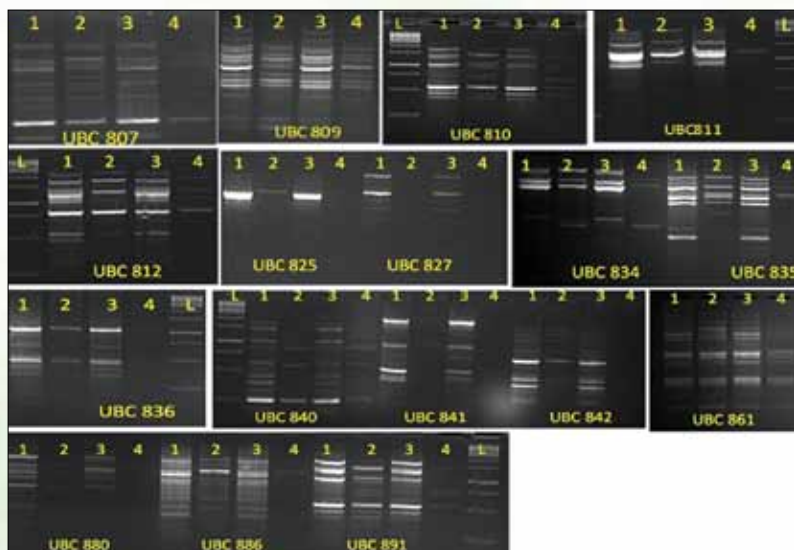


Fig. 1.27: Genetic fidelity testing of pomegranate samples using ISSR markers

Fidelity Testing under National Certification System of Tissue culture Raised Plant (NCS-TCP), a program run by Govt. of India through DBT and BCIL. This year we made an attempt to develop robust and reliable molecular marker system for genetic fidelity testing of guava and pomegranate plants through ISSR markers. Here, we screened six samples of guava and 4 samples of pomegranate plants. Our attempts revealed

that 13 ISSR primers amplified 2 to 11 bands in guava, and 17 primers amplified 1 to 9 bands in pomegranate after screening 100 ISSR primers. UBC primers UBC816, UBC822, UBC834, UBC840, UBC841 and UBC842 are most potent markers for genetic fidelity testing of guava (Fig. 1.26). UBC809, UBC810, UBC812, UBC835, UBC840, UBC842 and UBC891 are most potent markers for genetic fidelity testing of pomegranate (Fig. 1.27). After refinement with larger varietal screening, these can act as robust system for the genetic fidelity testing of guava and pomegranate.

2. GENETIC IMPROVEMENT OF WHEAT FOR ADAPTATION TO CLIMATE CHANGE INDUCED STRESSES



Analysis of homeologous transcripts of major genes related to nutrient reservoir (NR), carbohydrate metabolism (CM), and defence protein (DP) in hexaploid bread wheat and its diploid progenitors during grain development

Megha Kaushik, Shubham Rai, Sureshkumar Venkadesan, Subodh Kumar Sinha, Sumedha Mohan and Pranab Kumar Mandal

To elucidate the molecular functions of genes governing nutrient reservoir (NR), carbohydrate metabolism (CM), and defence protein (DP) in hexaploid wheat and its diploid progenitors during grain development, expression of homeologous transcripts was analysed and a heat map was constructed for all the major genes (Fig. 2.1). The analysis revealed that all the transcripts were differentially expressed. Avenin b1 had two transcripts from A and D homeologues of bread wheat. The expression of Avenin b1 was ~3-fold higher in

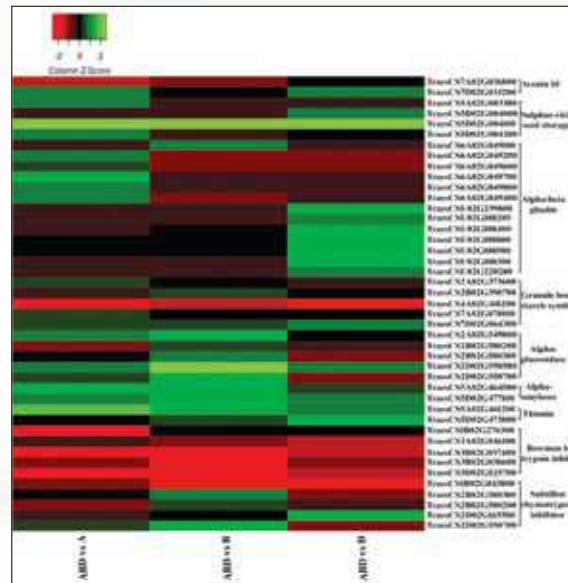


Fig. 2.1: Heat map representing different transcripts having similar functions (green: up-regulated in hexaploid; red: up-regulated in diploids)

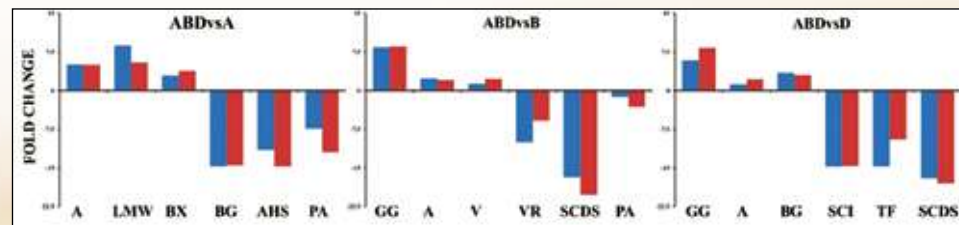


Fig. 2.2: Validation of differentially expressed genes (DEGs) (blue–qPCR; red–RNAseq)

D than that of A genome. Similarly, X6 isoform of high molecular weight glutenin also had two transcripts from B and D homeologues and their relative expression analysis showed 4-fold over expression of B in comparison to D homeologue.

Comparative transcriptome analysis of durum wheat vs. bread wheat during grain development

Megha Kaushik, Ekta Mulani, Subodh Kumar Sinha and Pranab Kumar Mandal

In an endeavour to identify genes related to anti-nutritional factors, RNAseq data of durum wheat and bread wheat was analysed and 20506 differentially expressed genes were identified. Among these genes 2892 (14.1%) were > 2 fold up-regulated and 7694 (37.5%) were < 2 fold down-regulated in tetraploid wheat. It suggests that majority of the genes during grain development are equally important in both the species.

Quantitative RT-PCR was performed for ten differentially expressed transcripts to validate RNA seq results (Fig. 2.2). The transcript expression of qRT-PCR of all selected genes was consistent with the RNA- seq results. We are in a process of identifying and analysing the immunogenic genes in two cultivated wheat species.

In silico characterization of ATIs in wheat

Megha Kaushik, Subodh Kumar Sinha and Pranab Kumar Mandal

Amylase tripsin inhibitors (ATIs) are present

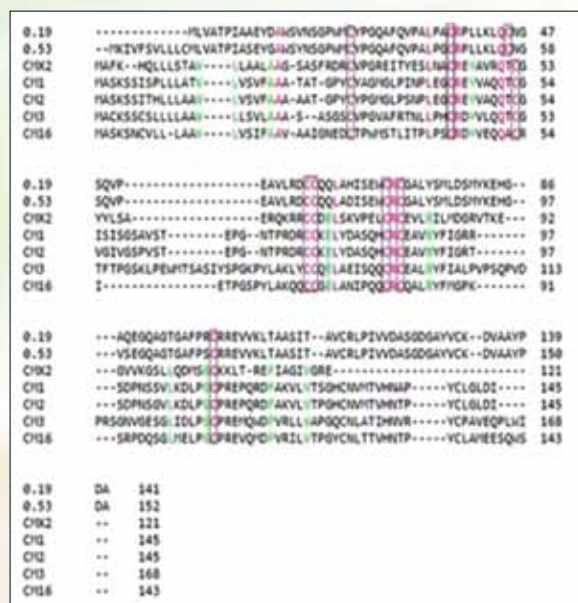


Fig. 2.3: Sequence alignment of distinct ATI variants from wheat. The highly conserved cysteine residues are boxed. Pink and green color denotes identity and homology.

in gluten containing seeds such as wheat. Different ATIs cloned from hexaploid wheat were analysed by Multiple Sequence Alignment. The analysis showed highly conserved ten cysteine

residues in all ATIs except CMX2 which is a characteristic feature of these inhibitors. These cysteine residues form disulphide linkages and an organized conserved motif that may have

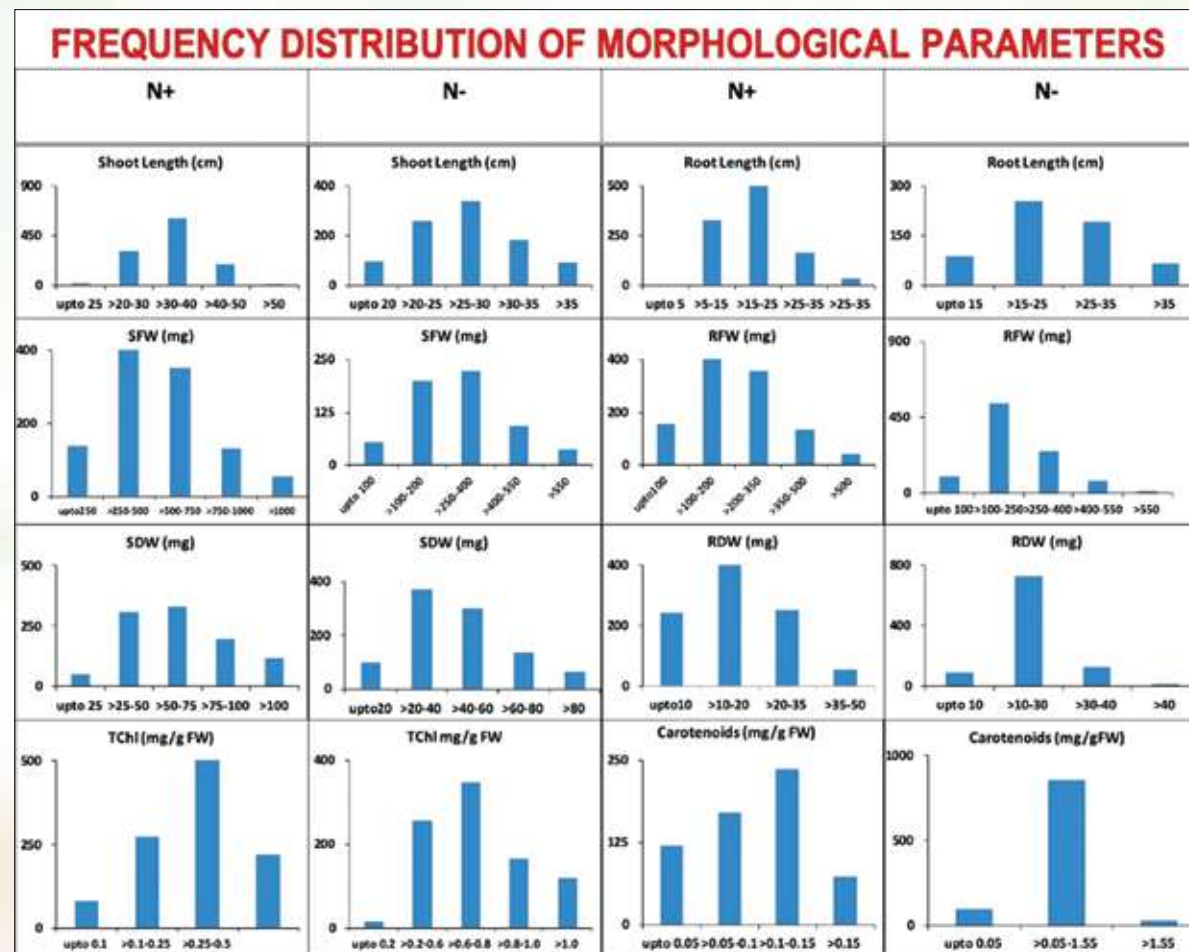


Fig. 2.4: Representative picture showing frequency distribution of different morphological parameters of 300 genotypes

an important role in making them resistant against denaturation and binding amylases and proteases. The comparative analysis of different wheat tetrameric amylase inhibitors subunits reveals highly conserved 40-50 amino acid residues in N-terminal signal peptide sequence. These conserved sequences may account for stable quaternary structure and activity of the tetrameric protein (Fig. 2.3).

Genome wide association studies under N-stress

Gayatri, Ashmita Rai, Subodh Kumar Sinha and Pranab Kumar Mandal

More than 300 wheat genotypes were phenotyped for Morphological (Shoot and Root length, Shoot and Root fresh and dry weight) and biochemical characters (Key N-metabolizing enzyme assay of NR, GS, GOGAT, ICDH and AlaT; soluble protein, Chlorophyll and Carotenoids). Based on the genotypic (SNP genotyping using 35K Breeder's array) and phenotypic data, preliminary level genome wide association studies (GWAS) were carried out.

The analysis revealed significant QTLs/MTAs for shoot length (Chr 2A), pigment content (Chr 1B), NR (Chr 5A), GS (Chr 1B, 2A, 2B, 3A, 4A and 7A) and Alt (Chr 3A) under N-stress condition; and for GS (Chr 5A) and GOGAT (Chr 2A, 3B, and 5B) under N optimum condition (Fig.2.4).

Gluten fractionation for immunological studies

Megha Kaushik, Daisy Masih, Subodh Kumar and Pranab Kumar Mandal

Gluten is responsible for wheat immunogenicity. To know the specific fractions of gluten responsible for immunogenicity, we standardized a protocol for fractionation of these proteins (Fig. 2.5). Gliadins were extracted in 70% ethanol while 50% propanol with dithiotritol (DTT) was utilized for glutenin

extraction. Both the proteins were analyzed through HPLC. Different fractions of gliadins were collected and reanalyzed on HPLC, however ω -gliadins were found to be present in every fraction. Since immunogenicity differ in A, B and D genome, we also studied the polypeptide profiles in mature seeds of ancient diploid progenitors [*T. monococcum* (AA); *Ae. speltoides* (BB); *Ae. squarrosa* (DD)] which were found to be quite different in different wheat species Fig. 2.6).

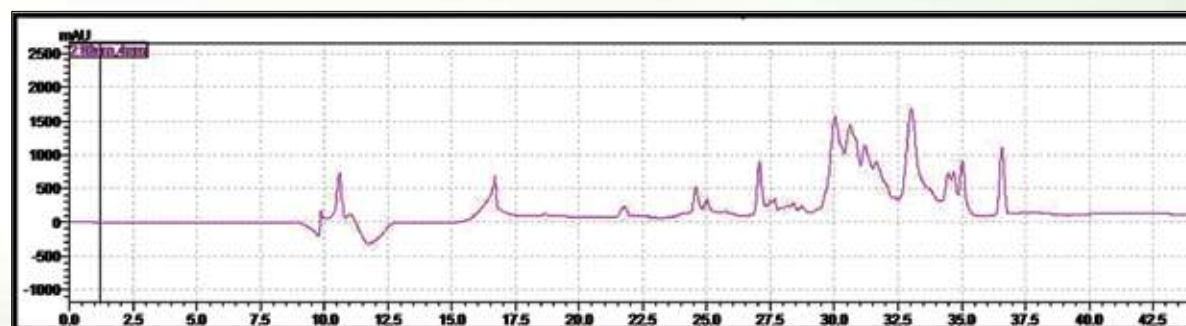


Fig. 2.5: Gluten analysis from hexaploid bread wheat. A) Gliadin and its different fractions; B) Glutenins

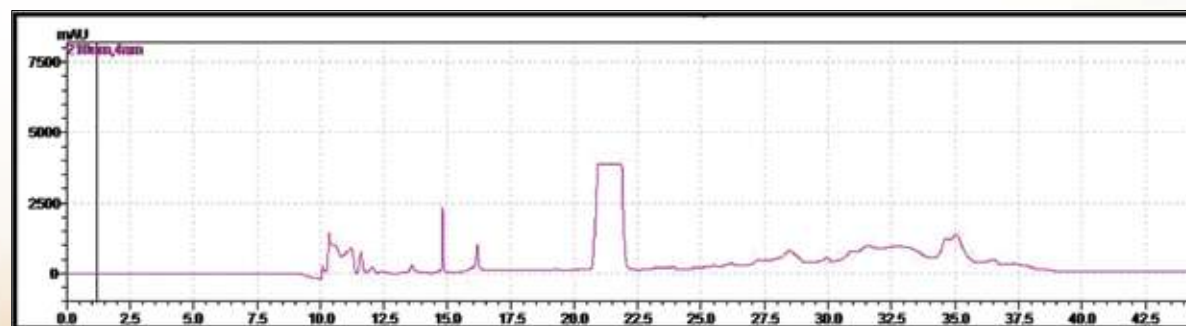


Fig.2.6: Gliadin profile of cultivated wheat species and their progenitors



Studies on effect of *Phalaris minor* on modulation of root system architecture and expression of *TaNPF6.4* of bread wheat under limited external nitrate concentration

Priyanka Kumari, Pranab Kumar Mandal and Subodh Kumar Sinha

We investigated impact of self and non-self root-root interaction on root system architecture and expression of putative nitrate

transceptor gene *NRT1.1* in bread wheat (*Triticum aestivum* L.) under nitrogen limited condition. Root biomass (length, fresh and dry weight) of wheat invariably increased under N-starvation condition which further increased in presence of *P. minor*. Both length-based root traits, e.g., TRS, MRP and LRS, as well as numbers-based traits, e.g., first and second order lateral root numbers, were increased by the presence of *P. minor*, especially under N-starved condition (Fig. 2.7). Out of 17 orthologs of *NRT1.1*, four putative nitrate transceptor (*TaNPF6.4*) on 4th, 5th and 7th chromosome of wheat were selected for further studies. The homeolog specific expression

of *TaNRT1.1* (*TaNPF6.4*) in wheat genotype K9107 revealed that sub-genome A is highest expressing homeolog in both seminal and lateral roots (Fig. 2.8). The overall relative expression gets reduced in seminal roots whereas lateral root showed up-regulation in the presence of *P. minor* under N-condition. In order to understand the mechanism of different expression pattern of these four homeologs under different conditions in the present study, 2 kb upstream sequences of three homeologues (A, B and D) were cloned and analysed bioinformatically.

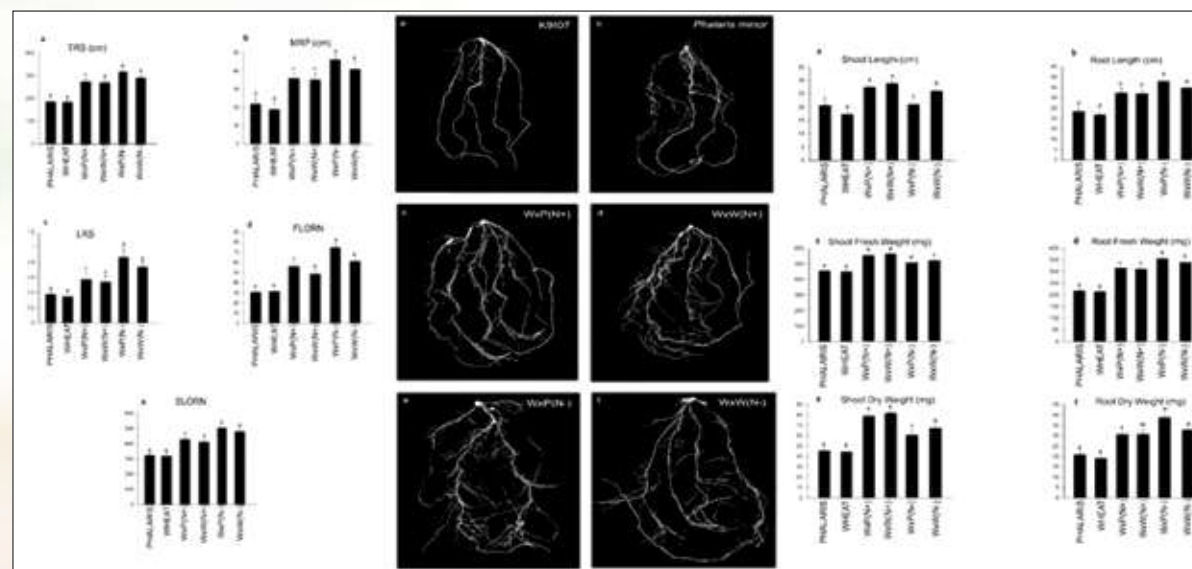


Fig. 2.7: Modulation in root system architecture (TRS: total root size; MRP: main root pathlength; LRS: lateral root size; FOLRN/SOLRN: first and second order lateral root number respectively) and biomass (length, fresh and dry weight of shoot and root) of wheat seedlings (W) in the presence of *P. minor* (P) under optimum-N (N+) and limited-N (N-) conditions

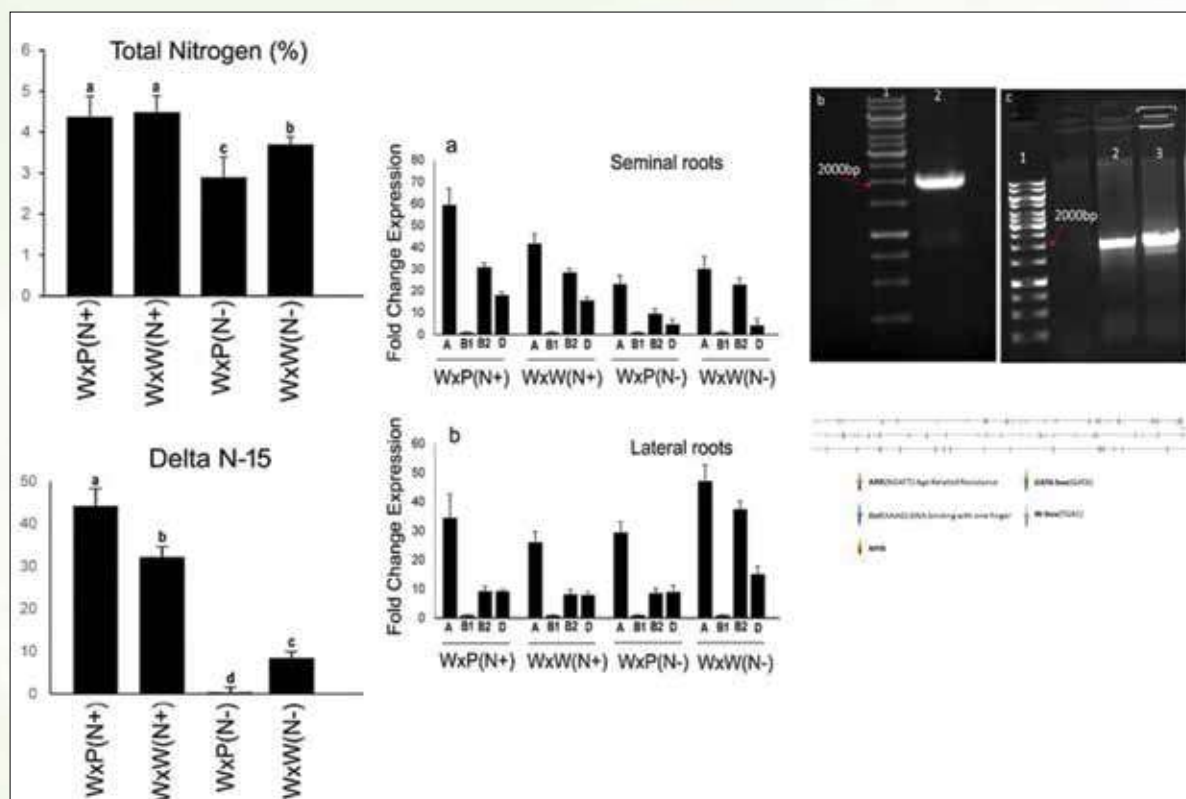


Fig. 2.8: Effect of presence of *P. minor* (P) on total-N, $\delta^{15}\text{N}$, homeolog specific relative transcript expression (*TaNP6.4*; A, B1, B2 and D sub-genome) of bread wheat (W) under optimum-N (N+) and limited-N (N-) conditions. Gel photograph depicts PCR amplification and analysis of upstream sequence of three homeologs, i.e., A (left panel), B and D (right panel)

***In-silico* identification and group specific expression studies of *NRT2.1* and *NAR2* family genes in bread wheat (*Triticum aestivum* L.)**

Amresh Kumar, Pranab Kumar Mandal and Subodh Kumar Sinha

Through *in silico* analysis, total 46 and 8 protein coding sequence of *TaNRT2* and *TaNAR2* respectively were identified in bread wheat genome (Fig. 2.9). Subsequent bioinformatic analysis revealed that *NRT2* genes were located in all wheat chromosomes except chromosome

number 4 and 5. Phylogenetic analysis revealed that *NRT2* and *NAR2* genes were distributed in 13 and 4 subgroups respectively. Group specific expressions of all identified *TaNRT2s* and *TaNAR2s* genes was studied under N-starvation and NO_3^- -inducible conditions in both root and shoot tissue in Chinese Spring (CS) cultivar and in root tissue of a N-uptake efficient wheat genotype (i.e., K9107; K9) at seedling stage. Similar condition was undertaken to study expression profiling of identified wheat *NAR2* genes as well. Among thirteen groups, only group 1, 4 and 6 were found to be highly expressed (~12 to ~44 folds) under N-starvation condition as compared to control condition in CS root tissue. These genes also expressed in N-inducible condition than that of other groups indicating the potential candidate for being *TaNRT2.1* type. Same expression profile was also observed in K9107 root tissue including group 2 genes which showed considerably higher expression under N-inducible condition as well. In case of *NAR2* genes, except group 3, all showed N-starvation induced expression in root tissue (Fig. 2.10).

Homeolog specific expression of these highly expressing genes was studied under N-starvation in root tissue of CS and K9. A homeolog expression bias was observed both at gene as well as at genotype level under N-starvation condition. While CS contributed maximum

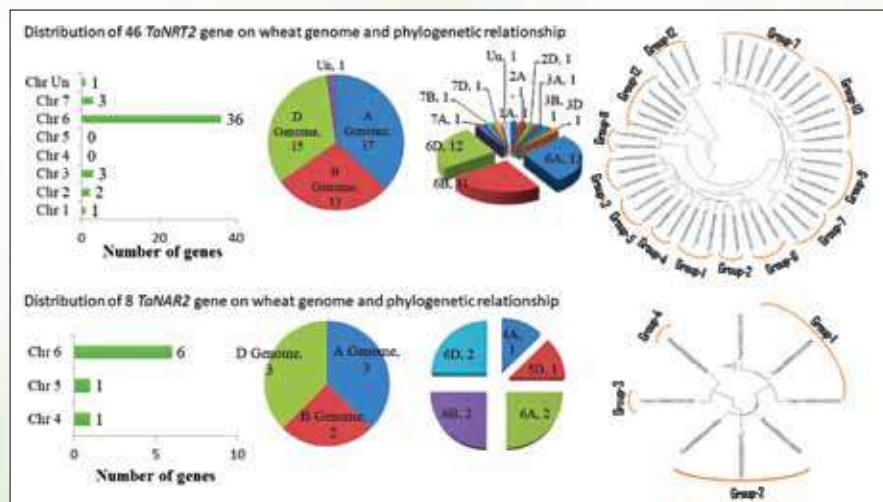


Fig. 2.9: Chromosomal distribution and phylogenetic relationship of wheat *NRT2* and *NAR2* genes. The distribution of 46 *TaNRT2* and 8 *TaNAR2* genes on A, B, and D sub-genomes are also shown.

expression from its A and D sub-genome; B and D sub-genomes expression was maximum for K9 in group 6 genes of *NRT2*. Similar trend of expression was observed for group 4 genes also, which was maximum for A genome in CS, whereas B genome in K9 (Fig. 2.10).

***In-vitro* confirmation of protein-protein interaction by split-ubiquitin assay**

Amresh Kumar, Pranab Kumar Mandal and Subodh Kumar Sinha

In order to check the possible interaction of component proteins, *TaNRT2.1* and *TaNRT2.2*; *TaNAR2.1* and *TaNAR2.2*, open reading frame

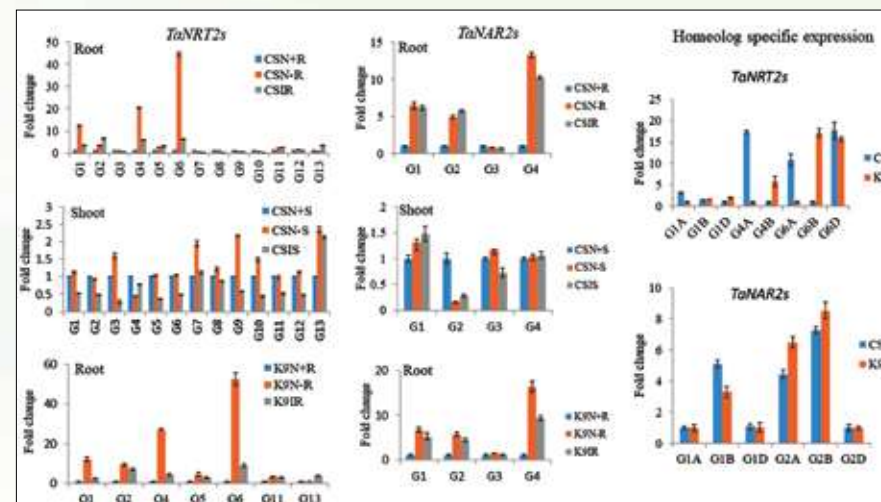


Fig. 2.10: qRT-PCR expression pattern of group specific *TaNRT2s* and *TaNAR2s* genes in two wheat cultivars, i.e., K9107 (K9) and Chinese Spring (CS) in shoot (CS) and root (CS and K9) under optimum N (N+), limited N (N-) and induced-N (NI) conditions.

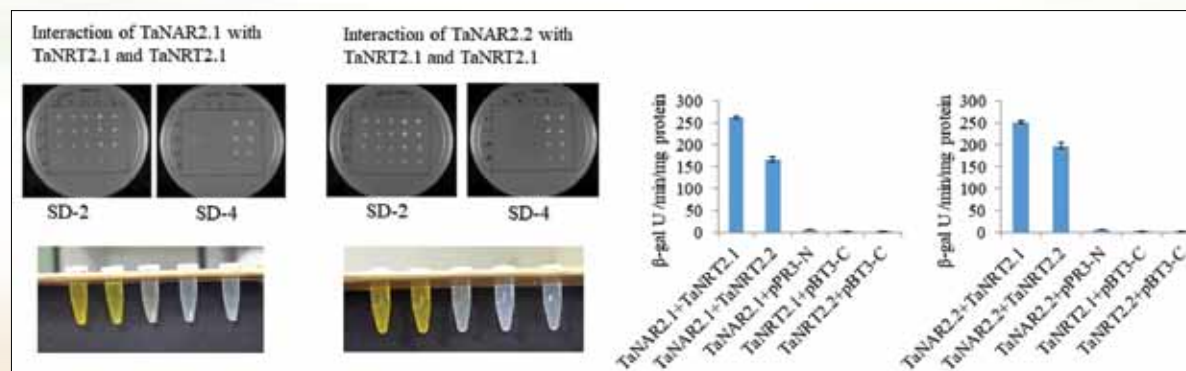


Fig. 2.11: Protein-protein interaction of two component proteins (*NAR2.1/NAR2.2* and *NRT2.1/NRT2.2*) in split-ubiquitin yeast system. Interaction was confirmed on SD-trp-leu-his-ade (SD -4) selection media as well as visual and β -galactosidase assays.

(ORF) cloned in prey and bait vectors were co-transformed in all combination in yeast strain (NMY51) under suitable selection nutrient media (SD-trp-leu-his-ade selection plates; SD -4), which showed positive physical interaction with them. Further, their interaction was also confirmed by β -galactosidase assays in the presence of ONPG as a substrate (Fig. 2.11). Our results suggest that both these partner proteins are capable of interacting in *in-vitro* split-ubiquitin yeast system.



Elucidating acquired and transgenerational thermotolerance in wheat vide hormonal and heat priming

Praful Jaiswal and Sharmistha Barthakur

The primability of plant stress response is a means to prevent the delay that a plant takes to respond to a stress. Thus, primability is the ability to become prepared or primed for improved inducible response to a subsequent stress. To enhance thermotolerance in wheat, seeds of wheat cv HD2967 were primed with IAA and GA3 alone or in conjunction with heat. Extensive phenotyping of few agronomic parameters was carried out in primed 2 (P2) and primed 3 (P3) generation for elucidating molecular mechanisms of heat stress memory in wheat (Fig. 2.12).

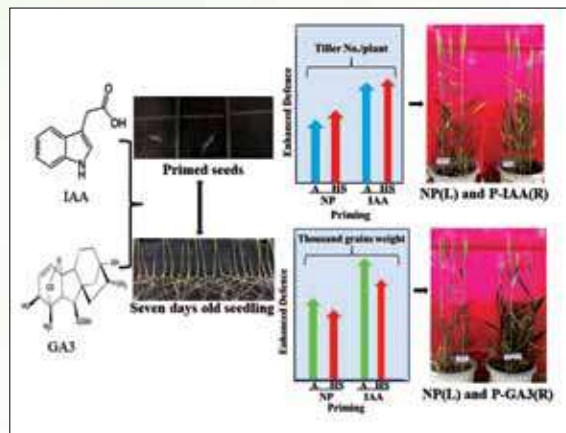


Fig.2.12: Improved seed parameters observed in primed and non-primed plants of wheat cv HD2967 grown in natural condition in net house after seed priming with IAA and GA3 respectively and exposed to high temperature at anthesis stage. A-ambient temperature; HS-heat stress; NP: no priming; P-primed. Representative plants in primed 2 (P2) generation are shown here.



Identification and cloning of genes for salinity tolerance in wheat

Kanika, Priyanka and Tamilarasi

Functional validation of salinity tolerance gene *TaNHX1* (vacuolar Na^+/H^+ antiporter) from Kharchia Local (a well-known salt tolerant Indian land race) was carried out using transgenic approach. The wheat transgenics harboring *TaNHX1* were evaluated for agronomic parameters under salt

stress treatment and exhibited improvement in agronomic traits like biomass and spike length. (Fig. 2.13). Additionally, Kharchia 65 and HD 2851 were phenotyped under salt stress conditions in

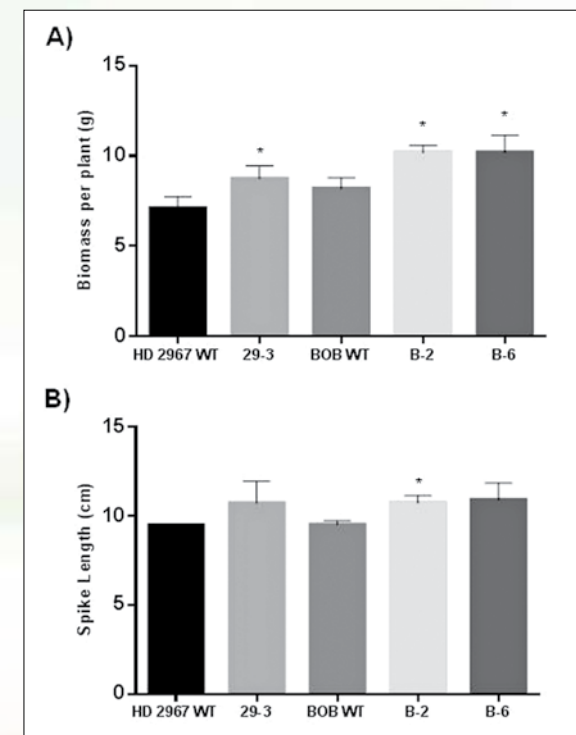


Fig. 2.13: Analysis of various agronomic parameters following salt stress in wild type and *NHX1* transgenics. Analysis of (A) Biomass per plant, (B) Spike length in WT and transgenics of HD 2967 (29-3) and Bobwhite (B-2 and B-6) following salt stress. Student's T-Test was conducted to ascertain the significant difference between means at a significant level of $P \leq 0.05$ and presented as mean $\pm$ standard error (S.E.)

hydroponics and various physio-biochemical, growth analysis parameters of root traits were studied. It was found that Kharchia 65 showed better performance as compared to HD 2851 under salt stress in terms of better maintenance of RWC, membrane stability, growth analysis parameters and root system architecture (Fig. 2.14A). The expression of salinity stress responsive gene (*vacuolar pyrophosphatase similar to AVP1*) was also higher in kharchia 65 as compared to HD2851 (Fig 2.14B) To identify the genomic regions (QTLs) and genes responsible for salt tolerance a mapping population of HD2851 x Kharchia 65 will be utilized.

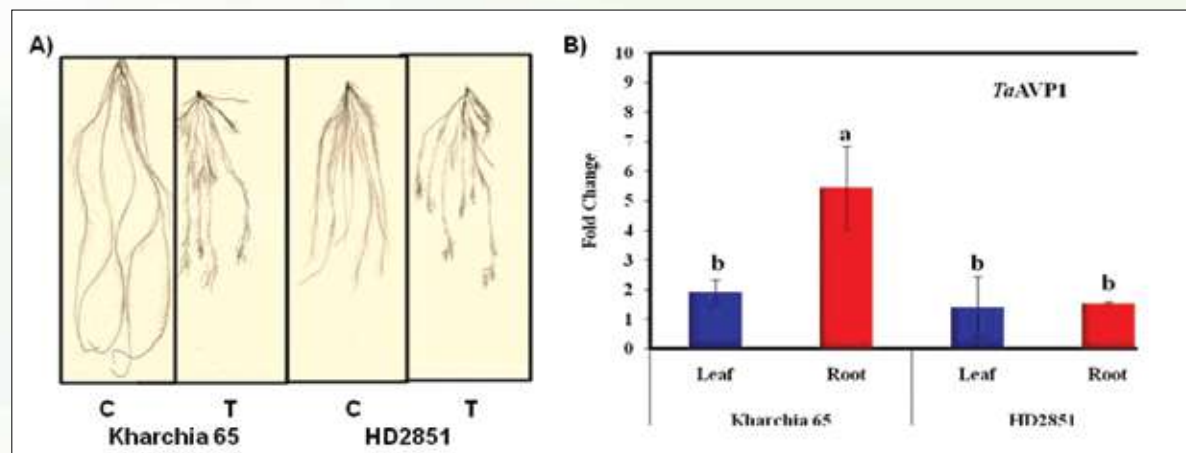


Fig. 2.14: (A) Effect of salt stress on root architecture of wheat seedlings under control and salt stress (10 ECwi) condition after 14 days of treatment. (B) Expression analysis of salinity related gene *vacuolar pyrophosphatase similar to AVP1* in leaf and root samples of Kharchia-65 and HD 2851 after 14 days of treatment. Error bars represent Standard Error (S.E).

3. IMPROVEMENT OF STRESS TOLERANCE AND QUALITY TRAITS IN CHICKPEA



Isolation and characterization of insecticidal genes from native *Bacillus thuringiensis* isolates

Sarvjeet Kaur, Anupma Singh, Mamta Gupta, Deepak

Kumar Rathore and Vinay Kalia

Anupma *et al.* 2019, Indian Journal of Entomology

Bacillus thuringiensis (Bt), an aerobic, spore-forming Gram-positive bacterium that produces insecticidal crystal proteins (Cry) and vegetative insecticidal proteins (Vip), which have specific toxicity towards insect pests of different insect orders while being safe for non-target species. Tobacco (*N. tabacum* cv. Petite Havana) was transformed with a codon-optimized synthetic version (GenScript, USA) of *vip3Aa44* gene (NCBI Accession No.HQ650163), from *B. thuringiensis* subsp. *thuringiensis* strain. Putative transgenics were developed and analyzed for expression, gene integration and efficacy towards *H. armigera* and *Spodoptera litura* (cotton leafworm). Mortality percentage after 72 h ranged from

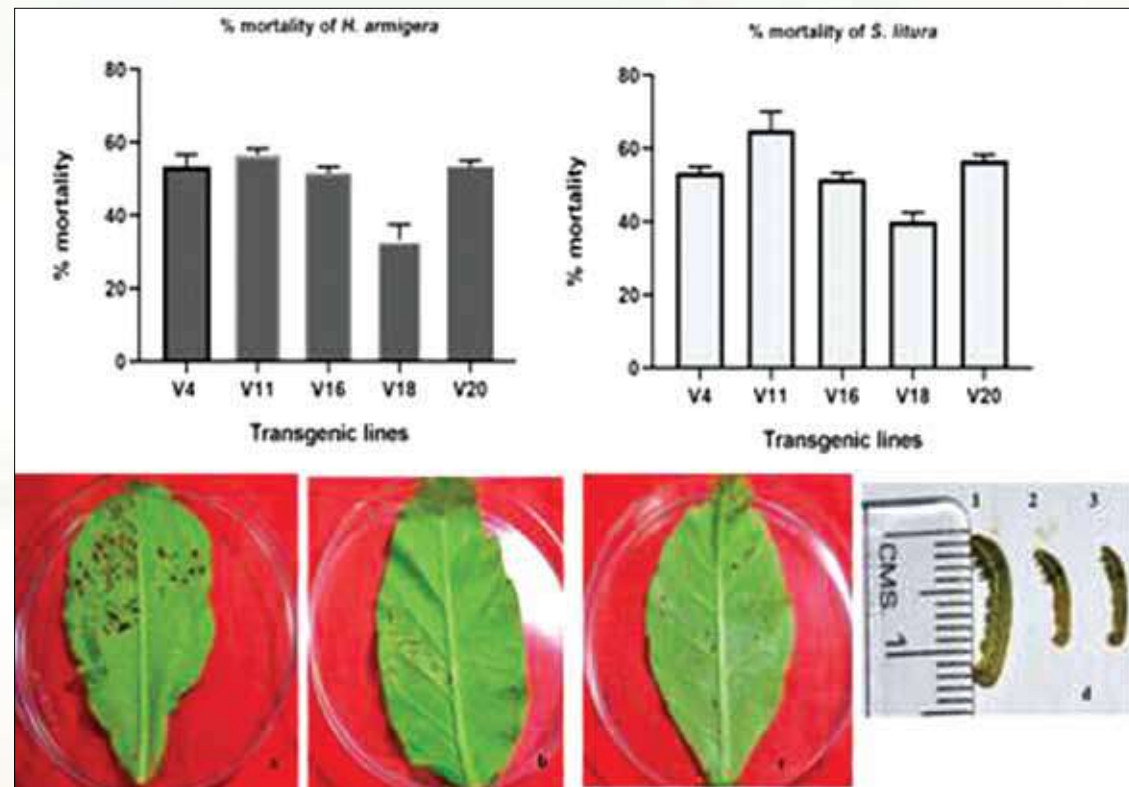


Fig. 3.1: Insect bioassay of *vip3a* transgenics A) Bar graph representing per cent mortality of *Helicoverpa armigera* on *vip3a* tobacco transgenic lines. B) Bar graph representing per cent mortality of *Spodoptera litura* on tobacco transgenic lines. C) Detached leaf bioassay-a-control leaf b, c-transgenic leaves D) Growth inhibition of test insect: *S. litura* larvae fed on transgenic leaves, control plant-fed larvae have normal morphology while transgenic-fed larvae have stunted growth and weight loss.

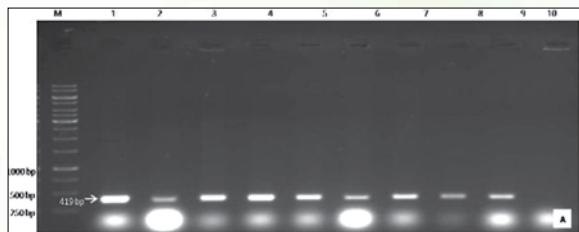


Fig. 3.2: Screening of native Bt isolates recovered from diverse soil habitats from different regions for the presence of *vip3*-type genes by PCR. Lane M: 1kb DNA Marker. Lane 1: Bt reference strain HD1 used as positive control; Lanes 2 -4: Chickpea soil isolates; 5-7: Cotton soil isolate; 8-10: Orchard soil isolates

between 30-56% for *H. armigera* and 40-60% for *S. litura*, indicating potential of *vip3Aa44* gene against these lepidopteran pests in transgenic development for crop protection (Fig 3.1).

Frequency of occurrence of *vip3A*-type genes from native Bt isolates recovered from soil habitats of different regions in India

Sarvjeet Kaur, Mamta Gupta and Deepak Kumar Rathore

Gupta et. al. 2020, Journal of Entomology and Zoology

The diverse repertoire of Bt isolates in India is a promising source for new kinds of toxic genes. Presence of *vip3*-type genes on 115 Bt isolates recovered from various soil habitats in India has been examined by PCR screening

with specially designed primers (Fig.3.2). The frequency of occurrence of *vip3*-type genes was found to vary in Bt isolates recovered from soil samples taken from different habitats of cropped, non-cropped and forest areas. Identification through PCR revealed 73.61%, 78.79% and 20% *vip3* gene frequency in Bt isolates of cropped area, non-cropped area and forest area, respectively (Figure 3.3).

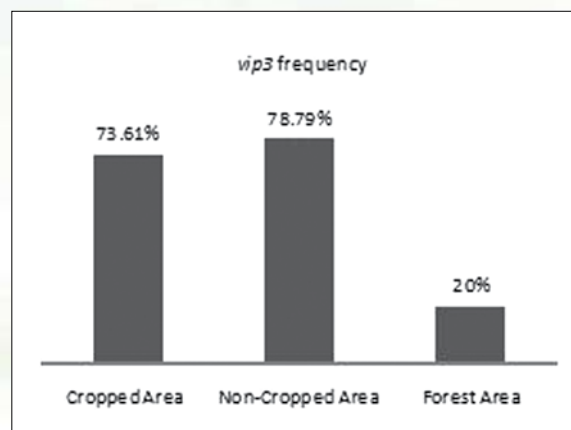


Fig. 3.3: Distribution of *vip3* genes in isolates recovered soils of diverse areas in India



Release of wilt resistant chickpea variety (in collaboration with IARI)

PK Jain and C Bharadwaj

In 2020, a chickpea variety Pusa Chickpea 20211 (Pusa Chickpea Manav)

was released by IARI and NIPB was involved as a Collaborator. Pusa Chickpea 20211 (Pusa Chickpea Manav) is a wilt resistant introgression line (WRIL-3) of Pusa 391, possessing the Wilt QTLs 1, 3, 4 and 5 on LG 2 locus introgressed from WR 315 through MABC. Pusa Chickpea 20211 (Pusa Chickpea Manav) gave an overall weighted mean yield of 2392 kg/ha and has an yield potential of 3915 kg/ha under wilt stress conditions over the recurrent parent Pusa 391 which yielded 1877 kg/ha. The VIC (Variety Identification Committee) in its meeting, held on 1st October 2020, identified BGM 20211 for central zone comprising Gujarat, Maharashtra and Madhya Pradesh. (CVRC notification no. SO No. 500 (E) dated 29/01/2021).



Fig. 3.4: A single plant view of Pusa Chickpea 20211, wilt resistant variety

Identification and validation of differentially expressed genes associated with seed protein content in chickpea

Deshika, Yashwant, Ila, MS Nimmy and PK Jain

Two chickpea genotypes contrasting in seed protein content viz., ICC8397 (30%) and FG212 (20%) were selected for transcriptome and miRNA analysis. The genes encoding seed storage

proteins (SSPs) namely, albumins, globulins (vicilins and legumins) and glutelins were found to be differentially expressed. The differential expression of eight SSPs was validated using qRT-PCR assay in the two chickpea genotypes which revealed that seven were upregulated in mature seeds of ICC8397 of which two were 7S globulins (vicilin 3.3 and 6), three belonged to

11S class of globulins (legumin 3.1, legumin 3.3 and legumin 4) while others were a glutamine and an albumin. Interestingly a single SSP encoding another member of albumin (albumin 7) was downregulated in ICC8397 genotype. Further, expression pattern of these genes was also evaluated at different seed development stages at 10 days after anthesis (DAA), 20 DAA, 30 DAA and 40 DAA besides mature seeds (Fig. 3.5). Amongst these, five genes that belongs to globulin class of SSPs invariably showed highest expression at initial stage of seed development i.e. 10 DAA.

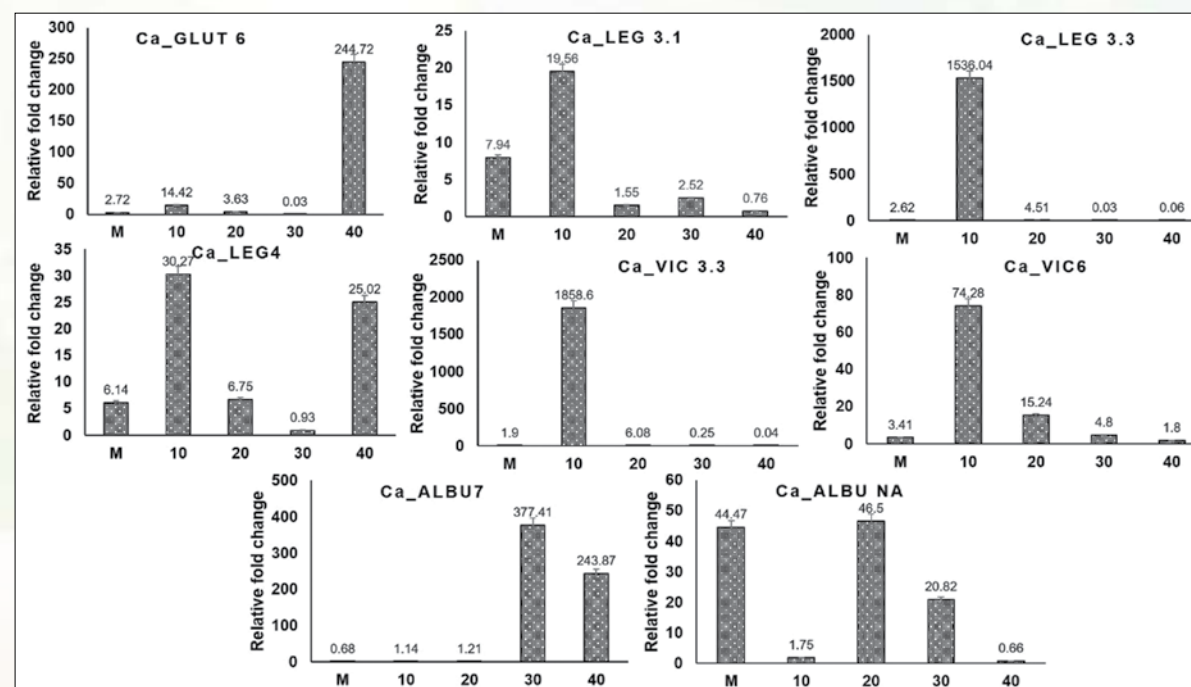


Fig. 3.5: Real-time qPCR based differential expression of eight seed storage protein genes in two chickpea cultivars with contrasting seed protein content viz., ICC8397 (High Protein Content) and FG212 (Low Protein Content). The gene expression in all the five stages of seed development of FG212, a low protein content genotype was considered as reference calibrator and hence assigned the value of 1 (and not depicted). The numbers above the bars indicates the respective fold change value obtained in ICC8397 genotype. M, 10, 20, 30 and 40 represent mature seed, 10-, 20-, 30- and 40-DAA (days after anthesis).

Identification of significant genes associated with seed protein content from both the transcriptome and miRNA data

Deshika and PK Jain

miRNAs validated in chickpea genotypes, contrasting in seed protein content, were found to target genes encoding SSPs and transcription factors that regulate promoter region(s) of SSP genes (Fig. 3.6). Among these, the highly conserved miRNA156 was found to target gene encoding globulins similar to that of miRNAs like miR159, miR319, miR393 and miR399 all of which target globulins. MicroRNAs miR166, miR1514 and miR398 targeted ABI-5 (ABSCISIC ACID-INSENSITIVE 5) gene which belongs to bZIP transcription factor family and is also reported to be significantly expressed in chickpea

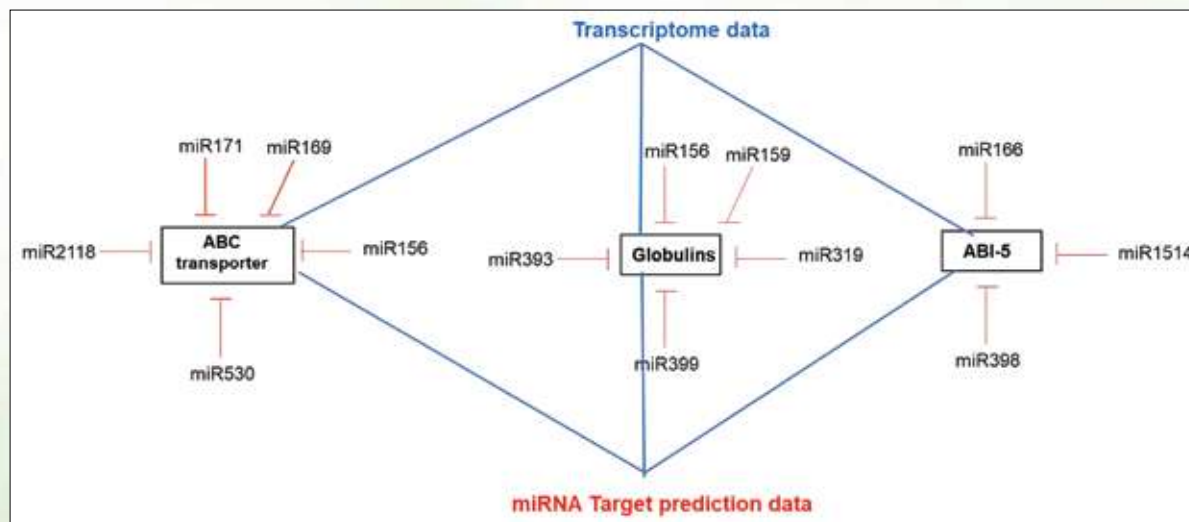


Fig. 3.6: Interaction of common DEGs found in transcriptome data and also predicted as one of the target genes of chickpea miRNAs. All the genes and miRNAs have been validated using qPCR. The genes have been marked in bold, miRNAs mediated suppression of target gene is indicated by bar.

genotype with low seed protein content. We found downregulation of miR166 and miR398 in FG212, a low protein content genotype in our study. Likewise, novel miRNAs also target SSP genes like vicilin and glutelin which are targets of miR263 and miR404, respectively. Among the transcription factors, that regulate SSP genes, different members of ABC transporter protein family were found and these are targeted by miR171, miR169, miR156, miR530 and miR2118.



Assembly and annotation of the mitochondrial genome of the *Fusarium udum*

R. Nagar, J. Vijayan, MS Nimmy, D. Pattanayak and PK Jain

The *Fusarium udum* is a causal agent of vascular wilt disease in *Crotalaria* spp. and pigeonpea. It is responsible for high economic losses of more than 70 million US dollars

annually in India alone. The fungal pathogen can attack any stage of the plant and can cause up to 100% yield loss if the infection happens at the pre-pod stage. Recently, with the advent of high-throughput sequencing technologies and bioinformatics tools, it has become relatively easy to assemble full-length mitochondrial genomes for whole-genome sequencing data. Since the *Fusarium udum* mitochondrial genome is not reported, hence an effort was made to assemble the mitochondrial genome from publicly available whole-genome short reads sequencing data. The full-length mitochondrial genome of *Fusarium udum* would likely facilitate a better understanding of its evolutionary relationship with other plant pathogenic *Fusarium* species in the *Fusarium* species complex.

The full-length mitochondrial genome of wilt pathogen, *Fusarium udum* was assembled from whole-genome Illumina sequencing data of the strain F-02845. The whole-genome sequencing data of the strain is available publicly on the NCBI's SRA database with accession number SRR7701182 which was submitted by the National Bureau of Agriculturally Important Microorganisms (NBAIM), Mau, Uttar Pradesh. To assemble the mitochondrial genome from whole-genome sequencing data, the short reads

data of the strain F-02845 was downloaded and converted into paired-end fastq files. The raw paired-end reads were quality processed and subsampled randomly to 20 % of the whole genome reads. These 20% reads were used for a reference assisted mitochondrial genome assembly using the MITObim tool. The reference mitochondrial genome used was of the *Fusarium oxysporum* f. sp. *lycopersici*. The MITObim produced a 38.1 Kb contig which was annotated using two independent mitochondrial genome annotation tools, GeSeq web browser, <https://chlorobox.mpimp-golm.mpg.de/geseq.html> and MITO web server using genetic code 4, <http://mitos.bioinf.uni-leipzig.de/index.py>.

Features of *Fusarium udum* mitochondrial genome

The annotation of the *Fusarium udum* mitochondrial genome obtained from the whole genome sequencing data of the *Fusarium udum* strain F-02845 revealed presence of all the essential protein-coding and non-coding genes typical for a fungal mitochondrial genome. Thereby indicating a full-length mitochondrial genome assembly. The mt-genome codes for fourteen protein-coding genes including three copies of the ATP synthase complex (*atp6*, *atp8* and *atp9*), seven subunits of the nicotinamide adenine dinucleotide

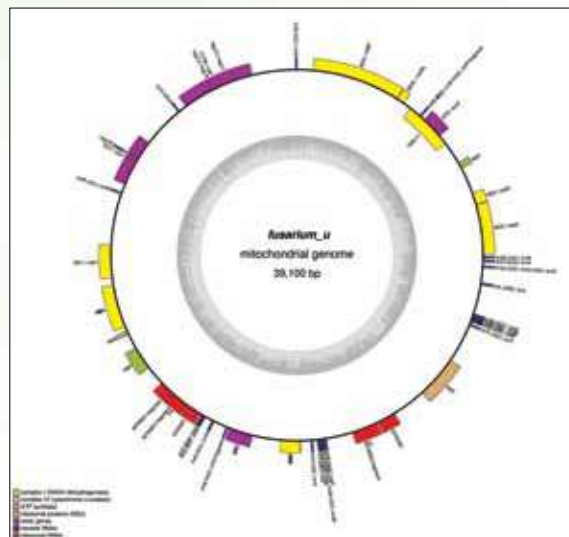


Fig. 3.7: Gene map of the *Fusarium udum* mitochondrial genome

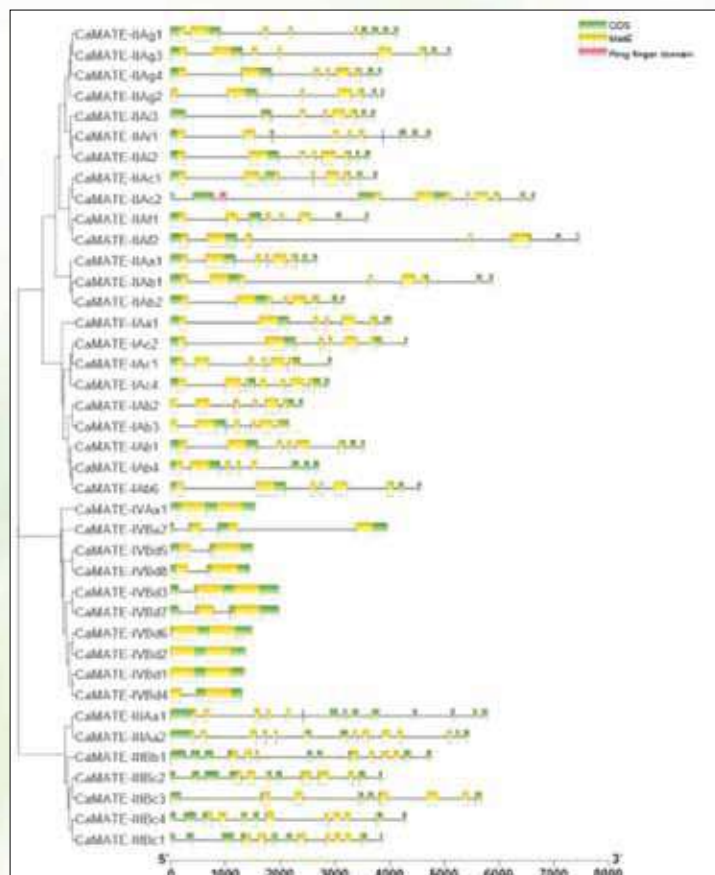
ubiquinone oxidoreductase complex (*nad1*, *nad2*, *nad3*, *nad4*, *nad4L*, *nad5* and *nad6*), one cytochrome b (*cob*), and three subunits of cytochrome c oxidase (*cox1*, *cox2* and *cox3*). Introns were also identified in two protein-coding genes *ned5* and *cob*. Also, the *Fusarium udum* mitochondrial genome encodes a set of 24 tRNA and one large and small subunit of rRNA gene (Fig.3.7). The mitochondrial genome would be manually curated for the gene order and submitted to the NCBI.



Characterization of the MATE (Multidrug and toxic compound extrusion) gene family in chickpea (*Cicer arietinum* L.)

MS Nimmy, V Kumar, P Krishnasamy, R Nagar and PK Jain

We have done a detailed analysis of MATE genes in chickpea with respect to their distribution, phylogeny and structure. A total of 48 MATE-domain containing proteins were identified from the CDC Frontier genome using a keyword search option (PF01554) in the Phytozome13 database. MATes were unevenly distributed in all of eight chromosomes of chickpea. Chromosomes 2 and 4 contained a maximum number of MATes ($n = 9-8$), chromosomes 6 and 7 contained a least number of MATes (2-3) while other chromosomes contained 4-6 MATes. The chromosomal position of six of 48 MATes was not known to-date and were mapped on scaffolds (Chr_Un). No correlation was observed between the chromosome length and the number of MATes. Syntelogs were identified using MCScanX (Fig. 3.8). In this study, seven CaMATE pairs were identified as syntelogs (i.e. the homologs retained in a similar genomic context) – among



which the full-length amino acid identity was ranged between 54–78%. To propose a systematic classification for plant MATEs, we have carried out unrooted NJ-derived phylogenetic tree reconstruction and four major phylogenetic groups were identified (Fig.3.8).

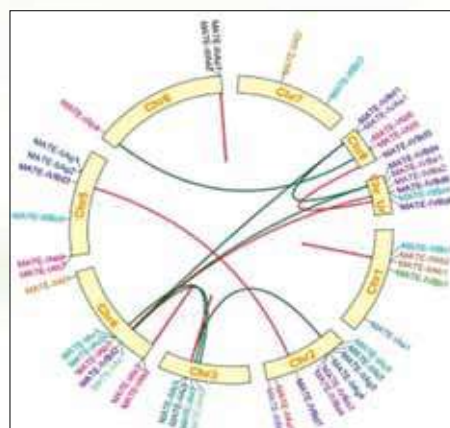


Fig. 3.8: Phylogeny and exon-intron organization of the MATE gene family in chickpea (L); Genomic synteny analyses of MATE genes. Each connecting line located inside the inverted circular gene tree indicates a synteny relationship between two MATE genes (syntelogs) (R)

4. STRESS TOLERANCE AND QUALITY IMPROVEMENT IN PIGEONPEA



Non-coding RNAs having strong positive interaction with mRNAs reveal their regulatory nature during flowering in a wild relative of pigeonpea (*Cajanus scarabaeoides*)

Antara Das, Swati Saxena, Kuldeep Kumar, Kishor U Tribhuvan, N K Singh and Kishor Gaikwad

Das *et al.* 2020 Molecular Biology Reports

In the present study, we have investigated the involvement of long non-coding RNAs (lncRNAs) and miRNAs during the process of flower development in *Cajanus scarabaeoides*, an important wild relative of pigeonpea. The transcriptome of floral and leaf tissues revealed a total of 1672 lncRNAs and 57 miRNAs being expressed during flower development (Fig.4.1). All the identified 57 miRNAs were novel, suggesting their genera specificity. Prediction of the secondary structure of lncRNAs and miRNAs followed by interaction analysis revealed that 199 lncRNAs could interact with 47

miRNAs where miRNAs were acting in the root of interaction. Expression analysis of identified transcripts revealed that higher expression of Csa-lncRNA_1231 in the bud sequesters Csa-miRNA-156b by indirectly mimicking the miRNA and leading to increased expression of flower-

specific SQUAMOSA promoter-binding protein-like (SPL-12) TF indicating their potential role in flower development. The present study will help in understanding the molecular regulatory mechanism governing the induction of flowering in *C. scarabaeoides*.

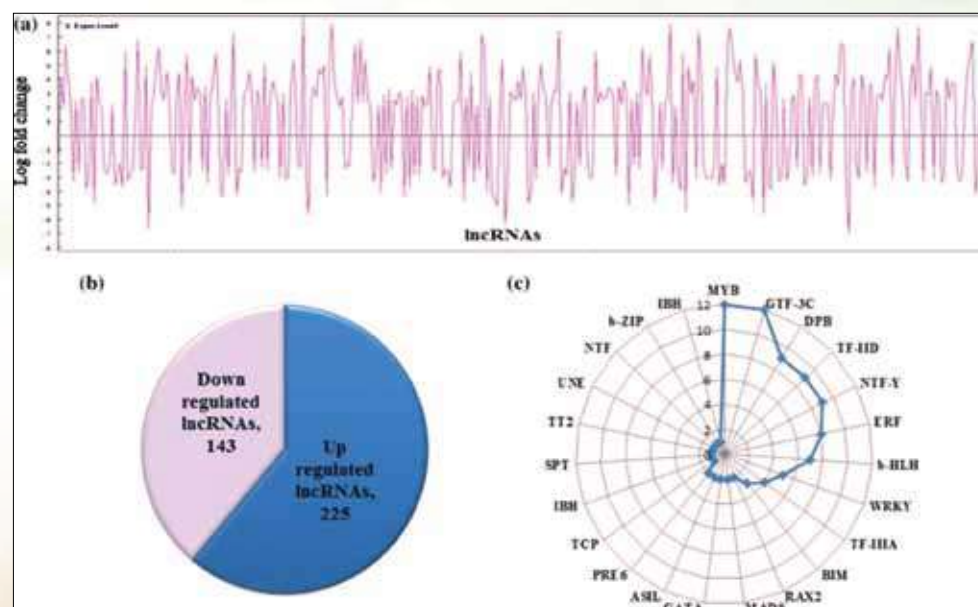


Fig 4.1: (a) Differentially expressed lncRNAs in *Cajanus scarabaeoides*, X-axis represent lncRNAs and Y-axis represent log2 fold change expression in the bud in comparison to leaf, (b) number of up and down-regulated lncRNAs in the bud compared to leaf, (c) number of transcription factors (TFs) targeted by lncRNAs

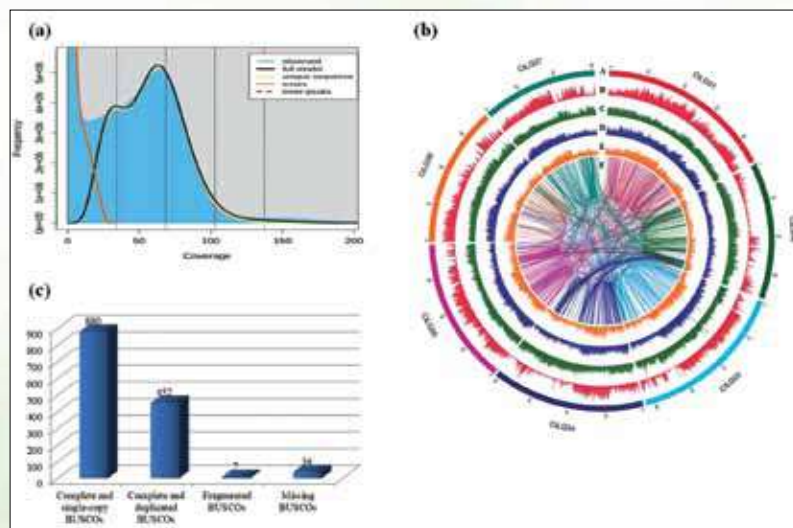


Fig. 4.2: (a) The 31-mer frequency distributions of the sequencing reads, (b) The cluster bean genome features. Track A represents 7 pseudochromosomes. Track B to E represent the distribution of protein-coding genes, Retrotransposons, DNA elements and Simple sequence repeats, respectively. Track F represents gene duplications in the cluster bean genome. (c) Benchmarking Universal Single-Copy Orthologs (BUSCO) analysis of cluster bean annotated genes

Chromosome scale reference genome of Cluster bean (*Cyamopsis tetragonoloba* (L.) Taub.)

Kishor Gaikwad, G. Ramakrishna, Harsha Srivastava, Swati Saxena, Tanvi Kaila, Anshika Tyagi, Priya Sharma, Sandhya Sharma, R Sharma, HR Mahla, SV Amitha Mithra, Amol Solanke, Pritam Kalia, AR Rao, Anil Rai, TR Sharma and NK Singh

We have developed a chromosome-scale reference genome assembly of cluster bean, from a high galactomannan containing popular guar

cultivar, RGC-936, by combining sequenced reads from Illumina, 10x Chromium and Oxford Nanopore technologies. The initial assembly of 1580 scaffolds with an N50 value of 7.12 Mbp was generated. Then, the final genome assembly was obtained by anchoring these scaffolds to a high-density SNP map. Finally, a genome assembly of 550.31 Mbp was obtained in 7 pseudomolecules corresponding to 7 chromosomes with a very high N50 of 78.27 Mbp (Fig. 4.2). We finally predicted 34,680 protein-coding genes in the guar genome. The high-quality chromosome-scale cluster bean genome assembly will facilitate understanding of the molecular basis of galactomannan biosynthesis and aid in genomics-assisted breeding of superior cultivars.



Screening of various pigeon pea genotype for quality traits (Anti-nutritional Factors TIA)

Sandhya Sharma, Kumar Durgesh, Kishor Gaikwad

A total of 75 pigeon pea

genotypes were screened for ANFs like Trypsin inhibitors, Tannin and Phenolic Content. Methods were standardised and standard curves were prepared for each of the components mentioned above. Trypsin inhibitor activity was measured indirectly by inhibiting the activity of trypsin and expressed as trypsin inhibitor units (TIU) per gram sample or per milligram protein. The maximum trypsin inhibitor activity was found in MC 56 i.e., 6.71mgTIU/g and minimum trypsin inhibitor activity was found in MC 32 i.e., 0.109 mgTIU/g.

Transcriptomic analysis in cluster bean at different pod development stages for identification of galactomannan pathway genes

Sandhya Sharma, Anshika Tyagi, Kishor Gaikwad

To gain insight into the mechanisms governing pod development, transcriptome profile of pod at three different stages (25DAF, 39DAF and 50DAF) were compared between two contrasting cluster bean genotypes RGC-936 (High gum) and M-83 (Low gum) using Illumina NextSeq500 platform.

Results showed stage specific expression of genes with 62.2% (4417) genes highly expressed at 39DAF as compared to 20.9% (1489) genes highly expressed at 25DAF and 16.8% (1198) genes highly expressed at 50DAF (Fig. 4.3). Comparative transcriptome data and qRT-PCR expression analysis revealed that the

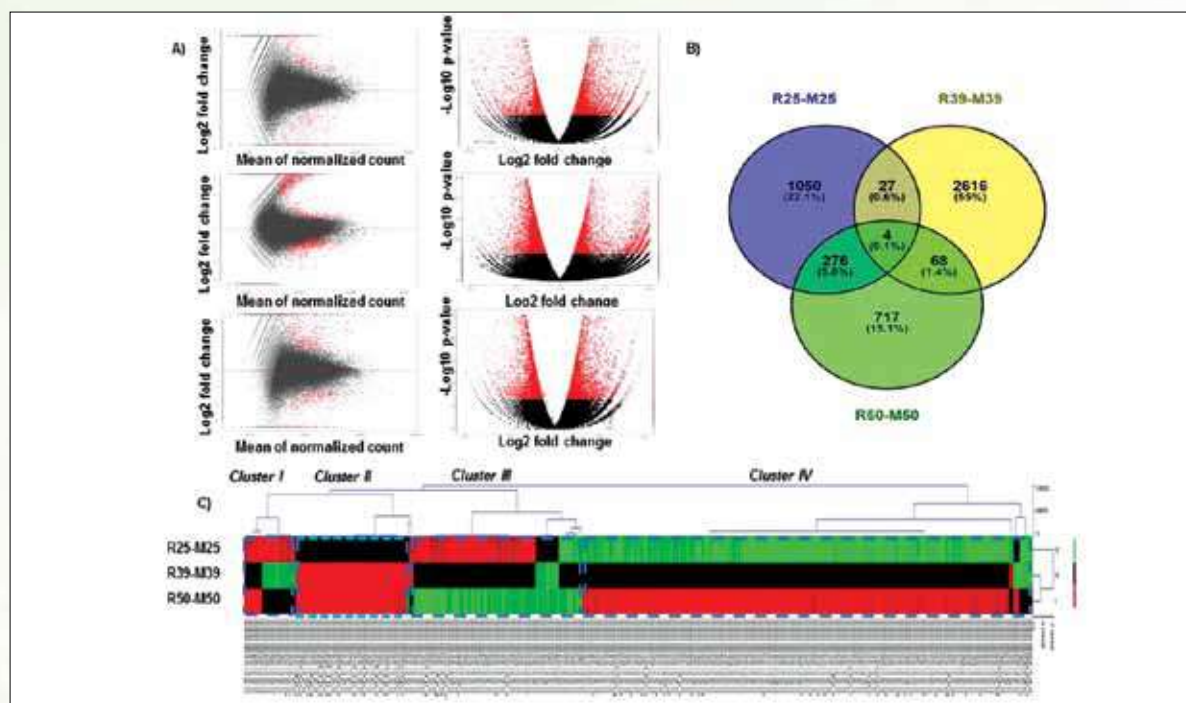


Fig. 4.3: Differentially expressed genes at three pod developmental stages (25DAF, 39DAF and 50DAF) between two contrasting genotypes RGC-936 and M-83 in cluster bean as shown by a) volcano plot, b) venn diagram and c) heat map diagram

transcript levels of galactomannan biosynthesis unigenes encoding mannan synthase, galactosyl transferase, UGE, cellulose synthase, phosphomannomutase were significantly increased in RGC-936, whereas the transcript levels of the unigenes encoding sucrose synthase was significantly increased in M-83. At 39DAF, when galactomannan accumulation is high, the master positive transcription factors (TFs) like *WRKY*, *BHLH*, *MYB*, *LBD*, showed significantly

higher expression in RGC-936 as compared to M-83 indicating an important role of these transcription factors in gum accumulation at 39DAF. On the contrary, expression of *NAC*, *G2-like* and *bZIP* were elevated in M-83 showing their indirect involvement in galactomannan biosynthesis. A systemic outline of gene and transcription factor involved in galactomannan biosynthesis have been shown in Fig. 4.4.

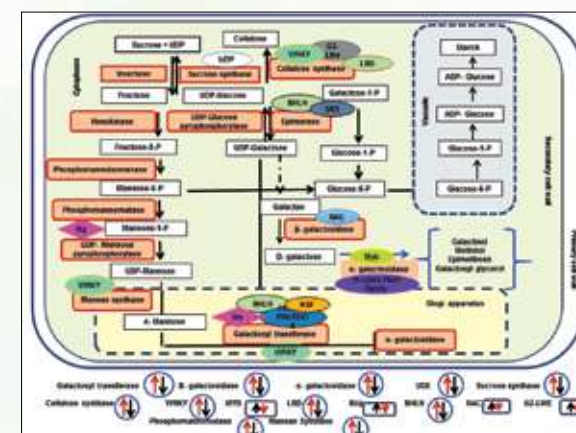


Fig. 4.4: Galactomannan biosynthesis pathway and galactomannan related gene expression and their correlation with transcription factors in cluster bean. Red arrow denotes relative expression in RGC-936 and black arrow denotes relative expression in M-83 genotype at 39DAF pod developmental stage.



Homology based identification and classification of Gibberellic Acid oxidases involved in Gibberellic Acid metabolism in pigeon pea (*Cajanus cajan*)

Joshitha Vijayan, Venkat Raman, Sandeep Jaiswal, Krishnayan Paul, Debasis Pattanayak

Gibberellic acid (GA) is a well-characterized phytohormone having profound impact on a wide range of biological processes. Level of bioactive GA forms (GA1 and GA4) in plants are tightly regulated by the action of anabolic

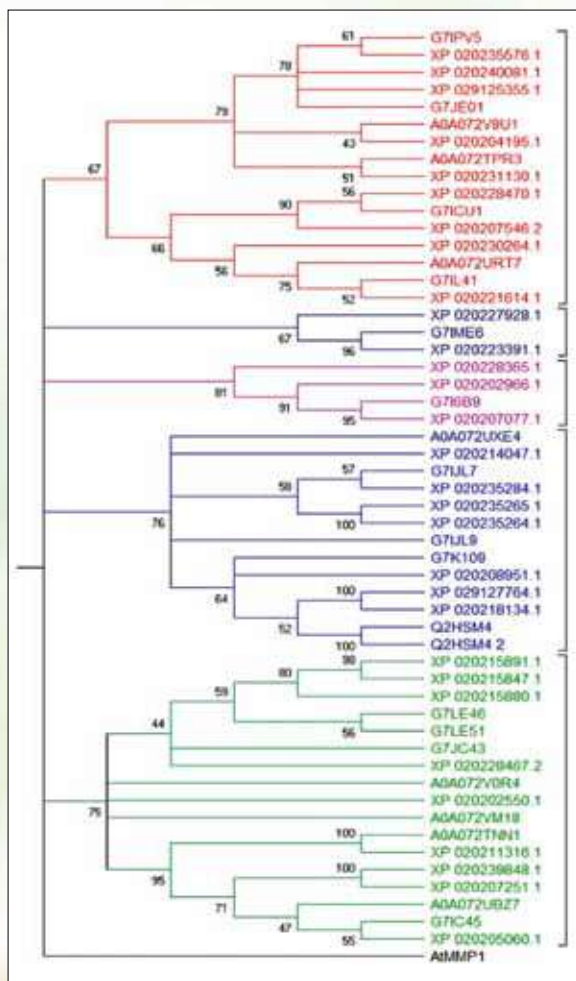


Fig. 4.5: Phylogenetic tree depicting homology relationship between GA2-, GA3- and GA20- oxidases of *Medicago truncatula* and *Cajanus cajan*. Proteins of *C. cajan* are represented by their Refseq identifier available at NCBI database while the *M. truncatula* proteins are represented by their Uniprot identifiers. Matrix metalloproteinase 1 of *Arabidopsis thaliana* (AtMMP1) has been used as outgroup to construct the phylogenetic tree.

enzymes GA3- and GA20- oxidases and catabolic enzyme GA2 oxidase. Hence, knockout of GA3 oxidase and GA20 oxidase leads to dwarfism in plants while knockout of GA2 oxidase leads to parthenocarpic fruits. A positive correlation has been observed between internal GA content, pod development and seed dry matter accumulation in *C. cajan* (*Cajanus cajan*). Besides, in *C. cajan* - a relatively recalcitrant crop to tissue culture - external supply of GA in growth media has been demonstrated to be effective in shoot development. Hence, considering its importance, we have made an effort to identify and classify the homologs of prime GA-metabolic enzymes; GA2 oxidase, GA3 oxidase and GA20 oxidase from the proteome of *C. cajan*. In order to do this, we took the previously studied and characterized proteins of

GA2 oxidase, GA3 oxidase and GA20 oxidase from *Medicago truncatula* and performed a protein blast search against *C. cajan* non redundant protein database. A total of 30 unique hits were obtained having both signature domains 2OG-FeII_Oxy and non-haem dioxygenase (DIOX_N). Among these, 16 hits belonged to GA2 oxidase, five were of GA3 oxidase and rest nine were of GA20 oxidase. Phylogenetic analysis was done to understand evolutionary relationship between GA oxidases of *Medicago* and *C. cajan*, taking AtMMP1 as an outgroup (Fig. 4.5). The obtained phylogenetic tree depicted five clear clusters (designated as A, B, C, D and E. Clusters 'A' and 'D' contain homologs of GA2 oxidases while Clusters 'B' and 'C' contain GA3 oxidases. GA20 oxidases formed a single cluster designated as 'E'.

5. BIOTECHNOLOGICAL APPROACHES FOR BRASSICA IMPROVEMENT



Transcriptome study of host response in *B. juncea* against different aphid species

Lianthanzauva, Murali Krishna Koramutla, Rohit Chamola, R.C. Bhattacharya

Duhlian *et al.* 2020, Scientific Reports

B. juncea leaves were independently infested with *L. erysimi* and *A. craccivora* and the whole transcriptome was analysed to identify the host-genes differentially regulated in case of successful and unsuccessful colonization. The high-quality reads were *de novo* assembled which led to the identification of 48775, 49646 and 42182 transcripts in case of uninfested (control), *L. erysimi*-infested (LE) and *A. craccivora*-infested (AC) samples (PRJNA576081). A common data set was generated for all the samples and comparative gene expression was assessed based on FPKM values. A total of 1307 genes were identified ($\log_2$ ratio ≥ 2 and $P > 0.05$) showing differential expression in the host when treated with two different aphid species (Fig. 5.1).

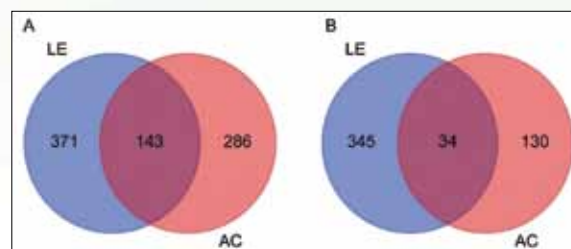


Fig. 5.1: Venn diagram showing the common and unique differentially expressed genes in response to infestation by *L. erysimi* (LE) and *A. craccivora* (AC). Up-regulated (A) and Down-regulated (B) DEGs in response to LE and AC infestations

In biological processes enriched GO terms of down-regulated transcripts in the case of LE-infested sample distinctly represented defence related mechanisms such as secondary metabolic process, response to oxidative stress, phenylpropanoid biosynthetic process, sulfur metabolic process, defence response and glucosinolates biosynthetic process, characteristically present in members of Brassicaceae. Further, to understand the mechanistic differences in AC and LE induced host responses differentially expressed genes were analyzed for biotic stress and secondary metabolism pathways (Fig. 5.2). Under biotic

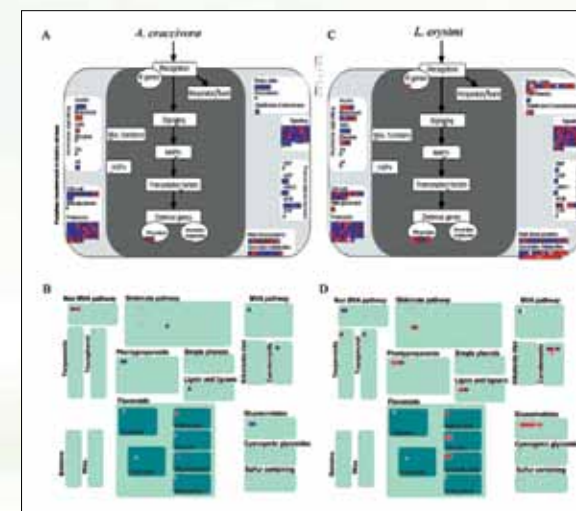


Fig. 5.2: Differentially expressed genes in response to AC and LE in *B. juncea* assigned to biotic stress (A and C) and secondary metabolism (B and D) categories based on MapMan software. The fold change in expression relative to the uninfested control sample is indicated by blue (≥ 2 -fold) and red (≥ 2 -fold) colours in the scale.)

stress, the transcripts belonging to hormone signalling, cell wall modification, proteolysis, redox state including glutathione S-transferase (GST), signalling, secondary metabolites, transcription factors, and heat shock protein categories were found to be up-regulated in

response to AC (Fig. 5.2 A and B). However, most of these pathways were suppressed within 24 h of successful host colonization by LE (Fig. 5.2C). Similarly, the transcripts belonging to biosynthetic pathways of secondary metabolites such as phenylpropanoid, carotenoids, flavonoids, lignin and glucosinolates were induced in response to AC (Fig. 5.2B), whereas attenuated by LE-infestation (Fig. 5.2D). The analysis empirically advocated the suppression of defence-related pathways in *B. juncea* in case of successful colonization by *L. erysimi*. The study generated insight into the basis of host susceptibility to aphids in Indian mustard.

Development of aphid resistance in Indian mustard through RNAi mediated gene silencing

Deepa Dhatwalia, Deepa Bhat, Muthuganeshan Annamalai, Raghavendra Aminedi, R.C. Bhattacharya

The initial studies indicated that the efficacy of the strategy in developing RNAi mediated insect resistance primarily depends on the importance of the target gene in aphid metabolism and survival. Therefore, we screened for the effect of transcript attenuation of several aphid genes by diet supplementation of dsRNAs homologous to the various target genes in insect bioassay experiments.

The experimental dsRNAs were supplemented in the diet for analysing their

effect on fecundity and survival of the feeding aphid - nymphs. Based on the outcome of diet-based insect bioassay, a gene encoding an alpha-glucosidase and primarily involved in osmoregulation in aphid gut was chosen.

A plant transformation vector construct was made for a host-mediated generation of dsRNA homologous to the target aphid gene (LeRIS dsRNA). Earlier we reported the generation of several transgenic lines using this plant transformation vector and their PCR screening for incorporation and expression of the dsRNA encoding domain. The expression of dsRNA encoding cassette and generation of siRNA molecules in transgenic *B. juncea* lines were verified by qRT-PCR, stem-loop RT-PCR and northern hybridization (Fig. 5.3).

Constitutive expression of the dsRNA encoding cassette ensures a continuous supply of dsRNA/siRNA to the aphids feeding on these plants and the persistent RNAi effect is contrary to oral delivery methods. The selected plants were subjected to insect bioassay using 3rd instar larvae of mustard aphid *L. erysimi*. After 9 days of insect release, most of the nymphs matured

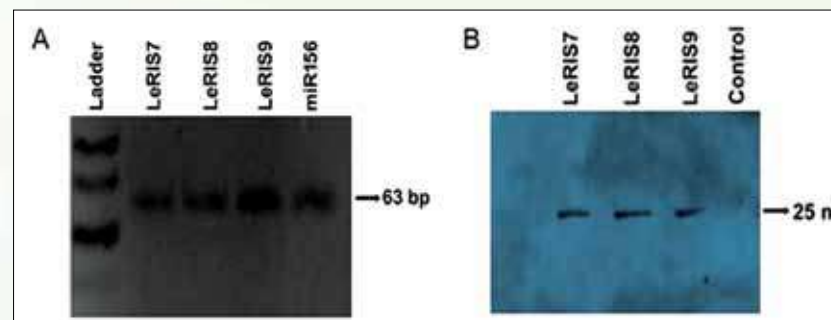


Fig. 5.3: Molecular analysis of LeRIS transgenic mustard **(A)** Detection of *LeRIS*-specific siRNAs by stem-loop RT and end-point PCR. DNA ladder of 50 bp followed by ~63 bp amplicon specific to *LeRIS* was detected in independent transgenic lines (7,8,9) and amplification of miR156 as a positive control. **(B)** Detection of *LeRIS*-specific siRNAs (~25 nucleotides) by northern hybridization. Small RNA (60 µg) isolated from control and independent transgenic lines (LeRIS7, LeRIS8 and LeRIS9) were loaded on a 15% denaturing acrylamide gel and hybridized with *LeRIS*-specific RNA probes. Lane 1-3, ~25-nt siRNAs were detected in transgenic lines (LeRIS7, LeRIS8 and LeRIS9). No signal was detected in vector transformed transgenic lines.

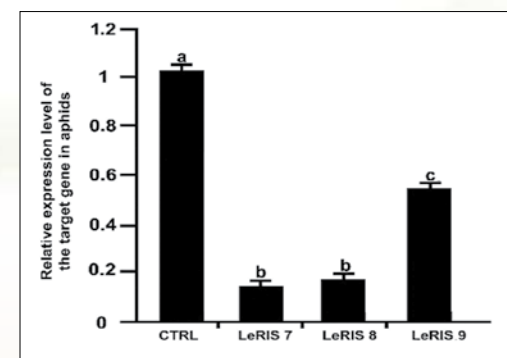


Fig. 5.4: Host delivered dsRNA mediated transcript attenuation in aphids. *LeRIS* transcript levels measured in aphids fed on the independent transgenic *B. juncea* lines (LeRIS7, LeRIS8 and LeRIS9). Data represent mean $\pm$ SD of expression values normalized to the reference gene (18S rRNA of *L. erysimi*) from at least three biological replicates. Significant differences between means were computed using Student's t-test ($P < 0.05$).

into adults and started parthenogenetic reproduction. In terms of induced mortality, no significant advantage of the transgenics was apparent when compared with the vector-transformed control plants. However, the number of newborn nymphs observed on the dsRNA transgenics was significantly lower compared to the control plants. When compared on the 9th-day post-inoculation, the average number of newborn nymphs was 22-40% less on the transgenic plants compared to their average number on the control plants. qRT-PCR based comparison of transcript level revealed a significant reduction in the endogenous transcript level of the target gene in the aphids collected from the transgenic plants (Fig. 5.4). The transcript levels detected in aphids fed on the independent transgenic lines varied from 15-54% of the average transcript level in aphids fed on the control plants.



Use of Insecticidal Genes from Legumes for Developing Aphid Resistant Mustard

Pawan S. Mainkar, Yamini Agarwal, Vinay K. Kalia, Rekha Kansal

The phylogenetic analysis of the 35 protease inhibitor genes, useful for insecticidal activity, identified earlier from the draft pigeonpea genome were divided into two major families;

cysteine protease inhibitor (9) genes and serine protease inhibitor (26) genes. Serine protease inhibitors were further divided into five major types based on the conserved domain, signature motif sequences and their target protease specificity, Kunitz PIs (16), BowmanBirk PIs (4), Potato protease inhibitor I (3), Kazal PIs (2) and Serpin (1). Potato protease inhibitor II and Squash type protease inhibitor genes were not found in the pigeonpea genome. The pigeonpea BowmanBirk protease inhibitors contained the highly conserved BowB (smart00269) domain in CcaBBPI029 and CcaBBPI026 while CcaBBPI027 and CcaBBPI029 contained the conserved BBI (cd00023) domain. The BowB PIs (CcaBBPI026-CcaBBPI029) lacked intron and showed 44-78% identity with the crystal structure of serine protease inhibitor from *Medicago scutellata* (1TAB, 1H34, 1MVZ and 2ILN).

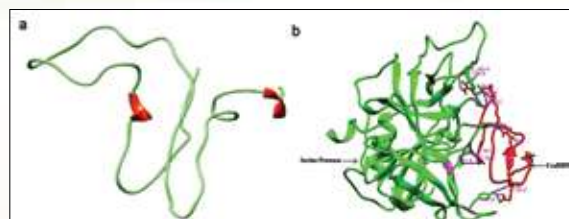


Fig. 5.5: The molecular structure of the protease inhibitor and their interaction with the receptor. (a) The three-dimensional structure of the serine protease inhibitor CcaBBPI0026 containing 114 amino acid residues with a molecular weight of 12.5 kDa, and (b) The molecular docking between Serine protease receptor and Pigeon pea protease inhibitor (CcaBBPI026)

CcaBBPI0026, a serine protease inhibitor has 114 amino acid residues with a molecular weight of 12.5 kDa, seven disulfide bridges and two reactive sites (Fig. 5.5a). The molecular docking between Serine protease receptor and CcaBBPI026 showed that the same amino acid residues present at different location interacted with the residues at a different position in the structures (Fig. 5.5b). The interacted amino acid

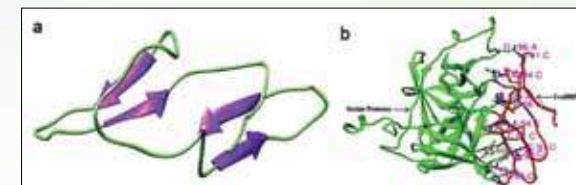


Fig 5.6: The molecular structure of the protease inhibitor and their interaction with the receptor. (a) The three-dimensional structure of the serine protease inhibitor CcaBBPI0027 containing 115 amino acid residues with a molecular weight of 12.6 kDa, and (b) The molecular docking of the amino acid residues of CcaBBPI027 protease inhibitor interacting with the amino acid residues of the serine protease receptor.

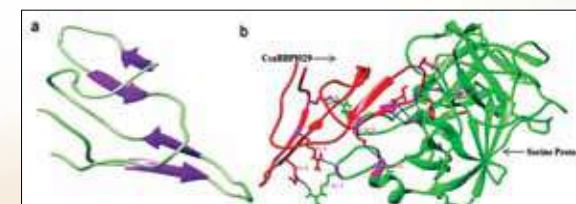


Fig. 5.7: The molecular structure of the protease inhibitor and their interaction with the receptor. (a) The three-dimensional structure of CcaBBPI029 containing 115 amino acid residues, and (b) The molecular docking of the amino acid residues of CcaBBPI029 interacting with the amino acid residues of the serine protease receptor.

residues formed a hydrogen bond length which varied between 1.6-2.4 Å and the minimum interaction energy is about -1262.7 Kcal/mol which makes it a good candidate for insecticidal protein. CcaBBPI027, a type of serine protease inhibitor contained 115 amino acid residues with a molecular weight of 12.6 kDa with seven disulfide bridges and two reactive sites (Fig. 5.6a). This inhibitor is most active against the serine proteases such as trypsin, chymotrypsin. The amino acid residues involved in the interaction of CcaBBPI027 and serine protease of *Helicoverpa armigera* through the Hydrogen bond of length varied from 1.6-2.1 Å. The interacted amino acid residues (Fig. 5.7b) were joined by the 15 hydrogen bonds and the minimum interaction energy of both the protein molecule complex was about -1108.5 Kcal/mol.

CcaBBPI029: The amino acid residues involved in the interaction of CcaBBPI029 with serine protease of *Helicoverpa armigera* formed the hydrogen bond varying from 1.7-2.1 Å. Interacting amino acid residues of CcaBBPI029 and that of serine protease receptor were identified (Fig. 5.7a and b). The interacted amino acid residues were joined by the 12 hydrogen bonds and the minimum interaction energy of both the protein molecule complex was about -835 Kcal/mol.

Cloning and *in vitro* expression of CcaBBPIs

Pawan S. Mainkar, Yamini Agarwal, Vinay K. Kalia, Rekha Kansal

Based on the molecular docking analysis three of the BBIs (CcaBBPI026, CcaBBPI0027 and CcaBBPI029) were found to have strong interchain hydrogen bonding with the midgut proteases of hemipteran and lepidopteran insects. These three BBPIs were cloned into a pET6XHN-C vector and transformed into the *E. coli* BL21 pLyseS cells (Fig. 5.8). The transformed PI genes were overexpressed using the IPTG inducer at 0.3, 0.5, 1 and 1.5mM concentration at different temperature 22°C, 28°C, 30°C and 37°C and different time intervals, 0, 3, 6 and 12h (Fig. 5.9 a and b).

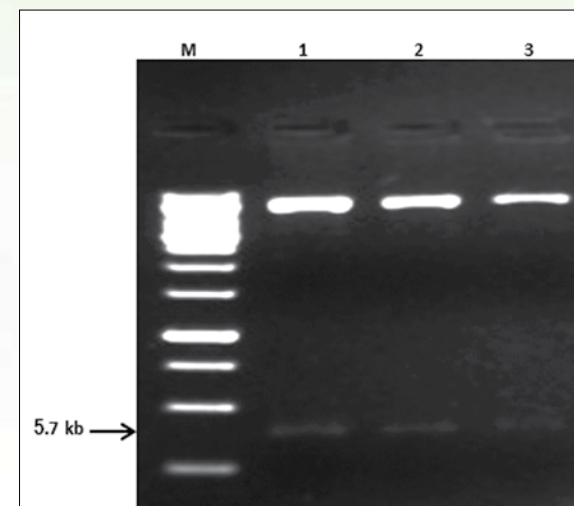


Fig. 5.8: Confirmation of the three BBPIs cloned into an expression vector pET6XHN-C and transformed into the *E. coli* BL21 pLyseS cells

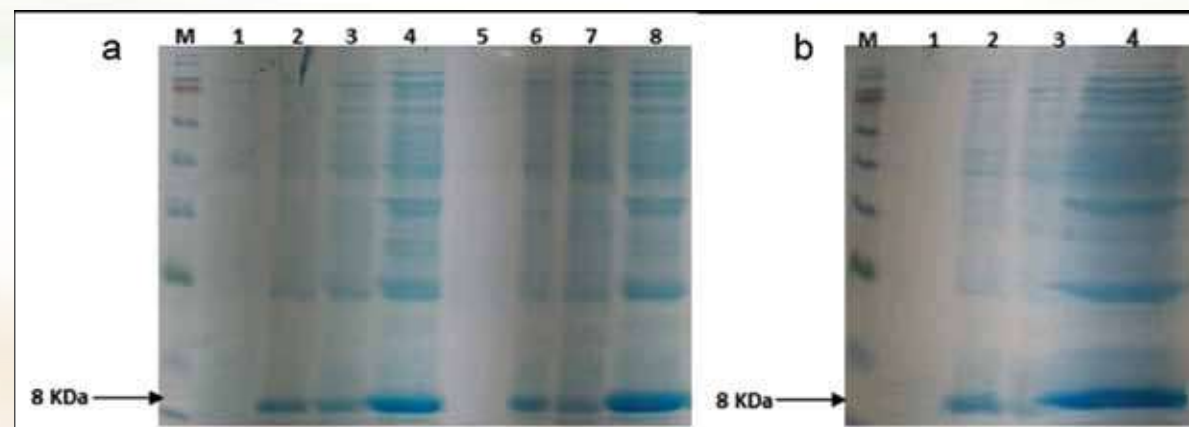


Fig. 5.9: SDS-PAGE of the CcaBBPI0026 and CcaBBPI0027 (a) and CcaBBPI029 (b) PI proteins overexpressed using the IPTG inducer at 0.5, 1 and 1.5mM concentrations at 37°C for 6h.

The overexpressed proteins revealed that 6h incubation time and 1mM IPTG concentration at 37°C produced a significant expression of CcaBBPI026, CcaBBPI0027 and CcaBBPI029 proteins (Fig. 5.10). The purified proteins were evaluated for their insecticidal activity against *Myzus persicae*, *H. armigera* and *Spodoptera litura*.

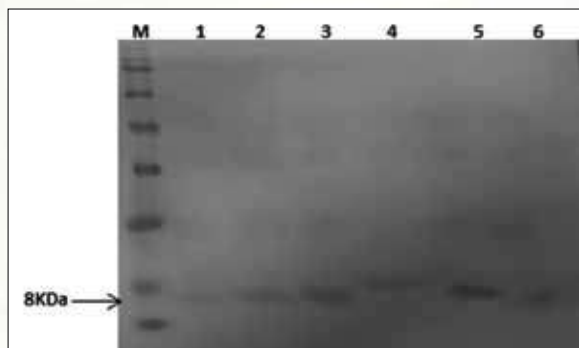


Fig. 5.10: The Immuno-blot analysis confirming the expressed PI protein of 8KDa

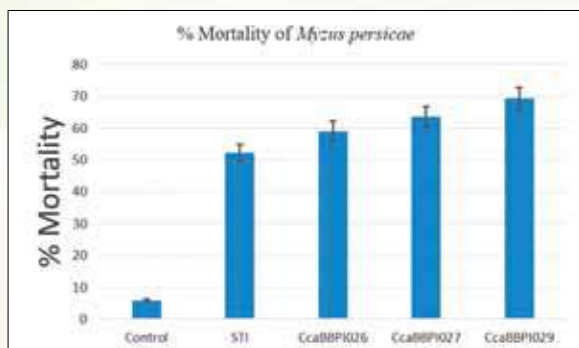


Fig. 5.11: Per cent mortality of *M. persicae* on an artificial diet supplemented with protease inhibitor proteins at 100ng/g concentration after 4th day of bioassay

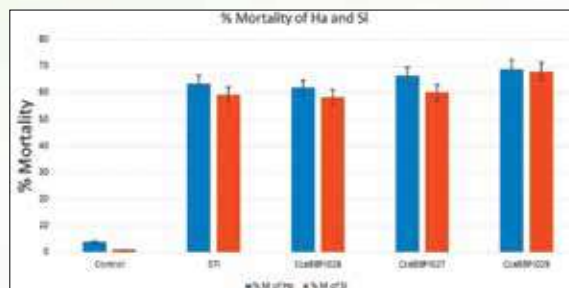


Fig. 5.12: Per cent mortality of *H. armigera* and *S. litura* on an artificial diet supplemented with protease inhibitor proteins at 100ng/g concentration after 8th day of bioassay

The purified and quantified proteins were also used in insect artificial diet bioassay to evaluate the efficacy against Lepidoptera (*Helicoverpa armigera* and *S. litura*) and Hemipteran (*Myzus persicae*).

Aphid bioassay was performed using the pigeonpea PPIs along with the reference protein (soybean trypsin inhibitor) at 100ng/ml of the total protein concentration. On the 4th day, the percent mortality of *M. persicae* was 63.3, 59.1 and 69.3% for CcaBBPI026, CcaBBPI027 and CcaBBPI029 mortality while that of soybean trypsin inhibitor was 52.3% (Fig. 5.11). In addition to higher mortality, CcaBBPI026, CcaBBPI027 and CcaBBPI029 proteins also showed reduced fecundity and reduction in the number of neonates. For the Lepidopteran bioassay, on 8th day of bioassay the larvae of both *H. armigera* and *S. litura* showed mortality of 59.33-68.22% (Fig. 5.12).



Understanding molecular mechanisms of *Alternaria* resistance in *Camelina sativa* and *Sinapis alba* and using them for developing *Alternaria*-resistant transgenic *Brassica juncea* plants

Rekha Kansal, Mahesh Rao, G Prakash, and Anita Grover

WRKY33 was shown to be required for resistance toward the necrotrophic pathogens such as *A. brassicicola* and *B. cinerea*. Cloning of the CsWRKY33 gene was done in the Gateway entry vector (Fig. 5.13) and then in the destination vector (Fig. 5.14). Cloning was confirmed through colony PCR amplification of ~ 3.812 kb size amplicon.

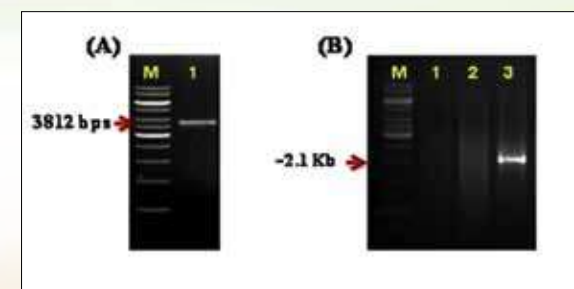


Fig. 5.13: Cloning of genomic fragment of CsWRKY33 gene including native promoter into Gateway entry vector pENTR/D-TOPO. (A) Amplification of full length CsWRKY33 genomic region including upstream promoter region. (B) Colony PCR confirmation of transformed XL1 blue colonies with recombinant CsWRKY33 genomic region with the native promoter in entry vector pENTR/D-TOPO.

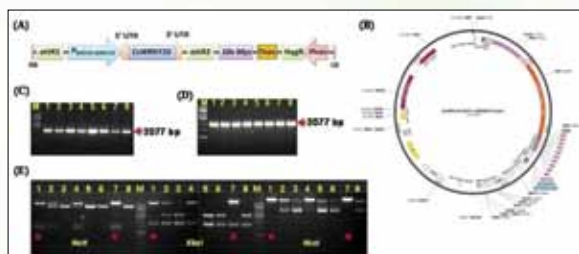


Fig. 5.14: A and B: Representation of gene construct *proCsWRKY33::pGWB519::CsWRKY33* used for plant transformation. Gateway promoterless vector pGWB519 was used as the backbone vector. (C) confirmation of transformed XL1 Blue colonies with recombinant *proCsWRKY33::pGWB519::CsWRKY33* gene construct by amplifying 2077 bp size, (D) Confirmation of transformed *Agrobacterium* GV3101 cells with recombinant *proCsWRKY33::pGWB519::CsWRKY33* gene construct by PCR, (E) Confirmation of recombinants by restriction digestion with *NotI* (size of obtained four bands -1267 bp, 1290 bp, 1533 bp, and 96433 bp), *XbaI* (four bands -190 bp, 1623 bp, 1814 bp, and 10106 bp) and *NcoI* (single band -13733 bp).

Transcriptional analysis of differentially induced proteins in *Alternaria* infected samples from *B. juncea*, *S. alba*, and *C. sativa*

Out of the initial 67 samples, a total of 40 proteins have been identified through MALDI-TOF analysis. Among these 40 differentially expressed proteins, fifteen proteins show marked up or down-regulation. Three of them have already been shown to be involved in disease resistance. Of the 15 differentially regulated proteins in *B. juncea*, *S. alba* and *C. sativa* after *Alternaria* infection, four of

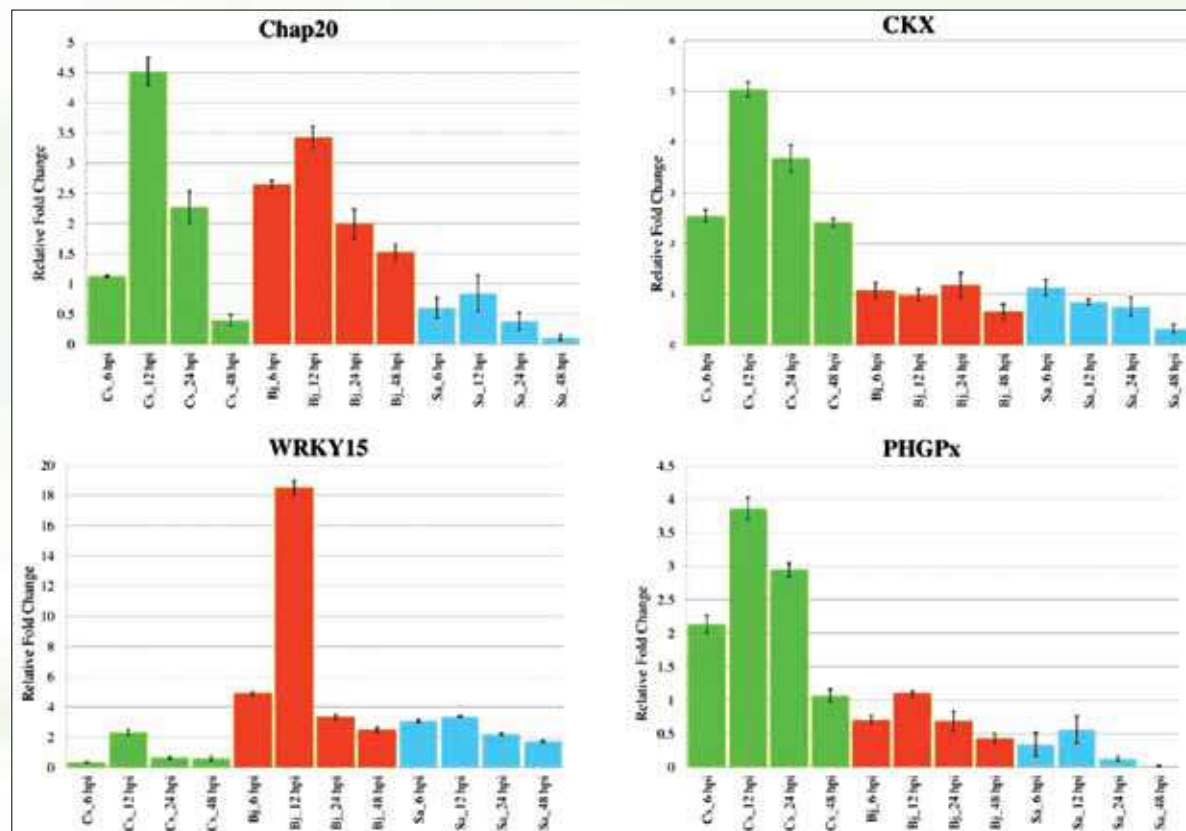


Fig. 5.15: Differential gene expression of *chaperonin*, *CKX*, *WRKY15* and *PHGPx* genes in the leaves of *C. sativa*, *B. juncea* and *S. alba* after *A. brassicae* infection at 6, 12, 24, and 48 hpi. Values are mean of three replicates $\pm$ SD and indicating a significant difference between control and infected leaves at $p < 0.05$.

the proteins (*WRKY15*, cytokinin oxidase, *chaperonin* and *phospholipid hydroperoxide glutathione peroxidase*) were studied at gene expression level (Fig.5.15).



Development of white rust resistance in mustard: Pathotyping of *Albugo candida* isolates and screening and evaluation of *Brassica* germplasm under

artificial inoculated conditions

Ashish Kumar, Jameel Akhtar, Anamika Kashyap, Mahesh Rao, N. C. Gupta and R. C. Bhattacharya

Ten different isolates of *Albugo candida* collected from major mustard growing locations were maintained on seedlings of susceptible cv. Varuna and pathotyped them separately. The incubation period for resulted infection varies with 8-12 days, 9-15 pinheads/100 mm² area, 1-5 mm pustule size and 20-90% disease severity

were observed on the inoculated plants.

Out of the 30 *Brassica* accessions obtained from NBPGR and evaluated against 10 different isolates, none of them were found to be resistant against any isolates of the pathogen. A *B. juncea* accession, IC265495 was found to be resistant against 8 isolates while a *B. carinata* accession, EC206641 (except for Ranchi) and 206642 were identified as immune for all the tested isolates of the pathogen. Of the total 160 advanced ILS evaluated against 10 isolates of *A. candida* under controlled environmental conditions, ERJ 40 was identified as immune (PDI=0) against 8 isolates namely, *Ac-Del*, *Ac-Ldh*, *Ac-Bhrtr*, *Ac-Mrt*, *Ac-Mrn*, *Ac-Rnc*, *Ac-Smstr* and *Ac-Pnt* except Gwlr and Skn isolates at both cotyledonary and true leaf growth stages of the crop (Fig. 5.17). The



Fig. 5.16: Screening of the BWR (Brassica wild relatives) to *Albugo candida* under the net house conditions

The BWR (*Brassica* wild relatives) lines of *B. fruticulosa*, *Camelina sativa*, *Diplotaxis assurgens*, *D. catholica*, *D. cretacia*, *D. erucoides*, *D. muralis*, *D. siettiana*, *D. tenuisiliqua*, *D. viminea*, *Erucastrum lyratus*, *E. abyssinicum*, *E. canariense*, *E. cardaminoides*, *Crambe abyssinica* and *Eruca sativa* accessions were found to be immune (PDI=0) against five isolates of the pathogen (Fig. 5.16).

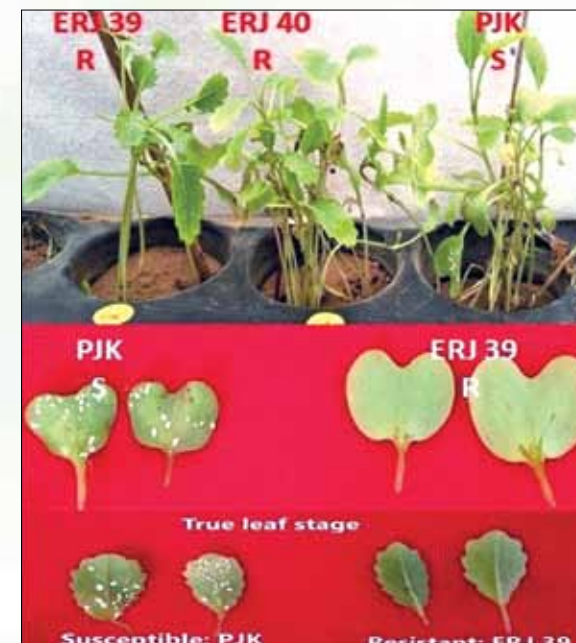


Fig. 5.17: The advanced introgressed lines showing resistance to *Albugo candida*

other introgressed lines such as ERJ-12, ERJ-14, ERJ-15, ERJ-39, ERJ-109, ERJ-110, ERJ-157, ERJ-159 and ERJ-160 were found to be completely resistant for 3-5 isolates which could be further utilized for resistance breeding programs. Among the 92 resynthesized *B. juncea* (RBJ) lines, RBJ 18 showed complete resistance to *Ac-Del*, *Ldh*, *Bhrtr*, *Morna* and *Bih* isolates at both the growth stages of cotyledonary and true leaf stages. The other RBJ lines namely 34, 38, 40, 59, 60, 76, 85 and 87 showed resistance reaction to 1-5 isolates.



Development of mapping population for Stem rot resistance by using the identified donor parent

Navin C Gupta, Pankaj Sharma, Mahesh Rao, Ashish Kumar and Ramcharan Bhattacharya

Gupta NC *et al.*, 2020 Journal of Microbiological Methods

The identified tolerance source for *Sclerotinia* stem rot disease in *Brassica* germplasm namely *B. carinata* (Bcar115; BBCC, 2n = 34), *B. napus* (Bnap 114; AACC, 2n = 38) and *B. juncea* (RH1222-28/RH-6; AABB, 2n = 36) were used as a donor parent to transfer the tolerance trait into the elite susceptible cultivars of *B. juncea* (Table 5.1). The seeds collected from the promising F₂ lines (lesion length <5 cm) during the cropping season of 2019-20, were planted for the rigorous phenotyping in F₃ and for advancement in the

generation of RILs to increase the homozygosity in the subsequent generation.

Genome-wide secretome analysis and effector prediction in *S. sclerotiorum* ESR-01 isolate

Navin C Gupta, Sunita Yadav, Shaweta Arora, Dwijesh C Mishra

Out of 9469 predicted protein-coding genes 529 genes encoding CAZymes (Carbohydrate-Active enzymes) capable of degrading the complex polysaccharides were predicted. The glycosyltransferase (GT) families were most abundant (49.71%) followed by glycoside hydrolase (GH; 23%). Among these GT2 (23%), GT4 (20%), and GH18 (11%) were prominently

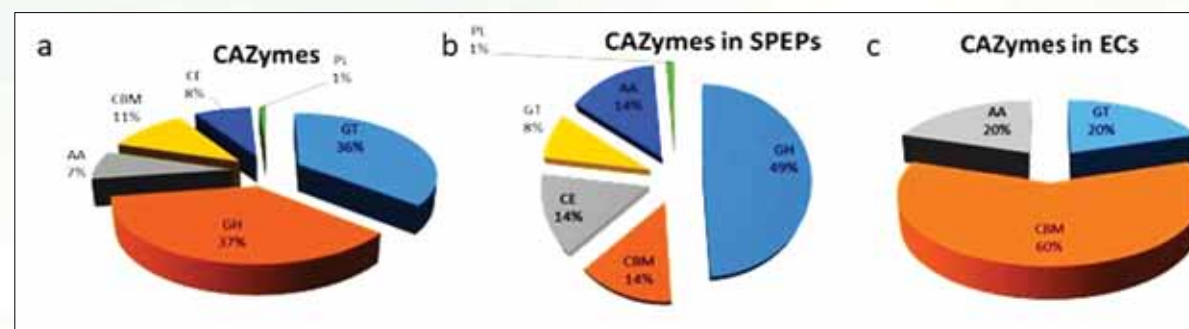


Fig. 5.18: CAZymes predicted in the secretome of *S. sclerotiorum* ESR-01 isolate. (a) Summary of the six CAZymes categories: carbohydrate-binding modules (CBMs), Glycosyltransferases (GTs), glycoside hydrolases (GHs) polysaccharide lyases (PL), carbohydrate esterase (CE), and auxiliary activities (AA) (b) Distinct summaries of the CAZyme present in SPEP (Secreted proteins expressed in planta) (c) Distinct summaries of each of the CAZyme present in (d) Distinct summaries of each of the CAZyme CBMs.

Table 5.1: The list of F₂ derived F₃ population and NILs sown in the current season for phenotyping and generation advancement.

S. No.	F ₂ Population	Parental lines	Number of F ₃ lines	Field
1	A5 (NC9026)	PJK X RH-6	250	IARI
2	A7 (NC9026)	PJK X RH-6	200	DRMR
3	A1 (NC316)	DRMRIJ31 X RH-6	200	DRMR
4	NIL-1 (BC2F1)	Varuna X RH-6	55	IARI
5	NIL-2 (BC2F1)	PJK X RH-6	60	IARI

present in the ESR-01 secretome (Fig. 5.18).

Out of 369 secretory proteins, 153 were categorized into all these above categories using PHI-base whereas 11 out of 57 predicted effectors were aligned to PHI-base (Fig. 5.19). The KEGG analysis results showed only 4 secretory effectors were involved in the distantly related pathway of pathogenesis and those were related to purine metabolism, glyoxylate and dicarboxylate metabolism, drug

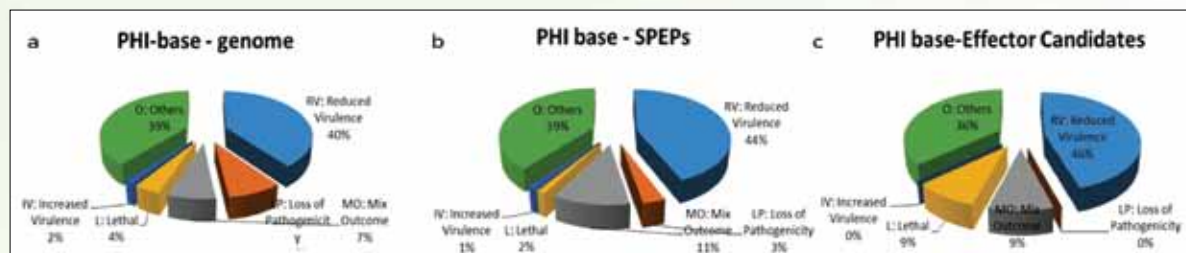


Fig. 5.19: Summary of different phenotypic categories of orthologs genes of *S. sclerotiorum* 'ESR-01' isolate based on the pathogen-host interactions (PHI-base) with the numbers and proportions of genes indicated in a. Genome b. SPEPs c. Effector Candidates

metabolism, T cell receptor signalling pathway along with riboflavin and thiamine metabolism. The effector analysis revealed 27 were having their analogues whereas the remaining 30 were the novel ones. Besides this, 156 genes essential for the pathogen-host interactions were also identified.

Transcriptional profiling in the tolerant and susceptible *B. juncea* lines for *S. sclerotiorum* infection in a time-specific manner

Navin C Gupta, Shaweta A, Pankaj Sharma, Dwijesh Mishra, Sunita, Y., Kishore Gaikwad, Mahesh Rao, Ramcharan Bhattacharya

To understand the differential defence response to *S. sclerotiorum* in a tolerant line (R-line) and a susceptible line (S-line) of *B. juncea* at 24, 48 and 96h post-inoculation transcriptomic analyses was performed. The gene ontology (GO) terms for transcripts were identified to be 4328 in molecular function, 5209

in biological process and 2980 in the cellular component category. Relative differential gene expression (RDEG's) is obtained when resistant is compared to susceptible excluding mock, a total

of 2042 DEG's were up-regulated. Among these up-regulated DEGs, 165, 45, 66 were common between T1-T2, T2-T3 and T3-T1 respectively, while 155 were common in all three time points. 14,609 DEGs (56.5%) were present at two or three time points. The remaining DEGs were time point specific, with 826, 381 and 317 DEGs unique for 24, 48 and 96hpi, respectively (Fig. 5.20a). Subsequently, a comparison of resistant to susceptible excluding mock gave a total of 2295 DEG's were down-regulated. Among these down-regulated DEGs, 339, 106, 52 were common between T1-T2, T2-T3 and T3-T1 respectively, while 337 were common in all three time points.

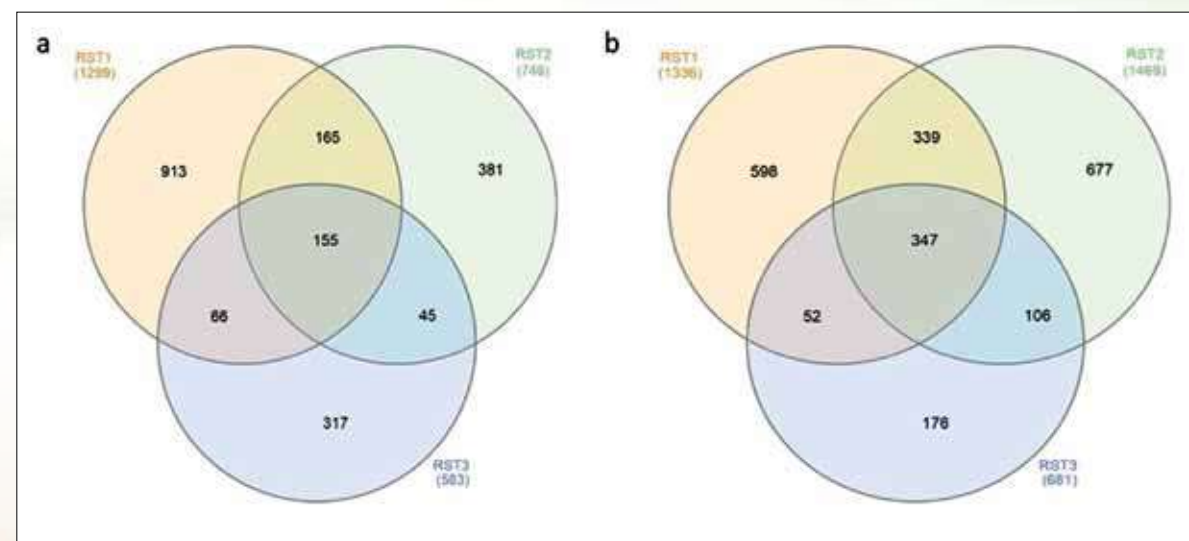


Fig. 5.20: The comparative DEGs of tolerant and susceptible Brassica lines at three time points 24 hr, 48 hr and 96 hr of *S. sclerotiorum* inoculation. (a) Up-regulated RDEG's: RST1 Vs RST2 Vs RST3_Up (b) Down-regulated RDEG's: RST1 Vs RST2 Vs RST3_Dn

The remaining DEGs were time point specific, with 598, 677 and 176 DEGs unique for 24, 48 and 96 hpi, respectively (Fig. 5.20b). Enriched GO terms belonged to diverse functions such as transporters, disease resistance proteins and any function that may play role in resistance.

Development of genome editing constructs for cytokinin modulation and *Agrobacterium*-mediated transformation in *B. juncea*

Navin C Gupta, Ambrish K Gautam, Mahesh Rao, Ramcharan Bhattacharya

Cytokinin oxidase/dehydrogenase regulates cytokinin (CK) level in plants. Evidence showed the knock-out effect of the CKX3 genes in rice and Arabidopsis leads to an increase in yield and abiotic stress tolerance, respectively. The sgRNA constructs were developed for the cloned CKX3 gene from *B. juncea* into a dual guide 'o' acceptor vector. Two sgRNA with minimum off-target effect were chosen from the cloned CKX3 gene sequence and three combinations of the two sgRNAs were made. The recombinant plasmid DNA of three different CRISPR/Cas9 constructs for the CKX3 gene namely A2, B1 and C1 in *E. coli* (Fig. 5.21a and b) were individually transformed into *Agrobacterium* (GV3101) cells (Fig. 5.21) and used to transform *Brassica juncea* L. cv. Varuna. With this procedure, the transformed *B. juncea* regenerants with A2, B1,

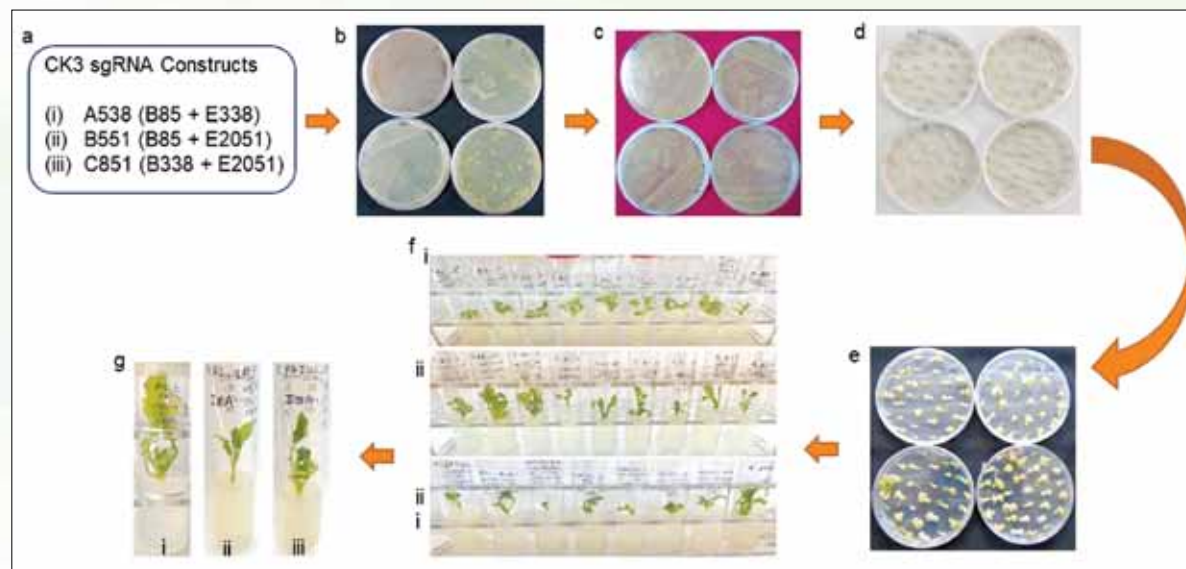


Fig. 5.21: CRISPR/Cas9 constructs for the CKX3 gene from *B. juncea*. (a) sgRNA and their combination (b) Recombinant *E. coli* cells carrying three CRISPR/Cas9 constructs in dual guide 'o' vector individually (c) *Agrobacterium* clones for these constructs, (d) Hypocotyl explants of *B. juncea* co-cultivated with *Agrobacterium* clones, (e) Regenerating transformants of *B. juncea*, (f) Putative transformants in shoot elongation medium, and (g) Selected transformants in rooting media.

C1 constructs and pBI121 vector are at their different developmental stage.



Introgression of the gene(s) for *Alternaria* tolerance from wild species into *Brassica juncea*

Anamika Kashyap, Pooja Garg, Navin C Gupta, Ashish Kumar, KK Singh, Usha

Pant Naveen Singh, Rohit Chamola, Lakshman Prasad, R C Bhattacharya and Mahesh Rao

The introgression lines (BC₂F₁₀) developed in *Brassica juncea* for *Alternaria* resistance from wild species *Diplotaxis erucoides* were subjected to screening at ICAR-NIPB, Delhi under control condition, at GBPUAT, Pantnagar and IARI-RS PUSA Bihar under natural condition (Fig. 5.22). The scoring was done and the resistant lines/plants were identified from all the locations. The stable lines for the resistance were identified for further studies on gene identification by genetic and RNA-seq approaches.

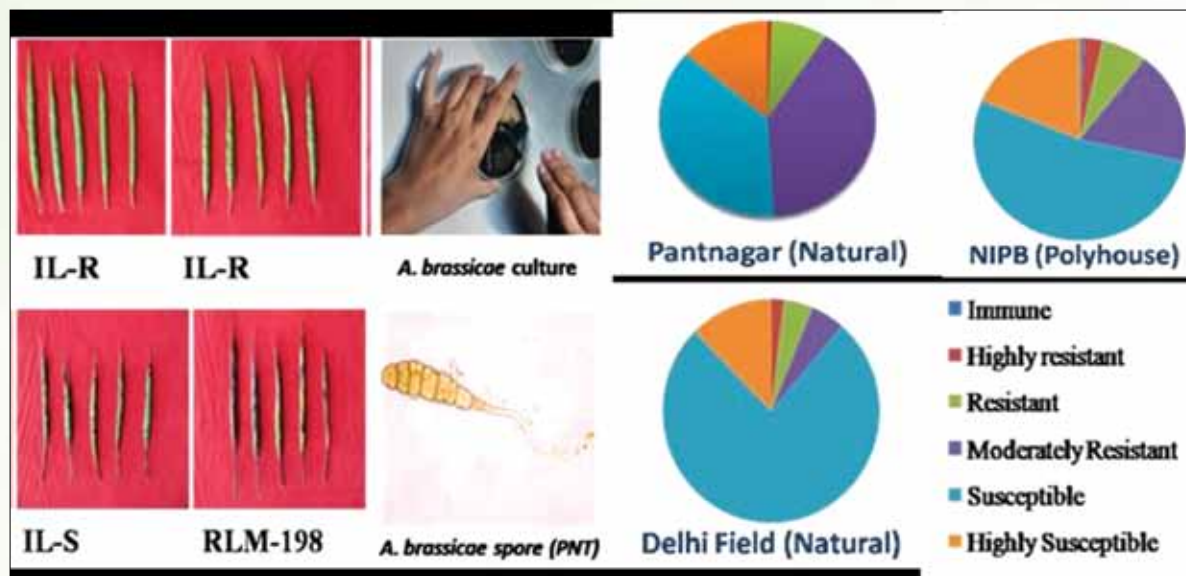


Fig. 5.22: Screening of the *Brassica juncea* introgression lines at GBPUAT, Pantnagar and Delhi under natural controlled conditions. (ILs-R: resistant ILs; ILs-S: susceptible ILs; RLM-198: *B. juncea*; *Alternaria brassicae* culture and spore of Pantnagar isolate)

Enlarging gene pool of *Brassica juncea* through resynthesis and generation of synthetic amphidiploids using wild species and *B. rapa*

Pooja Garg, Anamika Kashyap, Navin C Gupta, Rohit Chamola, Naveen Singh, R C Bhattacharya, KK Singh, Rashmi Yadav, Shashibhusan and Mahesh Rao

The resynthesized *B. juncea* lines (RBJ1 to RBJ-92) in S_6 generation were subjected for the multi-location evaluation for the agronomic trait. The samples were collected from these lines for further chromosomal as well as genomic studies.

New cross combinations were attempted and the new lines were generated which are in the early generation (S_2 generations). Further detailed chromosomal as well as genomic analysis being done using these lines to understand the changes in early as well as advanced generation in these lines. We are also developing the synthetic amphidiploids using wild species and the *B. rapa* (as bridge species) to transfer the gene(s) from wild species to cultivated species. We are using the NRCPB rapa 8, a promising wide compatible genotype, for the crossing followed by the chromosome doubling. Total

six synthetic amphidiploids using different wild species (*B. fruiticulosa*, *Erucastrum canariense*, *E. cardaminoides*, *E. gallicum*, *Enathocarpus lyratus*, *B. tournefortii*) were developed and further analysis is underway. These can be used for the development of introgression lines for the different traits.

Genetic Stock management of Rapeseed Mustard

Rohit Chamola, Anshul Watts, Navin C Gupta, Ashish Kumar, RC Bhattacharya and Mahesh Rao

At ICAR-NIPB, Delhi we are having a diverse set of genetic stocks of *Brassica* including the U-triangle species, wild relatives, CMS lines, synthetic amphidiploid. These are being used in the different research program at NIPB and also shared with other institutes/university as per the ICAR guidelines.



Transformation of *B. juncea* cultivar Varuna and RLM198, and *B. oleracea* var. botrytis cultivar Pusa Meghna with CENH3 CRISPR/Cas9 and GFP-tail swap cassette

Anshul Watts, Ritesh Kumar Raipuria and Komal Mehta

CRISPR/Cas9 vector targeting both *CENH3* paralogs of *B. juncea* and GFP-tail swap cassette were co-transformed in Varuna and RLM-198 variety of *B. juncea* using hypocotyl mediated transformation approach. Many of these plants are on shoot elongation medium. Vitrification is one of the major problems which was observed in RLM-198 derived transformants. Similarly, CRISPR/Cas9 and GFP-tail swap cassette were used to co-transformed Pusa Meghna variety of *B. oleracea* var. botrytis. However, none of the plants survived on shoot elongation medium till now.

B. oleracea is self-incompatible, cross-pollinated and highly heterozygous. In the *CENH3* gene of Pusa Meghna cultivar, 1310 and 1305 bp size *CENH3* alleles were identified. In the population of Pusa Meghna 1310/1310, 1303/1305 homozygous and 1310/1305 heterozygous plants were obtained. The expression analysis of both alleles showed that both were expressed in leaf and flower tissue. Further alternative splicing was identified in both alleles.

Understanding *Brassica-Orobanch* interaction for developing *Orobanch* resistance in *B. juncea*

Anshul Watts, Ritesh Kumar Raipuria, Komal Mehta and Ramcharan Bhattacharya

Around 100 different *Brassica* germplasms were screened for *P. aegyptiaca* infestation at the

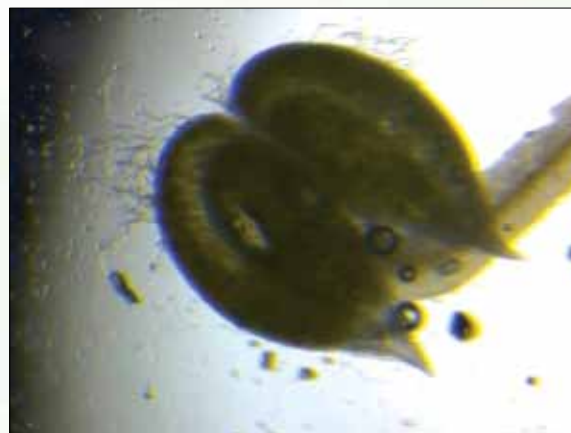


Fig. 5.23: Hairy anthers which are found in *P. aegyptiaca*

hot spot in the Charkhi Dadri district of Haryana. However, most of these lines were showing the infestation of *P. aegyptiaca*. The hairiness of anther is one of the key features which is used to identify *P. aegyptiaca* and distinguish it from its closely related species *P. ramosa*. Therefore, we checked our broomrape flower samples for anther hairiness. All the broomrape samples contained hairy anthers which indicated that all the samples belonged to the *P. aegyptiaca* species (Fig. 5.23). In March 2020, in the Mehra village of Charkhi Dadri district of Haryana near the mustard growing field the *P. aegyptiaca* were found growing among certain weed species. We identified that *P. aegyptiaca* attaches to the weed species *Lepidium didymum* (Fig. 5.24) which belongs to the Brassicaceae family. To the best of our knowledge, it is the first report that *L. didymum* is a host species for *P. aegyptiaca*.



Fig. 5.24: Attachment of *L. didymum* roots with *P. aegyptiaca*



NIPB-1: A promising CMS line of cauliflower (*Brassica oleracea* var. botrytis)

Rohit Chamola, S. R. Bhat, Deepak Bisht, Anshul Watts, R.C. Bhattacharya

At NIPB, male sterility from *Erucastrum canariense* was successfully transferred from *B. napus* background to cauliflower to yield a CMS cauliflower line called 'NIPB-1. Cauliflower variety Pusa Meghna ($CC - 2n=2x=18$) as a male parent was crossed with CMS- (*E. canariense*) *B. napus* ($AACC - 2n=4x=38$) to obtain F_1 plants. In subsequent backcrosses, Pusa Meghna was used as a recurrent male parent to recover the cauliflower genotype with male-sterile flowers. Embryo rescue technique/ ovary culture was done after some days of pollination to recover

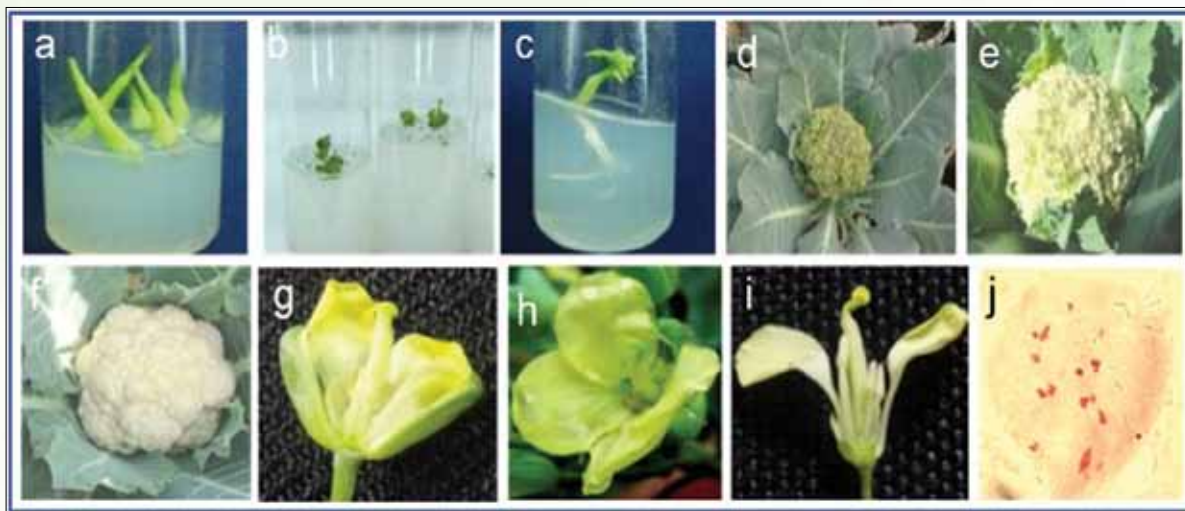


Fig. 5.25: (a), (b), (c) - embryo rescue stages. (d), (e), (f): curd formation in BC₁, BC₂, and BC₆ generation respectively. (g), (h), (i) - sterile flower at BC₁, BC₃ and BC₆ generation. (j)-chromosomes in BC₂ plant, 8II and 2 multivalent.

the succeeding backcross generations. Pistils, harvested at 14 days after pollination for embryo rescue (Fig. 5.25) gave the best results. Eventually, at the BC₄ stage, some seeds were

formed on hybrids plants. Plants obtained from such seeds were very healthy that formed perfect cauliflower curd. Subsequent advancement of generations never required embryo rescue.

NIPB-1 has dark green waxy leaves, light yellow sterile flowers, anthers are short with the absence of pollen grains, nectaries are good making it attractive to bees- an advantage for hybrid seed production. It belongs to the mid-early maturity group which bears compact big circular curd not covered with young leaves. NIPB-1 is a cauliflower CMS line which as such can be tested as a CMS female parent with other male combiners for production of hybrid variety or can be used as a source of male sterility for integration in other female combiners. Some incidence of cabbage butterfly larvae was seen during its early growth which was secured by insecticidal spray. It grows well under standard cultural practices normally followed for cauliflower farming. NIPB-1 has been registered by the Plant Germplasm Registration Committee (PGRC) of the Indian Council of Agricultural Research on December 2020 with registration no. INGR20092.

6. NATIONAL PROFESSOR -B. P. PAL CHAIR

Allele Mining for Agronomically Important Genes in Wild Rice Germplasm and Stress Tolerant Landraces of Rice Growing in the Hot Spots



Genomics-Assisted Introgression and Field Evaluation of Rice Varieties with Genes/QTLs for Yield under Drought, Flood and Salt stress

Dharitree Phulkan, Priyanka Jain, Manish Kumar, Neera Yadav, Vandna Rai, Nagendra Kumar Singh

Four varieties namely, Ranjit-Sub1, Bhadur-Sub1, DRR Dhan 50 and Samba Mahsuri-Sub1 in 2018 and CR Dhan 802 in 2019 were released and notified by the Central Sub-Committee on Crop Standards, Notification and Release of Varieties (CSCSNRV) from this project. A 50K SNP chip designed and patented was used for background selection of advanced backcross lines to identify lines with highest recurrent parent genome recovery in 246 samples. This year, Samba Mahsuri-Sub1-qDTY3.2 NIL was crossed with DRR Dhan 50 for pyramiding of three QTLs for drought tolerance (*qDTY2.1*, *qDTY3.1* and *qDTY3.2*) along with *Sub1A1* gene. Selection of lines homozygous for *qSubA1*, *qDTY2.1*, *qDTY3.1*,

and *qDTY3.2* in Samba Mahsuri background is in process. Sarjoo 52-qGN4.1 and Sarjoo 52-Sub1 NILs were crossed for combining yield with submergence tolerance. Foreground selection in F₂ for combining qGN4.1 with qSaltol 1 in Pusa 44 background is under progress. For fine mapping of the *qDTY3.2* locus, F₂ population (600 plants) were grown during kharif 2020 from cross of Samba Mahsuri Sub1- *qDTY3.2* and DRR Dhan 50 with *qSub1A1*, *qDTY2.1* and *qDTY3.1*. Selection of homozygous recombinants between flanking markers of *qDTY3.2* in the F₂/F₂ population and phenotyping and fine mapping

are underway (Fig. 6.1). For fine mapping of *qDTY3.1* QTL region ~10,000 F₂ seeds have been generated from cross between Swarna-Sub1 and Swarna-Sub1-*qDTY2.1+qDTY3.1*. Twenty-four INDEL markers were identified as informative markers for identification of recombinants for fine mapping (Fig 6.2).

Maintenance and utilization of wild rice germplasm available at ICAR -NIPB

Deepak Singh Bisht, Balwant Singh, Nitin Kumar, Kabita Tripathy, Amit Kumar, Tapan Kumar Mondal, Vandna Rai, Nagendra Kumar Singh

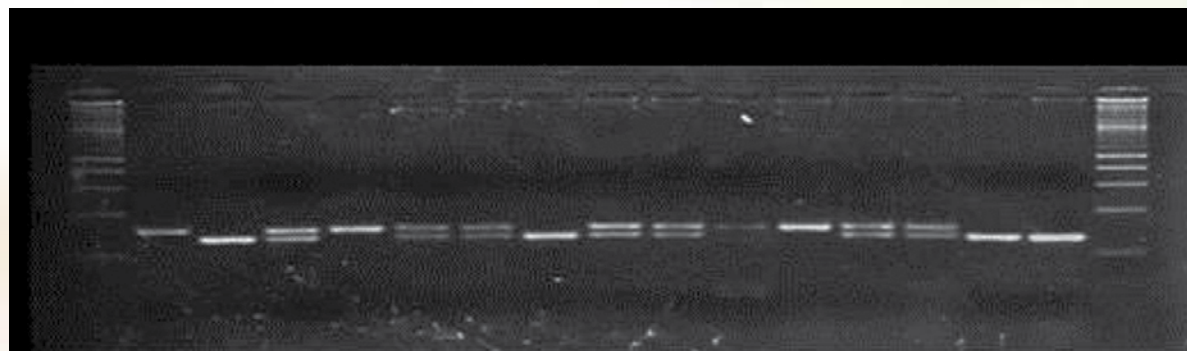


Fig. 6.1: Genotyping for the presence of *qDTY3.2* using marker RM523 (L-Ladder 100bp, P1-*qDTY3.2* SM Sub1 (Female), P2-DRR Dhan-50 (male), L3: (M+F), 1 to 12-BC₁F₁ progeny)

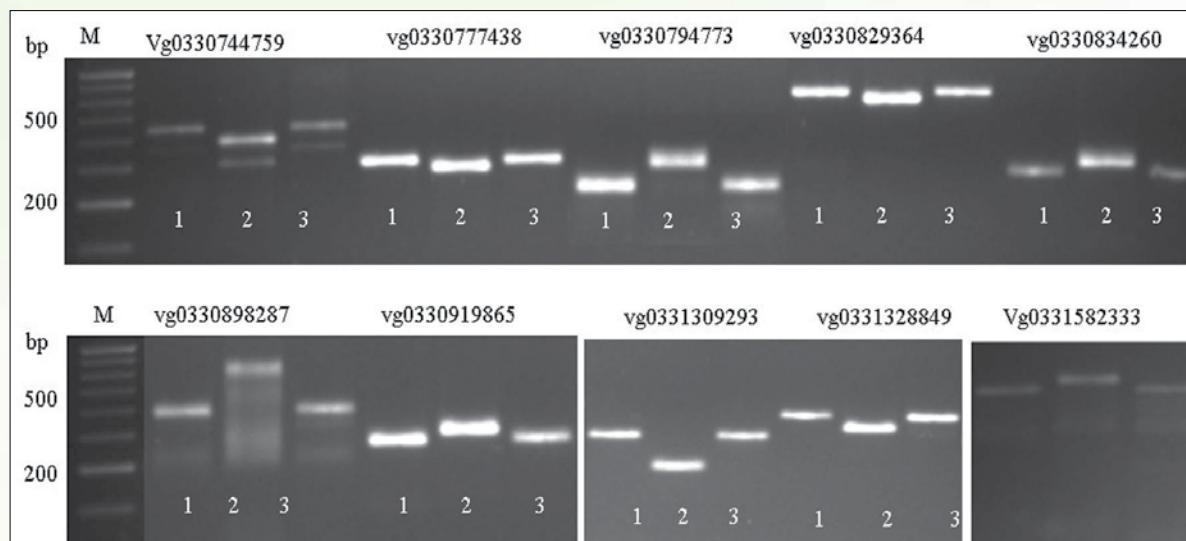


Fig.6.2: Representative informative markers identified between Swarna-Sub1 and Swarna-Sub1-NIL (*qDTY2.1* and *qDTY3.1*) in the *qDTY3.1* QTL region for fine mapping



Fig. 6.3 Multiplication process of Wild rice germplasm at IARI field, Delhi

India is a centre of diversity for rice wherein wild rice is still found in its natural habitat. This year, 791 wild rice germplasm seeds have been collected from 11 out of the 15 agro-climatic zones of India under the National Professor B.P. Pal chair project. Screening them under stress conditions has identified many accessions with high level of abiotic stress tolerance. Seeds of 354 accessions representing nine states, namely Assam, Bihar, Chhattisgarh, Goa, Gujarat,

Himachal Pradesh, Odisha, Uttar Pradesh and Uttarakhand were submitted to NBPGR germplasm database, of which, IC numbers have been granted to 249. Using wild rice germplasm information on geographical location like village, block, district, state, longitude and latitude and 46 morphological characters, a web portal “Indian Wild Rice” (IWR) Database has been developed (nksingh.nationalprof.in:8080/iwrdb). In Kharif 2020, 189 wild rice germplasm accessions were seeds multiplied and evaluated in field (Fig. 6.3).



Mapping and introgression of salt tolerance genes from wild rice into cultivated rice (*Oryza sativa* L.) for reproductive stage salt-tolerance

Tripathi Kabita, Satya Dhriti, Krishnmurthy SL, Nagendra Kumar Singh and Vandana Rai

A backcross population of 75 BC_1F_2 families from cross IR64/NKSWR173 were screened for reproductive stage salt-tolerance in salinity microplots and five best lines for yield related traits were identified (Fig. 6.4). Four QTLs for seedling stage salt tolerance on rice chromosomes 1 and 3 and a major QTL for reproductive stage salt tolerance on rice chromosome 11 were mapped. Analysis of genes present in these QTLs revealed a total of 541 genes which were classified into seven broad

Table 6.1: Number of genes under different functional categories co-located in the QTL intervals for reproductive stage salt tolerance in the IR 64*//NKSUR 173 BC_{1F2} mapping population.

Functional category	QTL name					Total
	qDTF2.1	qPHT8.1	qPHT8.2	qPTN2.1	qSTY11.1/ qPNL11.1	
Interval size (Mbp)	0.1	5.2	0.9	0.5	0.8	10.8
Salt tolerance genes	4	6	3	0	0	14
Other stress response genes	1	23	15	8	7	72
Transcription factors	1	4	2	3	0	26
Enzymes related	3	12	9	9	10	139
Other known proteins	3	72	31	24	16	319
Unknown function	6	94	37	20	30	276
Hypothetical	3	39	16	15	15	127
Total	21	250	113	79	78	973



Fig. 6.4: Field evaluation of parents and BILs for their reproductive stage salinity-tolerance

categories based on their annotated functions (Table 6.1). To know the progenies with QTLs of interest, a SNP chip is being designed.



Evaluation of sheath blight resistance in BC_{1F2-3} mapping population

Deepak Singh Bisht, Nitin Kumar, Tapan Kumar Mondal, Tilak Raj Sharma,

Nagendra Kumar Singh

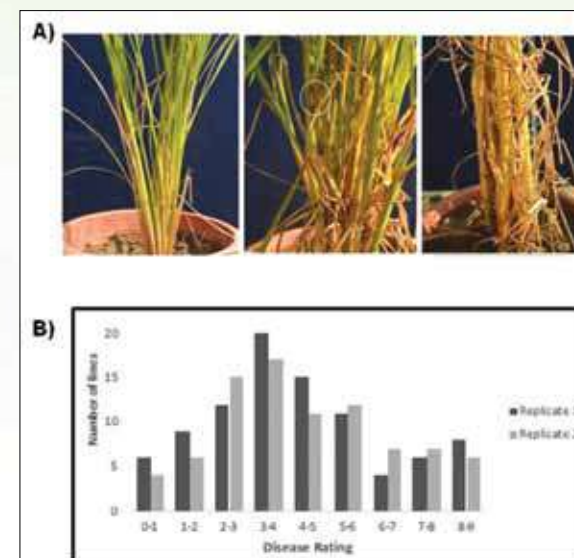


Fig. 6.5: Evaluation of mapping population segregating for sheath blight tolerance phenotype. (A) Disease phenotype in tolerant donor parent, and tolerant and susceptible individuals of the population; (B) Distribution of disease rating observed in replicates

Sheath blight is a major disease of rice for which resistant donor sources in primary gene pool are rare. We have earlier identified a tolerant source from the wild rice collection of our institute. During 2020 a BC_{1F2-3} mapping population segregating for sheath blight resistance phenotype was evaluated. Four phenotypic traits related to sheath blight tolerance viz., maximum lesion length, average lesion length, maximum disease rating and average disease rating were recorded (Fig. 6.5A). Phenotypic distribution of disease scoring

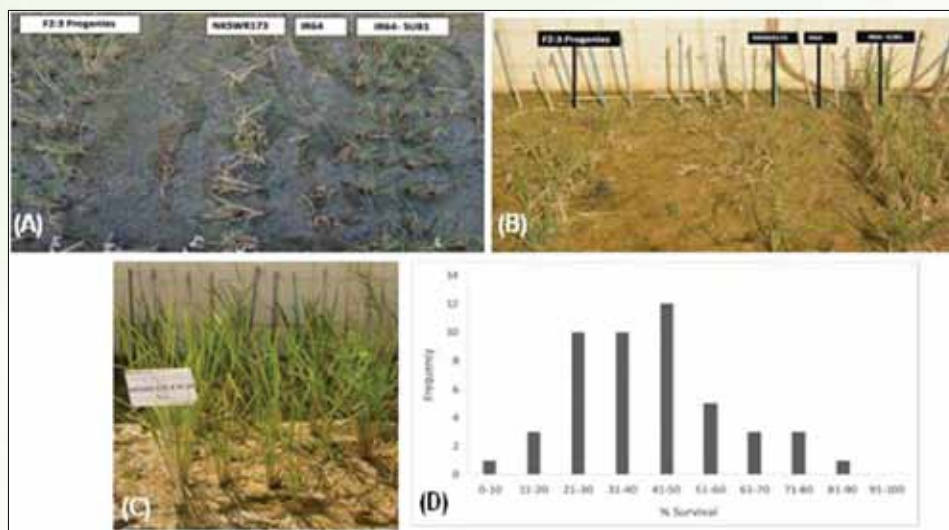


Fig. 6.6: Phenotypic evaluation of mapping population under submergence stress. (A) Immediately after draining the water from the tank; (B) Recovery after 21 days; (C) complete recovery after one month; (D) Average distribution of percent survival

is presented in Fig. 6.5B. The DNA from the individuals of mapping population has been isolated and genotyping is underway.

Utilization of wild rice germplasm in molecular mapping of submergence tolerance and anaerobic germination

Deepak Singh Bisht, Balwant Singh, Nitin Kumar, Kabita Tripathy, Amit Kumar, Tapan Kumar Mondal, Vandna Rai, Nagendra Kumar Singh

Two distinct mapping populations segregating for submergence tolerance and anaerobic germination trait developed during Kharif 2018 were advanced in Kharif 2020. The submergence screening of BC₁F_{2:3} mapping

population was done in the outdoor tank at NIPB phenotyping facility. Around 73 RILs and positive (SMSub-1 and IR64-Sub1) and negative check (IR64) were transplanted in water tank and after 14 days plants were completely submerged with a water depth of approximately 1 meter (Fig. 6.6a). Survival percentage was scored at 21 days after de-submergence, and percent survivability was computed for QTL analysis (Fig. 6.6 b-d).

Label-free non-invasive classification of rice seeds using optical coherence tomography assisted with deep neural network

Deepak Singh Bisht, Deepa Joshi, S. V. Amitha Mitra, Nagendra Kumar Singh

Joshi et al., 2020 Optics and Laser Technology

ICAR-NIPB in collaboration with IIT Delhi has developed a fast, non-contact and non-invasive

technique, deep learning assisted optical coherence tomography (OCT) technique of sub surface imaging to distinguish different rice varieties. The current technique (Fig. 6.7) can accurately classify seed varieties irrespective of the morphological similarities and can be adopted for the removal of varietal duplication and assessment of the purity of the seeds. The entire setup can not only classify rice seeds but can also be trained for any kind of seeds of crop plants.

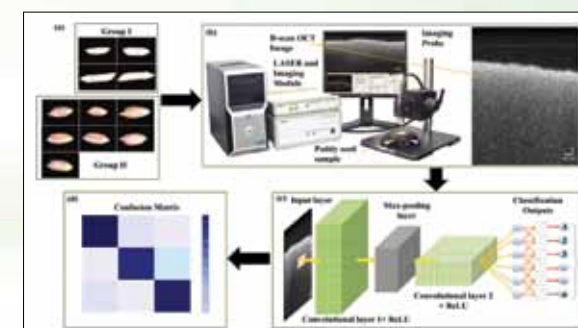


Fig. 6.7: Workflow diagram showing the steps for classification of OCT images of different seeds (a). Swept-source optical coherence tomography (SS-OCT) set-up is used to acquire B-scan OCT images of the rice seed (b). All B-scan images were taken as an input of (c) Inception-ResNet-V2 architecture is trained with 160 seeds (40 for each class) from Group I and 210 seeds (30 for each class) from Group II datasets. The testing results of the network are shown in terms of the confusion matrix (d). Diagonal elements of the matrix represent correct classification while the off-diagonal elements are wrong classifications.

Assembly of an association mapping panel of 495 pigeonpea cultivars varying in grain and nutritional quality traits

Nisha Singh, Nagendra Kumar Singh

A total of 800 diverse genotypes of pigeonpea have been obtained from NBPGR, New Delhi and 95 varieties from IARI, New Delhi, India for the identification of QTLs/genes for agronomically useful traits such as seed weight, seed size, seed color, total protein content, amino acid profiling and resistant starch. We have developed a set of 400 mini-core collections from 800 diverse genotypes of pigeonpea for GWAS analysis in grain and nutritional quality traits. The in-house developed 62K genotyping chip was used for association mapping and QTL mapping for identification of QTLs/genes for agronomically useful traits. We have successfully standardized the high-throughput phenotyping methods to measure the total protein content using two different methods BSA and Kjeldahl methods. Other nutritional traits such as amino acid estimation, resistant starch estimation, specific gravity, antioxidant content and anti-nutritional factors of a total 495 mini-core association panel are under progress.

Analysis of phenotyping traits of seed size, length, weight, width and seed coat color

Phenotyping of seed quality traits were evaluated for GWAS. We have calculated the seed length and width (in cm), weight (in

gm/100 seeds), seed coat color, size of seed, shape of seed by using a measurable tool. Out of 400 mini-core collections 5 pigeonpea genotypes (UP 7-I, MC-109, MC-119, MC-99) had larger seed length and ICP-20092, PDF-EE, U-11-34, had lower seed length. Genotypes UP 12-I, UP 7-I, MC-109, UP 26-II, ICPL-7035 had more seed weight and DG(RG)-53, ICP-20092, Pusa Dwarf, Pusa 2001 and Asha possessed reduced seed weight. The most common seed coat color, seed size and seed shape were cinnamon, medium and semi-spheroid respectively. The rarest seed coat color, seed size and seed shape were black and falsa, small size and ovoid shape respectively found in pigeonpea (Fig. 6.8).

Multi-omics strategies and prospects to enhance seed quality and nutritional traits in pigeonpea

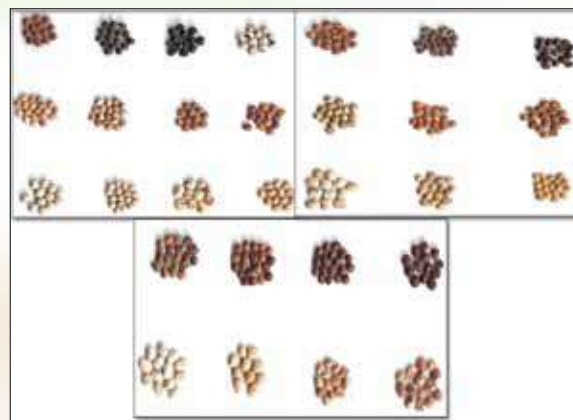


Fig. 6.8: Differences in pigeonpea seed dimensions, seed size, length, weight, width and seed coat color among randomly chosen minicore collection.

Nisha Singh, Vandna Rai, Nagendra Kumar Singh

Identification of potential genes/alleles governing complex traits of seed quality and

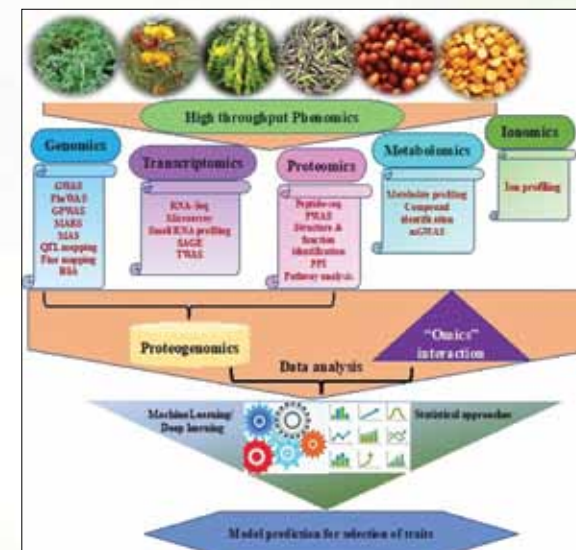


Fig. 6.9: Systematic representation of the development of nutritionally dense pigeonpea varieties through integrated Omics approaches. Integrate multi-omics science approach such as (1) genomics: genome-wide association study (GWAS), phenotype wide association studies (PheWAS), genome phenotype wide association study (GPWAS), marker assisted recurrent selection (MARS), marker assisted selection (MAS), QTL mapping, fine mapping and bulk segregant analysis (BSA). (2) Transcriptomics: RNA-Sequencing, microarray, small RNA profiling, serial analysis of gene expression (SAGE) and transcriptome-wide association studies (TWAS). (3) Proteomics: peptide sequencing, Proteome wide association study (PWAS), protein-protein interaction (PPI) and pathway analysis. (4) metabolite profiling and metabolite-GWAS (mGWAS). (5) Ionomics: ion-profiling. These are the multi-omics data sets that can be utilized to boost the genetic gains in pigeonpea.

nutritional content such as seed weight, seed size, seed color, total protein content, amino acid, antioxidant, resistant starch and disease resistance are essential in genomic selection for quality trait improvement of pigeonpea. (Fig.6.9).

Genome wide exploration of Sugar transporter proteins (STPs) in 12 diverse legumes for sustainable protein and carbon source

Nisha Singh, Megha Ujinwal, Nagendra Kumar Singh

Sugar transporter proteins (STPs) are membrane proteins that are required for sugar transport throughout cellular membranes. They

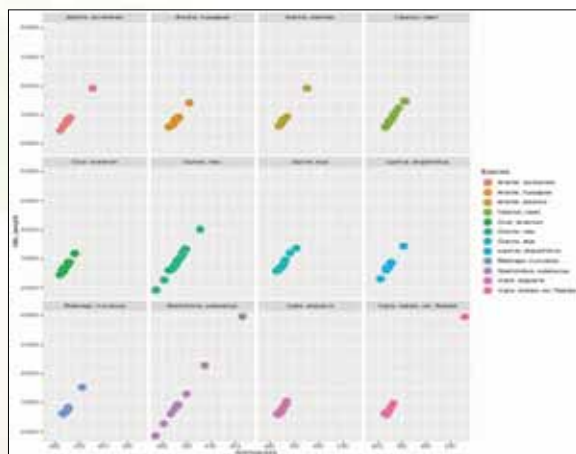


Fig. 6.10: Distribution of total amino acid length and molecular weight of 154 STPs

play an imperative role in sugar transmission across the plant and determinants of crop yield. The analysis of this important STPs family was performed to analyze its cellular, molecular and structure composition in 12 diverse legume species. Here, we performed a comprehensive and systematic analyses of 154 sugar transporter protein across 12 legumes species (*Cajanus cajan*, *Glycine max*, *Vigna radiata*, *Vigna angularis*, *Medicago truncatula*, *Lupinus angustifolius*, *Glycine soja*, *Spatholobus suberectus*, *Cicer arietinum*, *Arachis ipaensis*, *Arachis hypogaea*, *Arachis duranensis*). The amino acid composition revealed that STPs are rich in polar amino acid i.e., alanine, leucine, serine and phenylalanine while glutamine, tryptophan, cystine and histidine are less abundant with an average 255

amino acid length and 286-22 KD weight (Fig. 6.10 & 6.11). Secondary structure analysis revealed that these 154 sugar transporter proteins are richer in alpha helix (40%) than extended strand (30%) and random coil (25%) respectively. A total of four main motifs that are conserved were found in all 154 STPs in legumes. These motifs are profuse in leucine, valine and isoleucine while lacking in tryptophan and histidine amino acid content. Overall, this analysis provided a detailed information of primary, secondary, tertiary structure and phylogenetic analysis of STPs for further studies to infer their mechanism at structural and functional level which play essential roles in plant development and stress responses. A comprehensive overview of the legume STPs family will offer new insights into the structural and functional features as well as facilitate further research on the roles of STPs genes in response to abiotic and biotic stresses. Furthermore, this analysis will be useful to examine photosynthetic productivity, embryo sugar content, seed quality and yield enhancement.

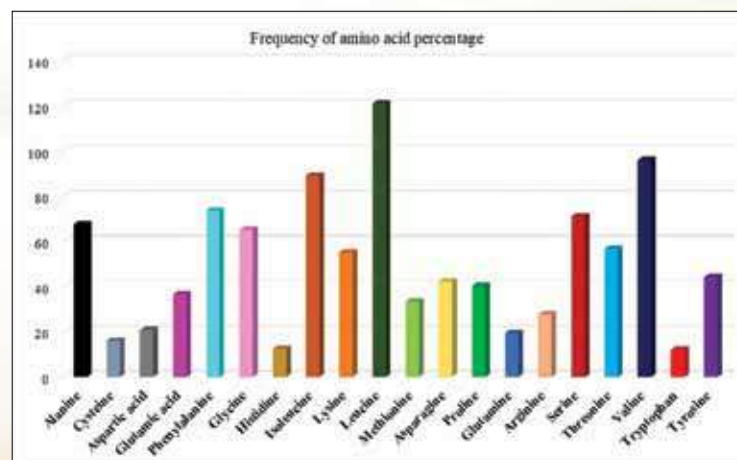


Fig. 6.11: Distribution of amino acid frequency across 154 STPs

- 1. POST-GRADUATE TEACHING
PROGRAMME**
- 2. TRAINING AND CAPACITY
BUILDING**
- 3. TRAINING PROGRAM
ORGANIZED**

HUMAN RESOURCE DEVELOPMENT

POST-GRADUATE TEACHING PROGRAMME

ICAR-National Institute for Plant Biotechnology (NIPB) has been actively engaged in Human Resource Development in the area of plant Molecular Biology and Biotechnology since its inception. Currently 32 Ph. D. and 22 M. Sc. students are registered in the discipline of Molecular Biology and Biotechnology at the Institute. Seven Ph. D. and five M. Sc. students were awarded with doctoral and master's degrees in the 58th Convocation 2020, respectively.

Students on roll in the discipline of Molecular Biology and Biotechnology during the Academic Session 2020-21

Ph.D. students

S. No.	Student Name and Roll No.	Name of the Chairperson	S. No.	Student Name and Roll No.	Name of the Chairperson
1.	Ms. Shruti Sinha (10497)	Dr. Kishor Gaikwad	17.	Mr. Krishnayan Paul (11546)	Dr. Debasis Pattanayak
2.	Mr. Sandeep Jaiswal (10639)	Dr. Debasis Pattanayak	18.	Mr. Muhammed Shamnas V. (11547)	Dr. Subodh Kumar Sinha
3.	Ms. Priyanka Singh (10697)	Prof. N. K. Singh	19.	Mr. Naresh Kumar Samal (11548)	Dr. R. C. Bhattacharya
4.	Ms. Sharani Choudhury (10836)	Dr. R. C. Bhattacharya	20.	Mr. Zaherul Islam (11549)	Dr. Amolkumar Solanke
5.	Mr. Mahendra C. (10844)	Dr. Kanika	21.	Mr. Deepesh Kumar (11550)	Dr. S. V. A. C. R Mithra
6.	Ms. L. Ashakiran Devi (11070)	Dr. R. C. Bhattacharya	22.	Mr. Dhivyanandham K. (11551)	Dr. Monika Dalal
7.	Ms. Bablee Kumari Singh (11071)	Dr. Monika Dalal	23.	Mr. Gopal (11552)	Dr. Pradeep Kumar Jain
8.	Mr. Sachin (11074)	Dr. Pranab Kumar Mandal	24.	Mr. Ankur Poudel (11646)	Dr. Pranab Kumar Mandal
9.	Mr. Suhas G. Karkute (11288)	Dr. Amolkumar Solanke	25.	Mr. Samar Deb (11816)	-
10.	Mr. Bipratip Dutta (11290)	Dr. S.V.A.C.R. Mithra	26.	Ms. Mareyam Mukhtar (11817)	-
11.	Ms. Sheel Yadav (11291)	Dr. Pradeep Kumar Jain	27.	Mr. Jeet Roy (11818)	-
12.	Mr. Akash Paul (11293)	Dr. Kishor Gaikwad	28.	Ms. Vibha Kamati (11820)	-
13.	Mr. Saakre Manjesh (11294)	Dr. Debasis Pattanayak	29.	Ms. Priya (11821)	-
14.	Mr. Mawuli K Azameti (11362)	Dr. Jasdeep C. Padaria	30.	Ms. Nitasana Rajkumari (11822)	-
15.	Ms. Shaziya Sultana (11367)	Dr. Sharmistha Barthakur	31.	Mr. Anuj Kumar (11823)	-
16.	Mr. W. Sandesh Tulsiram (11370)	Dr. Anita Grover	32.	Ms. Renu Kumari (11824)	-

M.Sc. Students

S. No.	Student Name and Roll No.	Name of the Chairperson	S. No.	Student Name and Roll No.	Name of the Chairperson
1	Mr. Akash Maity (21308)	Dr. Monika Dalal	12	Ms. Pyla Bhuvaneswari (21491)	-
2	Ms. Rekha Mahato (21309)	Dr. Tapan Kumar Mondal	13	Mr. Yogesh Kumar S (21492)	-
3	Mr. Mutawar Ashfaq S. (21310)	Dr. Sarvjeet Kaur	14	Mr. Manish Dev Pratap (21493)	-
4	Mr. Mahamed Ashiq I. (21311)	Dr. Sharmistha Barthakur	15	Ms. Nuzat Banu (21494)	-
5	Ms. Shwetha R (21312)	Dr. Jasdeep C. Padaria	16	Mr. Ramesh R (21495)	-
6	Ms. Ankita Vilasrao Chinche (21313)	Dr. Kanika	17	Mr. Anik Basak (80007)	-
7	Ms. Anindita Barua (21314)	Dr. N. C. Gupta	18	Ms. Lerissa Sweetie Dsilva (80008)	-
8	Mr. Gowtham T. P. (21315)	Dr. P. K. Dash	19	Mr. Sampatirao Dilip (90006)	-
9	Mr. Sagnik Chanda (21488)	-	20	Mr. Sri Harsha Malempati (90007)	-
10	Mr. Adil Rahim Magray (21489)	-	21	Ms. Olivia Niang Lun Hoih (90008)	-
11	Ms. Rishika K S (21490)	-	22	Ms. Pragati Sudhakar Gajbhar (90009)	-

Degrees awarded in the discipline of Molecular Biology and Biotechnology during the convocation-February, 2020

S No.	Name of the Student and Roll No.	Chairperson	Thesis Title
Ph.D. Students			
1.	Mr. Rakesh Bhowmick (10152)	Prof. N. K. Singh	Fine mapping of QTL for salt tolerance on rice chromosome 8.
2.	Ms. Antara Das (10154)	Dr. Kishor Gaikwad	Genome wide identification and characterization and long non-coding RNAs during seed and pod development in pigeon pea (<i>Cajanus cajan</i>)
3.	Mr. Prashant Yadav (10155)	Dr. Anita Grover	Proteomic analysis of <i>Brassica juncea</i> , <i>Sinapis alba</i> and <i>Camelina sativa</i> in response to <i>Alternaria</i> infection
4.	Mr. Vinod Kr. Jangid (10318)	Dr. Anita Grover	Identification, isolation and expression analysis of defense transcription factor gene <i>WRKY33</i> in <i>Brassica coenospecies</i> and validation of best expressing gene in <i>Arabidopsis</i> for <i>Alternaria</i> resistance
5.	Ms. Anupma (10319)	Dr. Sarvjeet Kaur	Validation of <i>in planta</i> efficacy of <i>cry1Ac34</i> and <i>vip3Aa44</i> genes against <i>Helicoverpa armigera</i>

S No.	Name of the Student and Roll No.	Chairperson	Thesis Title
6.	Ms. Shikha Dixit (10495)	Dr. Anita Grover	Exploring the cross-talk among SA, JA and ABA at molecular level in response to <i>Alternaria brassicae</i> in Brassica and its wild relatives
7.	Ms. Priyanka (10501)	Dr. Kanika	Functional validation of <i>NHX</i> antiporter gene associated with salt tolerance in wheat



Dr. Shikha Dixit was awarded IARI Merit Medal for her Doctoral research work in 58th Convocation, 2020

Degrees awarded in the discipline of Molecular Biology and Biotechnology during the convocation-February, 2020

S No.	Name of the Student and Roll No.	Chairperson	Thesis Title
M. Sc. Students			
1.	Mr. Mohammad Shamnas V. (20994)	Dr. Subodh K. Sinha	Identification, cloning and expression studies of <i>AtNRT1.5</i> orthologue in wheat (<i>Triticum aestivum</i> L)
2.	Mr. Johan Ajnabi (20995)	Dr. Monika Dalal	Identification of cis-regulatory regions regulating the expression of <i>PM19</i> gene in wheat
3.	Mr. Krishnayan Paul (20996)	Dr. D. Pattanayak	Heterologous expression and characterization of acetylcholinesterase-1 of <i>Helicoverpa armigera</i>
4.	Mr. Naresh Kr. Samal (20997)	Dr. R. C. Bhattacharya	Molecular and biochemical basis of differential aphid infestation in wild species of Brassica
5.	Mr. Zaherul Islam (20999)	Dr. T. K. Mondal	Study of ureide pathway gene-allantoinase in halophytic species <i>Oryza coarctata</i> Roxb.

TRAINING AND CAPACITY BUILDING

Details of training by ICAR-NIPB staff during 2020

S.No.	Name	Subject Area	Duration	Host Institute
1.	Dr. Yuvaraj.I.	Foundation training	January-March, 2020	NAARM, Hyderabad
2.	Dr. Yuvaraj.I.	Online Orientation training	May-June, 2020	NIPB, New Delhi
3.	Dr. Yuvaraj.I.	Bioinformatics	September- November, 2020	IISc, Bangalore
4.	Dr. Yuvaraj.I.	ABC of Scientific Writing'	August- September 2020	ICAR-NRRI, Cuttack
5.	Dr. Rekha Kansal	Training Management Information System	May, 2020	IASRI, New Delhi-110012
6.	Dr. Rekha Kansal	Vigilance Officers	August,2020	NAARM, Hyderabad
7.	Ms. Shbana Begam	Foundation training	January-March, 2020	NAARM, Hyderabad
8.	Ms. Shbana Begam	Online Orientation training	May-June, 2020	NIPB, New Delhi
9.	Ms. Shbana Begam	Professional attachment Training	September- November, 2020	ICAR-IASRI, New Delhi
10.	Ms. Shbana Begam	Omics in Agriculture	07-09 October, 2020	RPCAU, PUSA, Bihar
11.	Ms. Shbana Begam	Kosambi International Webinar Series on Plant genomics	31 July -2 August 2020	SPPU, Pune

HRD Fund allocation and utilization

Total HRD fund allocation for 2019-2020 (Rs. in Lakh)	Actual expenditure for 2019-2020 (Rs. in Lakh)	% utilization
1.5	1.04	69.33

TRAINING PROGRAM ORGANIZED

NAHEP-CAAST Training program on “Pre-breeding and molecular breeding approach: Two important pillars for the vegetable and crop improvement”

A training on “Pre-breeding and Molecular breeding Approach: Two important pillars for the vegetable and crop improvement” was conducted for training the M. Sc. and Ph. D. during 17th February 2020 to 01 March 2020 at ICAR-NIPB, Delhi. The training was sponsored



by CSAUA&T Kanpur, U.P. under ICAR-World bank Funded NAHEP-CAAST Project. Dr. Sanjay Singh was course Co-ordinator and Dr. Mahesh Rao, Dr. Nimmy MS and Dr. Sandhya were the course Co-ordinators. The training program was designed with the lecture series on different aspects of prebreeding and molecular breeding followed by the hands-on training.

DBT-BCIL sponsored Training program for Accredited Testing Laboratories (ATLs) on “Hands on Training and Proficiency Testing for Genetic Fidelity and Virus Indexing of Tissue Culture Raised Plants”

A DBT-BCIL sponsored Training program for Accredited Testing Laboratories (ATLs) on “Hands on Training and Proficiency Testing for Genetic Fidelity and Virus Indexing of Tissue Culture Raised Plants” was conducted at the two referral centres: ICAR-National Institute for

Plant Biotechnology, Pusa Campus, New Delhi for Genetic Fidelity Testing and Advanced Centre for Plant Virology, Division of Plant Pathology, IARI for Virus Indexing from 28th January-7th February, 2020. Dr Amolkumar Solanke and Dr Amitha Mithra Sevanthi were Course Directors for the training. The participants were from the ATLs ICAR-IISR, Lucknow and GKVK Campus, Bangalore. Trainees were trained about the genetic fidelity testing protocols of five main tissue culture grown crops banana, sugarcane, date palm, bamboo and potato using ISSR and IRAP marker system.



EXTENSION AND OUTREACH ACTIVITIES

ICAR-National Institute for Plant Biotechnology organizes extension and outreach activities through two major initiatives of government of India namely Mera Gaon Mera Gaurav (MGMG) and Scheduled Castes Sub Plan (SCSP). Since year 2020 was marred by Coronavirus pandemics, all the institute's outreach programs were conducted through online mode.

Online Farmers-Scientist interaction (December 19, 2020)

ICAR-National Institute for Plant Biotechnology Delhi organized a Scientist-Farmer interaction session on 19th September 2020 through online mode. The Chief Guest Dr. Ashok Kumar Singh DDG (Agriculture Extension), ICAR, Delhi gave an outline of the different farmer-oriented program of the ICAR and also appreciated the extension work done by ICAR-NIPB in catering the needs of the farmers. The entire panel of experts discussed

on newly released Rice, wheat, mustard and chickpea varieties, protected cultivation, Integrated farming system (IFS), GM crops,



and plant disease management. The major emphasis of the program was to brief farmers with new technologies, and broaden their potential source of income for achieving self-sufficiency.

Kisan Diwas Celebration (December 23, 2020)

Kisan Diwas was observed at ICAR-NIPB on December 23, 2020. The Director (Incharge), Dr. Sarvajeet Kaur interacted with farmers including village women and briefed them about the enormous contribution of late Chaudhary Charan Singh Ji in the upliftment of livelihood of farmers. She also emphasized the importance of cleanliness in our daily lives and called on participants to maintain a clean and healthy environment in their houses and workplaces. All the staff members of NIPB present during the program.



Seed and planting material distribution during Kharif 2020

Due to the outbreak of the pandemic novel coronavirus (COVID-19), the 2020 Kharif season was full of challenges for farmers and extension workers. However, ICAR-NIPB Delhi continued its extension work during this difficult period to transfer technology to farmers. ICAR-NIPB New Delhi along with different KVKs of the Uttar Pradesh and NGO's jointly organized the seed distribution program and Vriksharopan program during Kharif 2020 following all the Covid-19 guidelines of GOI and ICAR. Development of fruit orchards was also undertaken as a prime objective.

Seed distributed during Kharif 2020

1. Rice: DDR Dhan 50 & Pusa Basmati 1637
2. Pigeon Pea: Pusa Arhar 16

Planting material distributed during Kharif 2020

1. Mango : Amrapali, Dasher, lagda
2. Lichi
3. Guava : L-49
4. Citrus : Kagji Nibu



1. INSTITUTIONAL PROJECTS
2. EXTERNALLY FUNDED PROJECTS
3. TECHNOLOGY
COMMERCIALIZATION AND IPR
4. AWARDS AND HONOURS
5. VISITS ABROAD
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OTHER INSTITUTIONAL ACTIVITIES

INSTITUTIONAL PROJECTS

Project Title	Date of Start	Date of Completion	Principal Investigator	Name of Associates
Search and deployment of genes for stress tolerance and grain quality in rice.	1 st April, 2017	31 st March, 2021	Dr. T. K. Mandal	Dr. Vandna Rai, Dr. P. K. Dash Dr. Rhitu Rai, Dr. SVACR. Mithra Dr. Amolkumar U. Solanke Dr. Deepak S. Bisht Dr. Pankaj Kumar, Ms. Rita
Genetic improvement of wheat for adaptation to climate change induced stresses.	1 st April, 2017	31 st March, 2021	Dr. Sanjay Singh	Dr. J. C. Padaria, Dr P. K. Mandal Dr. S. Barthakur, Dr. Kanika Dr. Monika Dalal, Dr. Subodh Sinha
Improvement of stress tolerance in chickpea.	1 st April, 2017	31 st March, 2021	Dr. Sarvjeet Kaur	Dr. P. K. Jain, Mr. R. A. Nagar Dr. M. S. Nimmy, Mr. Deepak K. Rathore
Stress tolerance and quality improvement in pigeonpea.	1 st April, 2017	31 st March, 2021	Dr. Debasis Pattanayak	Dr. Kishor Gaikwad, Dr. Rohini Sreevathsa Dr. A. Dinabandhu, Dr. Sandhya Dr. R.S. Niranjana, Ms. Megha Mr. Anshul Verma
Biotechnological approaches for Brassica improvement.	1 st April, 2017	31 st March, 2021	Dr. R. C. Bhattacharya	Dr. Anita Grover, Dr. Rekha Kansal Dr. Ashish Kumar, Dr. Navin C. Gupta Dr. Mahesh Rao, Mr. Anshul Watts Mrs. Seema Dargan, Dr. Rohit Chamola Mrs. Sandhya Rawat

EXTERNALLY FUNDED PROJECTS

S.No.	Title of the project	Funding agency	Budget (Rs. Lakhs)	Principal Investigator
1.	Functional genomics of rice, wheat	ICAR-NPTC	500	Prof. Nagendra K. Singh
2.	Improvement of Pigeon pea for plant type, early maturity, pod borer resistance and moisture stress tolerance	DBT-ISCB	640	Prof. Nagendra K. Singh
3.	From QTL to Variety: breeding for abiotic stress tolerance in rice-Phase I	DBT	1000	Prof. Nagendra K. Singh
4.	High through put approach to discover of novel salt tolerant genes from wild halophytic species <i>Oryza coarctata</i>	DST	42	Dr. T. K. Mondal
5.	Structural and functional genomics study of deepwater adaptation of local rice landraces of Assam	DBT	24.47	Dr. T. K. Mondal
6.	Generation and comparative analysis of salinity responsive miRNAs and miRNA-mediated pathways among halophyte (<i>Oryza coarctata</i>), tolerant glycophyte (<i>Oryza sativa</i> cv Nona bokra) and susceptible glycophyte (<i>Oryza sativa</i> cv IR-64) genotypes	DBT	60.60	Dr. T. K. Mondal
7.	Decoding tea (<i>Camellia assamica</i>) genome for identification of commercially important genes	Tea Board, Kolkata	116	Dr. T. K. Mondal
8.	Gene Discovery for Reproductive Stage salt tolerance in rice	DBT	80	Dr. Vandana Rai
9.	Mainstreaming sesame germplasm for improvement through genomics assisted core development and trait discovery	DBT	125	Dr. Vandna Rai
10.	Identification of QTLs for subcomponent of WUE through strategic utilization of whole genome sequences and accurate phenotyping rice.	ICAR-NASF	28.72	Dr. Prasanta K Dash
11.	Genetic improvement of rice for yield, abiotic and biotic stress tolerance through RNA guided genome editing (CRISPR-Cas9)	ICAR-NASF	78.10	Dr. Prasanta K. Dash
12.	Flax genomics	ICAR-NPFGGM	10.00	Dr. Prasanta K. Dash

S.No.	Title of the project	Funding agency	Budget (Rs. Lakhs)	Principal Investigator
13.	Sequencing of <i>Xanthomonas oryzae</i> pv <i>oryzae</i> pathotypes for better management of bacterial blight disease in rice	DBT	64.87	Dr Rhitu Rai
14.	Referral centre for Genetic Fidelity testing of Tissue culture raised plants (NCS-TCP)	DBT	71.37	Dr. Amolkumar U Solanke
15.	RiceMetaSys: Understanding rice gene network for biotic and abiotic stress management through system biology approach	ICAR-CABin	13.49	Dr. Amolkumar U. Solanke
16.	CRISPR-Cas9 based genome editing of multiple negative regulators for blast resistance in rice	DST-SERB	31.93	Dr. Amolkumar U. Solanke
17.	Maintenance, Characterization and use of EMS Mutants of upland variety Nagina 22 for Functional Genomics in Rice- Phase II	DBT	258.98	Dr. Amitha Mithra Sevanthi/ Dr NK Singh
18.	Molecular Dissection of mechanism of resistance to Sheath Blight in Rice	DST	36.36	Dr. Deepak S. Bisht
19.	Exploiting alien genetic resources for developing climate resilient wheat and understanding mechanism of heat tolerance	ICAR	25.64	Dr. Jasdeep Chatrath Padaria
20.	Genetic modifications to improve biological nitrogen fixation for augmenting nitrogen needs of cereals	ICAR- Incentivizing Research in Agriculture	60.8	Dr. Pranab Kumar Mandal
21.	Development of Low Immunogenic Wheat	ICAR-CRP on Biofortification	28.23	Dr. Pranab Kumar Mandal
22.	Development of transgenic wheat for heat stress tolerance	ICAR-NPFGGM	35	Dr. Sharmistha Barthakur
23.	Deciphering the transcriptional regulator(s) of hydrotropism in wheat	DST-SERB	40.7	Dr. Monika Dalal
24.	Identification and functional validation of partner proteins of two-component high affinity nitrate transporter of wheat	SERB-DST	36.32	Dr. Subodh Kumar Sinha
25.	Characterization, mapping and transcriptome analysis of seed protein, β -carotene and mineral contents in chickpea (<i>Cicer arietinum</i> L.)	NASF	102	Dr. P.K. Jain
26.	Study on methylation patterns under drought stress for identification of key genes and microRNAs involved in drought stress tolerance in chickpea	DST	38	Dr P. K. Jain
27.	Amelioration of cold-induced sweetening in potato	ICAR-NPTC	10	Dr. Debasis Pattanayak

S.No.	Title of the project	Funding agency	Budget (Rs. Lakhs)	Principal Investigator
28.	Decoding the genomes of guar and black gram for the identification of important genes	ICAR-CRP on Genomics	145.00	Dr. Kishor Gaikwad
29.	Molecular mapping and identification of candidate gene(s) responsible for the cleistogamous trait in pigeonpea <i>Cajanus cajan</i> Millsp	DST-SERB	36.63	Dr. Kishor Gaikwad
30.	Functionality of Methionine sulfoxide reductase gene(s) in redox homeostasis-mediated resistance to <i>Helicoverpa armigera</i> in pigeonpea wild relative <i>C. platycarpus</i>	SERB	42	Dr. Rohini Sreevathsa
31.	Herbicide tolerance in cotton	ICAR-NPTC	47	Dr. Rohini Sreevathsa
32.	Genomics-led improvement of biotic and abiotic stress tolerance in mustard rape for economic and environmental sustainability	DBT	108	Dr. R. C. Bhattacharya
33.	Development of haploid inducer line, and enhancement of seed-meal quality in <i>Brassica juncea</i> through CRISPR/Cas mediated genome editing	DBT	50	Dr. R. C. Bhattacharya
34.	A way forward to developing aphid-resistance in Indian mustard (<i>Brassica juncea</i>): Host- and virus-mediated gene silencing of parthenogenetic genes in mustard aphid	SERB	38	Dr. R. C. Bhattacharya
35.	Development of Aphid Resistant Transgenic Brassica	ICAR-NPTC	70	Dr. Rekha Kansal
36.	Identification of effector candidates from necrotrophic plant pathogen, <i>Sclerotinia sclerotiorum</i> and its target proteins in Indian mustard (<i>Brassica juncea</i>)	DBT	38.16	Dr Navin C Gupta
37.	Identification and functional characterization of the key resistance/susceptible determinants for <i>Sclerotinia</i> stem rot disease in oilseed Brassica	DST	32.0	Dr Navin C Gupta
38.	Broadening genetic diversity of Indian mustard (<i>Brassica juncea</i>) through resynthesis of amphidiploids genome by crossing parental diploid species	DST-SERB	33.64	Dr. Mahesh Rao
39.	Induced mutation of synthetic B. juncea haploids for isolation of climate resilient genotypes	BRNS	31.456	Dr. Mahesh Rao
40.	Strengthening cultivar diversity of rapeseed mustard to manage climate related risks and foster productivity in stress prone areas of Central and Western India	Bioversity International	29.742	Dr. Mahesh Rao
41.	Development of haploid inducer line in <i>Brassica oleracea</i> var. botrytis through CRISPR-Cas mediated engineering of centromeric histone H3 gene	DST-SERB	40.20	Dr. Anshul Watts

TECHNOLOGY COMMERCIALIZATION AND IPR

The mandate of the Institute Technology Management Unit relates to registration of patents, facilitation of contract research projects and commercialization of IPR enabled technologies of the institute through Public- Private Partnership. The following activities were undertaken by the ITMU during the year 2019.

I. Patent Granted:

S. No.	Title of patent application	Patent Application Number & date of filling	Date of Grant	Granted Patent Number
1.	Pathogen inducible promoter from rice and uses thereof.	783/DEL/2014 18.03.2014	13.02.2020	331964
2.	A single copy gene based 50K SNP rice DNA chip and uses thereof.	2280/DEL/2014 11.08.2014	25.06.2020	339438
3.	Nucleotide sequence of rice responsible for resistance to <i>Magnaporthe Oryzae</i> and uses thereof.	782/DEL/2014 18.03.2014	15.09.2020	346774

II. MTA/MoUs signed:

MTA	For transfer of seeds of 541 EMS induced mutants of Nagina 22 to Amity University Uttar Pradesh, Noida on 28.02.2020 for research purpose
MOUs	MOU Signed between ICAR-NIPB, New Delhi and GLA University, Mathura on 04.08.2020, for inter – institutional research for staff and students' in cutting edge areas
	MOU signed between RARS, Assam Agricultural University, North Lakhimpur for genetic improvement of deepwater rice of Assam through marker assistant selection”
	MOU was signed between UBKV, Cooch Behar and NIPB, New Delhi was done for, “Development of semi-dwarf high yielding short grain aromatic traditional rice varieties of North Bengal through marker assisted breeding”
	MOU was signed between JNKVV, Balakot and NIPB, New Delhi for “Development of semi-dwarf high yielding short grain aromatic traditional rice varieties (Chinoor) of Madhya Pradesh through marker assisted breeding”

AWARDS AND HONOURS

Dr Rhitu Rai was awarded Indo-US Genome Editing Technology Initiative Fellowship (IUSSTF-DBT)

Dr. P. K. Mandal is Faculty Council Member of Uttar Banga Krishi Viswavidyalaya, Coochbehar

Dr. P. K. Mandal nominated as IBSC member of BAYER Crop Science Limited

Dr. P. K. Mandal became editorial board Member for SAARC Journal of Agriculture

VISIT ABROAD

Dr Rhitu Rai, Principal Scientist visited Cornell University, USA during May 2019 to Sept 2020

Dr. P.K. Dash, Principal Scientist visited USA during 11th to 15th January 2020

RECRUITMENTS/PROMOTIONS/RETIREMENTS

Recruitments

Ms. Shbana Begam joined as Scientist, ICAR-NIPB on 04th April, 2020.

Mr. Yuvaraj Iyyapan joined as Scientist, ICAR-NIPB on 04th April, 2020.

Dr. Joshitha Vijayan joined as Scientist, ICAR-NIPB on 05th Oct, 2020.

Mr. Harshit Agarwal joined as Senior Administrative Officer, ICAR-NIPB on 18th July, 2020.

Mr. Rahul Kumar joined as Finance & Accounts Officer, ICAR-NIPB on 02nd Nov, 2020.

Promotions

Dr. Subodh Kumar Sinha promoted from Senior Scientist to Principal Scientist.

Dr. Rohini Sreevathsa promoted from Senior Scientist to Principal Scientist.

Smt. Sangeeta Jain promoted from Assistant-to-Assistant Administrative Officer.

Retirements

Sh. Krishan Dutt, Assistant Administrative Officer superannuated from 31st May, 2020.

Dr. N.K. Singh, Principal Scientist superannuated from the post of Principal Scientist from 30th November, 2020.

Transfers:

Mr. Suresh Kumar Sharma, Finance & Accounts Officer transfer from NIPB to ICAR HQ.

OTHER ACTIVITIES

Hindi Activities 2020

Hindi Chetna Maas

Competitions in Hindi essay writing, technical work writing, poetry recitation and noting and drafting were conducted for the staff of NIPB as a part of Hindi Chetna Maas from 14th September, 2020 to 13th October, 2020. Due to corona pandemic, all the activities were conducted online.



Hindi Workshops: Three hindi workshops were conducted during the Year 2020

S. No.	Date	Quarter	Guest Speaker
1.	17-02-2020	January-March	Ms. Sunita

S. No.	Date	Quarter	Guest Speaker
2.	03-09-2020	July-September	Ms. Sunita
3.	28-12-2020	October-December	Mr. V.N. Tiwari

Mahila Kisan Divas

On October 13, 2020, Mahila Kisan Divas was organized at the institute through online mode. A total of 45 people including 15 women participated from the villages adopted by ICAR-NIPB under Mera Gaon Mera Gaurav program. The program was formally inaugurated by Director (Incharge) NIPB Dr. Sarvajeet Kaur. The chief guest of the function Ms. Neelam



Tyagi, Secretary, Laxmi Jan Kalyan Sevak Sansthan, Ghaziabad shared her views on the efforts towards making women farmers independent and self-reliant. Dr. Alka Singh, Professor and Head of the Division, Agricultural Economics Division, IARI delivered a lecture on the contribution of women in agriculture and women empowerment.

Swachh Bharat Abhiyan – Swachhta Pakhwada (16-31 December, 2020)

Under the aegis of Swachh Bharat Abhiyan, the Swachhta Pakhwada was organized at ICAR – NIPB during 16-31 December, 2020. During





the two-week long program many events were organized at the institute, nearby residential areas and market places. The main theme of all the programs was to sensitize the residents of the campus to the importance of cleanliness

and encouraging them to adopt an environment friendly behavior in their daily routine.



Vigilance Awareness Week (27 October - 2 November, 2020)

Vigilance Awareness Week (VAW) was observed at ICAR-NIPB from October 27, 2020 to November 02, 2020.

All the programs were held via online mode. On the first day the entire staff of ICAR-NIPB took pledge to promote integrity, transparency and accountability. The Director (Incharge), Dr. Sarvjeet Kaur, emphasized and spoke about the importance of Vigilance in our daily office routine. The SAO, Mr. Harshit Agarwal, spoke



on the genesis of vigilance awareness, office procedures, Institute activities and the need of being vigilant in routine follow ups. He also apprised the staff of why it is important and how to be vigilant. Under the focus theme Vigilant India-Prosperous India, Mr. S. K. Sinha, CAO, IARI, New Delhi delivered a talk on preventive vigilance. He also said that preventive vigilance is the best governance.

LIST OF PUBLICATIONS

Research Articles

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2. Awakale P, Tiwari S, Vinod, Dalal M (2020) Osmotic stress induced root growth and regulation of root development related genes in progenitors of wheat. *Indian Journal of Genetics* 80(3): 334-338.
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12. Carpenter SCD, Mishra P, Ghoshal C, Dash PK, Wang L, Midha S, Laha GS,Bogdanove AJ, Rai R (2020) An xa5 resistance gene-breaking Indian strain of the rice bacterial blight pathogen *Xanthomonas oryzae* pv. *oryzae* is nearly identical to a Thai Strain. *Frontiers in Microbiology* 11: 579504.

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